Clinical Infectious Diseases

IDSA GUIDELINES

Infectious Diseases Society of America 2026 Guidance on the Treatment of Antimicrobial-Resistant Gram-Negative Infections

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Background. The Infectious Diseases Society of America (IDSA) is committed to delivering timely, evidence-informed guidance on the management of antimicrobial-resistant (AMR) infections. This updated IDSA AMR Guidance document provides treatment suggestions for infections caused by extended-spectrum β-lactamase–producing Enterobacterales (ESBL-E), AmpC β-lactamase–producing Enterobacterales (AmpC-E), carbapenem-resistant Enterobacterales (CRE), *Pseudomonas aeruginosa* with difficult-to-treat resistance (DTR *P. aeruginosa*), carbapenem-resistant *Acinetobacter baumannii* (CRAB), and *Stenotrophomonas maltophilia*. This update replaces earlier versions of the IDSA AMR Treatment Guidance.

Methods. A panel of 6 infectious diseases specialists with expertise in AMR convened to address important questions regarding the treatment of infections due to ESBL-E, AmpC-E, CRE, DTR *P. aeruginosa*, CRAB, and *S. maltophilia*. Given geographic variability in AMR epidemiology and antibiotic availability, the IDSA AMR Guidance focuses on clinical practice in the United States.

Results. Preferred and alternative treatment strategies are outlined with accompanying rationales, based on pathogen identification and in vitro susceptibility data. Suggestions addressing empiric therapy, transition to oral therapy, duration of treatment, and other management considerations are briefly discussed. While applicable to both adult and pediatric populations, specific dosing guidance is only provided for adults.

Conclusions. The field of AMR is highly dynamic. This document is current as of 1 March 2026 and will be updated periodically. The most up-to-date version of this document, including date of publication, is available at www.idsociety.org/practice-guideline/amr-guidance/.

Keywords. ESBL; AmpC; carbapenem-resistant Enterobacterales; *Pseudomonas aeruginosa*; CRAB; *Stenotrophomonas maltophilia*.

Antimicrobial resistance (AMR) represents an escalating global health crisis. In 2019, an estimated 4.95 million deaths were associated with AMR pathogens worldwide, including 1.27 million deaths directly attributable to resistant infections [1]. Projections suggest that, without effective intervention, AMR could result in up to 10 million deaths annually by 2050 [2]. In the United States alone, over 2.8 million AMR infections occur each year, leading to more than 35 000 deaths and imposing an economic burden of approximately $20–35 billion [3].

Recognizing the limitations of traditional practice guidelines in rapidly evolving areas such as AMR, the Infectious Diseases Society of America (IDSA) has developed narrowly focused guidance documents. This AMR Treatment Guidance document is meant to address important clinical questions where high-quality evidence may be limited or emerging. Developed by a small panel of infectious diseases experts, it is informed by a comprehensive (though not necessarily systematic) review of the literature, combined with clinical expertise and consensus opinion.

This document provides suggestions on the treatment of infections caused by extended-spectrum β-lactamase–producing Enterobacterales (ESBL-E), AmpC β-lactamase–producing Enterobacterales (AmpC-E), carbapenem-resistant Enterobacterales (CRE), *Pseudomonas aeruginosa* with difficult-to-treat resistance (DTR *P. aeruginosa*), carbapenem-resistant *Acinetobacter baumannii* (CRAB), and *Stenotrophomonas maltophilia*. These pathogens are responsible for a broad spectrum of infections encountered in hospitals across the United States and are associated with substantial morbidity and mortality [4].

Received 28 July 2026; published online 8 August 2026

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Clinical Infectious Diseases®

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https://doi.org/10.1093/cid/ciag481

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Management suggestions are presented in a question-and-answer format for each pathogen. In general, preferred and alternative antibiotic options are listed in alphabetical order in the "suggested approach" sections and more nuance regarding the ranking of agents among the preferred and alternative options is provided in the "rationale" sections. While concise summaries of relevant clinical trials, resistance mechanisms, and antimicrobial susceptibility testing (AST) considerations are included, this document is not intended to serve as a comprehensive review of these topics. Furthermore, GRADE methodology (Grading of Recommendations, Assessment, Development, and Evaluation) is not employed. Given international variability in resistance mechanisms and antibiotic availability, treatment suggestions are specific to the US context. This guidance applies to both adult and pediatric populations. Suggested adult antibiotic dosing for patients with normal renal and hepatic function is summarized in Table 1; pediatric dosing suggestions for select antibiotic agents are available elsewhere [5]. Clinical and Laboratory Standards Institute (CLSI) susceptibility criteria are displayed in Table 2. This document reflects the evidence and expert consensus as of 1 March 2026; the most current version is available at www.idsociety.org/practice-guideline/amr-guidance.

GENERAL MANAGEMENT SUGGESTIONS

The management suggestions outlined in this guidance assume identification of the causative pathogen and confirmation of in vitro susceptibility to the proposed agents. When multiple antibiotics demonstrate comparable efficacy, selection should also be informed by factors such as safety and tolerability profile, cost and third-party payer coverage, ease of administration, and institutional formulary availability.

Empiric Therapy

While empiric treatment suggestions are beyond the scope of this guidance, initial antibiotic selection should be informed by the most likely pathogens, the severity of illness, the suspected source of infection, and relevant patient-specific factors (eg, history of severe β-lactam allergy, profound immunosuppression, and underlying renal disease). Additional considerations that should inform empiric therapy include: (1) prior microbiologic data, including organisms and associated AST results within the past 12 months; (2) antibiotic exposure in the preceding 3 months; and (3) local epidemiology and cumulative AST data for the most probable pathogens [9–11]. Empiric regimens should be reassessed and refined once the pathogen has been identified, incorporating AST data and the presence of clinically relevant β-lactamases, when available.

Distinguishing infection from colonization remains essential for all pathogens, as inappropriate antibiotic use promotes AMR and places patients at risk for unnecessary harm [12, 13]. Notably, commonly selected empiric regimens often lack activity against CRAB and *S. maltophilia*. Decisions to include agents targeting these organisms should be individualized and based on a careful assessment of prior culture results, host-specific risk factors, clinical presentation, and the potential for antibiotic-related toxicity.

Duration of Therapy and Transitioning to Oral Therapy

This document does not provide specific suggestions regarding duration of therapy. However, clinicians are advised that treatment durations for infections caused by AMR pathogens generally need not differ from those caused by more susceptible pathogens [14]. Once AST results become available, they may reveal that the empirically selected regimen lacked in vitro activity against the causative organism. This information may influence treatment duration decisions.

For example, uncomplicated urinary tract infections (uUTIs; defined as UTIs confined to the bladder in afebrile women or men) are typically mild and self-limited [15]. If an empiric regimen lacked activity against the uropathogen, but the patient nonetheless exhibited clinical improvement, it is generally not necessary to repeat urine cultures, modify the regimen, or extend the planned treatment course. In contrast, for other infection types, including complicated UTIs (cUTIs), if AST results indicate that the empiric therapy is inactive, a switch to an agent to which the pathogen tested susceptible is suggested, and the treatment duration should be counted from the initiation of active therapy. Of note, a cUTI is defined as an infection of the urinary tract that extends beyond the bladder and includes conditions such as pyelonephritis, febrile or bacteremic UTI, catheter-associated UTI, or prostatitis [15].

As with all infections, host-specific factors—including immune status, adequacy of source control, and clinical response to therapy—should be considered when determining treatment duration. Finally, when feasible, transitioning from intravenous (IV) to oral therapy should be pursued, provided the following criteria are met: (1) the pathogen is susceptible to an appropriate oral agent; (2) the patient is hemodynamically stable; (3) an oral agent anticipated to concentrate at the site of infection is available; and (4) gastrointestinal absorption is expected to be sufficient [16].

INFECTIOUS DISEASES SOCIETY OF AMERICA DISCLAIMER

The IDSA AMR Treatment Guidance is intended to support clinical decision-making but cannot account for all individual patient circumstances. Treatment suggestions represent a synthesis of current scientific and clinical evidence and are provided as an educational resource. However, the content is not continuously updated and may not incorporate the most recent

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Table 1. Suggested Dosing of Antibiotics for the Treatment of Antimicrobial-Resistant Infections in Adults, Assuming Normal Renal and Hepatic Function[a,b,c,d]

Amikacin	uUTI: 15 mg/kg IV as a single dose cUTI: 15 mg/kg IV once; subsequent doses and dosing interval based on pharmacokinetic evaluation Additional information in Supplementary Material
Ampicillin–sulbactam	Total daily dose of 9 g of sulbactam via one of the following regimens: 9 g of ampicillin–sulbactam (6 g ampicillin, 3 g sulbactam) IV every 8 h, infused over 4 h **OR** 27 g of ampicillin–sulbactam (18 g ampicillin, 9 g sulbactam) IV as a continuous infusion over 24 h Additional information in Supplementary Material
Aztreonam–avibactam	2.67 g of aztreonam–avibactam (2 g aztreonam, 0.67 g avibactam) IV once, infused over 3 h as a loading dose **THEN** 2 g of aztreonam–avibactam (1.5 g aztreonam, 0.5 g avibactam) IV every 6 h, infused over 3 h as maintenance dosing
Cefepime	uUTI: 1 g IV every 8 h, infused over 30 min All other infections: 2 g IV, infused over 30 min as a loading dose **THEN** 2 g IV every 8 h, infused over 3 h
Cefepime–enmetazobactam	2.5 g (2 g cefepime, 0.5 g enmetazobactam) IV every 8 h, infused over 2 h CrCL ≥130 mL/min: 2.5 g IV every 8 h, infused over 4 h
Cefiderocol	2 g IV every 8 h, infused over 3 h CrCL ≥120 mL/min: 2 g IV every 6 h, infused over 3 h
Ceftazidime–avibactam	2.5 g (2 g ceftazidime, 0.5 g avibactam) IV every 8 h, infused over 3 h
Ceftazidime–avibactam PLUS aztreonam	Ceftazidime–avibactam: 2.5 g (2 g ceftazidime, 0.5 g avibactam) IV every 8 h, infused over 3 h **PLUS** (administered simultaneously via Y-site administration) Aztreonam: 2 g IV every 8 h, infused over 3 h Additional information in Supplementary Material
Ceftolozane–tazobactam	uUTI: 1.5 g (1 g ceftolozane, 0.5 g tazobactam) IV every 8 h, infused over 1 h All other infections: 3 g (2 g ceftolozane, 1 g tazobactam) IV every 8 h, infused over 3 h
Ciprofloxacin	uUTI: 400 mg IV every 12 h OR 500 mg PO every 12 h All other infections: 400 mg IV every 8 h OR 750 mg PO every 12 h
Colistin	Refer to international consensus guidelines on polymyxins[e]
Eravacycline	1 mg/kg IV every 12 h
Ertapenem	1 g IV every 24 h, infused over 30 min Additional information in Supplementary Material
Fosfomycin	uUTI: 3 g PO as a single dose cUTI: 6 g IV every 8 h, infused over 1 h
Gentamicin	uUTI: 5 mg/kg per dose IV as a single dose cUTI: 7 mg/kg IV once; subsequent doses and dosing interval based on pharmacokinetic evaluation Additional information in Supplementary Material
Gepotidacin	uUTI: 1.5 g PO every 12 h
Imipenem–cilastatin	uUTI: 500 mg imipenem IV every 6 h, infused over 30 min All other infections: 500 mg imipenem IV every 6 h, infused over 3 h (if feasible) OR 1000 mg imipenem IV every 8 h, infused over 1 h CrCL ≥90 mL/min: 1 g imipenem IV every 6 h, infused over 1 h Dosing is based on the imipenem component. An equivalent amount of cilastatin is present when dosing (ie, 500 mg imipenem is formulated with 500 mg cilastatin) Additional information in Supplementary Material
Imipenem–cilastatin–relebactam	1.25 g (500 mg imipenem, 500 mg cilastatin, 250 mg relebactam) IV every 6 h, infused over 30 min
Levofloxacin	750 mg IV/PO every 24 h
Meropenem	uUTI: 1 g IV every 8 h, infused over 30 min All other infections: 1–2 g IV every 8 h, infused over 3 h (if feasible); 2 g preferred for obesity, central nervous system infections, and extracorporeal membrane oxygenation or CrCL ≥130 mL/min Additional information in Supplementary Material
Meropenem–vaborbactam	4 g (2 g meropenem, 2 g vaborbactam) IV every 8 h, infused over 3 h
Minocycline	200 mg IV/PO every 12 h
Nitrofurantoin	Macrocrystal/monohydrate (Macrobid®): 100 mg PO every 12 h Oral suspension: 50 mg PO every 6 h
Pivmecillinam	uUTI: 370 mg pivmecillinam PO every 8 h (185 mg pivmecillinam is equivalent to 200 mg pivmecillinam hydrochloride)
Plazomicin	uUTI: 15 mg/kg IV as a single dose cUTI: 15 mg/kg IV once; subsequent doses and dosing interval based on pharmacokinetic evaluation Additional information in Supplementary Material
Polymyxin B	Refer to international consensus guidelines on polymyxins[e]
Sulbactam–durlobactam	2 g (1 g sulbactam, 1 g durlobactam) IV every 6 h, infused over 3 h CrCL ≥130 mL/min: 2 g (1 g sulbactam, 1 g durlobactam) IV every 4 h, infused over 3 h Additional information in Supplementary Material

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Table 1. Continued

Sulopenem etzadroxil–probenecid	500 mg sulopenem etzadroxil and 500 mg probenecid PO every 12 h
Tigecycline	200 mg IV once as a loading dose **THEN** 100 mg IV every 12 h
Tobramycin	uUTI: 5 mg/kg/dose IV as a single dose cUTI: 7 mg/kg IV once; subsequent doses and dosing interval based on pharmacokinetic evaluation Additional information in Supplementary Material
Trimethoprim–sulfamethoxazole	uUTI: 160 mg (trimethoprim component) IV/PO every 12 h All other infections: 8–15 mg/kg/day (trimethoprim component) IV/PO divided every 8 to 12 h Additional information in Supplementary Material

Abbreviations: AMR, antimicrobial-resistant; CrCl, creatinine clearance; cUTI, complicated urinary tract infection; IDSA, Infectious Diseases Society of America; IV, intravenous; PO, orally; uUTI, uncomplicated urinary tract infection.

[a]Dosing suggestions limited to organisms and infectious syndromes discussed in the 2026 IDSA AMR Treatment Guidance document.

[b]Dosing suggested for several agents may differ from dosing recommended by the US Food and Drug Administration.

[c]For suggested pediatric dosing of newer beta-lactam agents refer to Lockowitz et al [5].

[d]For suggested dosing of central nervous system infections refer to Hsu et al [6].

[e]Tsuji et al [7].

data available at the time it is accessed, as new evidence may emerge after development and prior to publication or review.

This guidance is not intended to be comprehensive or to represent the only appropriate approach to treatment. Clinical judgment remains paramount, and application of this guidance should be individualized based on the specific clinical context and patient characteristics.

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SECTION 1: EXTENDED-SPECTRUM β-LACTAMASE–PRODUCING ENTEROBACTERALES

ESBLs are enzymes that inactivate most penicillins, cephalosporins, and aztreonam. ESBL-Es generally remain susceptible to carbapenems. ESBLs do not inactivate non-β-lactam agents (eg, ciprofloxacin, trimethoprim–sulfamethoxazole [TMP-SMX], gentamicin, and doxycycline). However, organisms carrying ESBL genes often harbor additional genes or mutations in genes expanding their resistance to a broad range of antibiotics.

Any gram-negative organism has the potential to carry ESBL genes; however, they are most prevalent in *Escherichia coli*, *Klebsiella pneumoniae*, and *Klebsiella oxytoca* [17–19]. CTX-M enzymes, particularly CTX-M-15, are the most common ESBLs in the United States [18]. ESBLs other than CTX-M enzymes with unique hydrolyzing abilities are also present, including variants of TEM and SHV β-lactamases with amino acid substitutions [20–23]. Routine ESBL testing is not performed by most clinical microbiology laboratories and is not considered compulsory by the CLSI [8]. Rather, *E. coli*, *K. pneumoniae*, or *K. oxytoca* isolates resistant to ceftriaxone (ie, ceftriaxone minimum inhibitory concentrations [MICs] ≥4 μg/mL) is a threshold commonly acknowledged as a surrogate for ESBL production [18, 24–26]. Treatment suggestions for ESBL-E infections assume that in vitro activity of preferred and alternative antibiotics has been demonstrated.

Question 1.1: What are Preferred Antibiotics for the Treatment of Uncomplicated Urinary Tract Infections Caused by Extended-Spectrum β-Lactamase–Producing Enterobacterales?

Suggested Approach

Aminoglycosides (as a single dose), gepotidacin, nitrofurantoin, pivmecillinam, sulopenem, and TMP-SMX (listed alphabetically and in no preferential order) are preferred treatment options for uUTIs caused by ESBL-E. Oral fosfomycin is an alternative option for uUTIs caused by ESBL-producing *E. coli*. When resistance, availability, or intolerance preclude the use of preferred or alternative agents, fluoroquinolones (eg, ciprofloxacin and levofloxacin), cefepime–enmetazobactam, or carbapenems (eg, ertapenem, meropenem, and imipenem) may be used.

Rationale

Because treatment courses for uUTI are brief and access to susceptibility testing, particularly for newer agents, remains limited, therapy for uUTI is often initiated (and completed) before ESBL production is known and in the absence of susceptibility data. This underscores the importance of reviewing prior urine culture results from individual patients and understanding local ESBL-E epidemiology to determine if ESBL-E coverage may be warranted. The prevalence of ESBL-producing *E. coli* among UTIs in the United States is increasing and estimated at approximately 17%, with substantial regional variability [27]. Of note, for several preferred and alternative agents for the treatment of ESBL-E uUTI, surveillance data indicate that susceptibility exceeds 90% making them reasonable for empiric use in the absence of confirmatory susceptibility data.

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Table 2. 2026 Clinical and Laboratory Standards Institute Susceptible Breakpoints for Select Gram-Negative Organisms and Antibiotic Combinations as Suggested in the 2026 IDSA AMR Guidance Document[a]

Antibiotic	Enterobacterales (μg/mL)	*Pseudomonas aeruginosa* (μg/mL)	Carbapenem-Resistant *Acinetobacter baumannii* (μg/mL)	*Stenotrophomonas maltophilia* (μg/mL)
Amikacin	≤4	≤16[b]	…	…
Ampicillin–sulbactam	…	…	≤8/4	…
Aztreonam	≤4	≤8	…	…
Aztreonam–avibactam	≤4/4	…	…	…[c]
Cefepime	≤2[d]	≤8	…	…
Cefepime–enmetazobactam	≤8/8[e]	…	…	…
Cefiderocol	≤4	≤4	≤4	≤1
Ceftazidime	≤4	≤8	…	…
Ceftazidime–avibactam	≤8/4	≤8/4	…	…
Ceftolozane–tazobactam	≤2/4	≤4/4	…	…
Ciprofloxacin	≤0.25	≤0.5	…	…
Colistin or polymyxin B	…[f]	…[f]	…[f]	…
Ertapenem	≤0.5	…	…	…
Fosfomycin (intravenous)	≤8[e,g]			
Fosfomycin tromethamine (oral)	≤64[h]	…	…	…
Gentamicin	≤2	…	…	…
Gepotidacin	≤16[b,e]	…	…	…
Imipenem	≤1	≤2	…	…
Imipenem–relebactam	≤1/4	≤2/4	…	…
Levofloxacin	≤0.5	≤1	…	≤2
Meropenem	≤1	≤2	…	…
Meropenem–vaborbactam	≤4/8	…	…	…
Minocycline	≤4	…	≤1	≤1
Nitrofurantoin	≤32[b]	…	…	…
Piperacillin–tazobactam	≤8/4[i]	≤16/4	…	…
Pivmecillinam (mecillinam)	≤8[b]	…	…	…
Plazomicin	≤2	…	…	…
Sulbactam–durlobactam	…	…	≤4/4	…
Sulopenem	≤0.25[e]	…	…	…
Tigecycline	≤2[e]	…	…	…
Trimethoprim–sulfamethoxazole	≤2/38	…	…	≤2/38
Tobramycin	≤2	≤1	…	…

Abbreviations: AMR, antimicrobial-resistant; CLSI, Clinical and Laboratory Standards Institute; *E. coli*, *Escherichia coli*; FDA, Food and Drug Administration; IDSA, Infectious Diseases Society of America; *K. pneumoniae*, *Klebsiella pneumoniae*; MIC, minimum inhibitory concentration.

[a]For full details of antibiotic susceptibility testing interpretations refer to Clinical and Laboratory Standards Institute [8]. CLSI M100 document is updated annually; susceptibility criteria subject to changes in 2027.

[b]Breakpoints only available for urinary tract infections.

[c]Neither CLSI nor FDA breakpoints are available.

[d]Cefepime MICs of 4–8 μg/mL are susceptible dose dependent.

[e]CLSI breakpoints are not available. FDA breakpoints are listed instead.

[f]No susceptible category for colistin or polymyxin B; MICs ≤2 μg/mL are categorized as intermediate.

[g]The *E. coli* breakpoint is ≤8 μg/mL; the *K. pneumoniae* breakpoint is ≤32 μg/mL.

[h]Applies to *E. coli* urinary tract isolates only.

[i]Piperacillin–tazobactam MICs of 16/4 μg/mL are susceptible dose dependent.

Aminoglycosides are nearly exclusively eliminated by the renal route. A systematic review of 13 studies including 13 804 patients demonstrated a pooled microbiologic cure of approximately 95% with single-dose aminoglycoside therapy, with no recurrences in 73% of patients by day 30 [28]. This was further supported by an observational study of 13 patients receiving single-dose aminoglycoside therapy [29]. Although robust clinical trial data are lacking, given the relatively high likelihood of treatment success and low likelihood of nephrotoxicity with single dose treatment, the panelists consider single-dose aminoglycoside therapy a preferred approach for the treatment of ESBL-E uUTI.

Gepotidacin, the first triazaacenaphthylene antibiotic, was approved in March 2025 for the treatment of uUTI. It is an oral antibiotic that inhibits bacterial DNA replication via unique binding sites on DNA gyrase and topoisomerase IV, such that single target-specific mutations that may cause resistance to agents such as ciprofloxacin or levofloxacin do not confer resistance to gepotidacin [30]. Gepotidacin demonstrates activity against more than 90% of ESBL-E isolates

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[31, 32]. In 2 randomized clinical trials comparing 5 days each of gepotidacin and nitrofurantoin, treatment success among the subgroup of women with ESBL-producing *E. coli* uUTI was numerically more favorable for gepotidacin (70% [35/50] vs 63% [25/40] in one trial; 68% [23/34] vs 40% [10/25] in the second trial) [33]. Collectively, these data suggest gepotidacin is at least as effective as another preferred agent for the treatment of ESBL-E uUTI. The most common adverse event with gepotidacin is mild-to-moderate diarrhea [33, 34]. Given gepotidacin's activity against *Neisseria gonorrhoeae*, clinicians should remain cognizant that excessive use could potentially contribute to the selection of resistant *N. gonorrhoeae* strains [35].

Nitrofurantoin retains high activity against ESBL-E, with susceptibility generally exceeding 90% [36–38]. It is well tolerated, with mild gastrointestinal symptoms being the most common adverse event. In a randomized trial, a 5-day course of nitrofurantoin resulted in significantly higher treatment success than single-dose oral fosfomycin among patients with *E. coli* uUTI (70% [171/244] vs 58% [139/241]) [39]. Although ESBL status was not specified, these findings are likely also applicable to ESBL-E uUTI.

Pivmecillinam, an oral penicillin and prodrug of mecillinam that has been used internationally for decades, was approved in the United States in April 2024. Its distinctive structure renders it a poor substrate for most β-lactamases [40]. Pivmecillinam demonstrates activity against over 90% of ESBL-E isolates [41, 42]. Three-day regimens achieve treatment success in approximately 75%–95% of women with uUTI [43, 44]. While observational studies suggest increased failure risk for ESBL-E compared with non-ESBL-E uUTIs, this difference has not been observed with higher dosing (400 mg 3 times daily) [45]. Pivmecillinam is associated with a favorable safety profile, most commonly mild gastrointestinal adverse events [46]. Its narrow spectrum and low propensity for resistance emergence support its role as a preferred agent. Suggested dosing for pivmecillinam, which differs from the Food and Drug Administration (FDA)-approved label, is provided in Table 1.

Sulopenem, administered as the oral prodrug formulation sulopenem etzadroxil, which is coformulated with probenecid, is a novel thiopenem antibiotic that achieved FDA-approval in October 2024. It is active against over 95% of ESBL-E isolates [47]. When administered for 5 days for uUTI, sulopenem had similar treatment success (ie, overall clinical and microbiological response) as a 3-day course of ciprofloxacin for patients with ESBL-E uUTIs (56% [41/73] vs 47% [34/72]) [48]. Mild diarrhea is the most common adverse event associated with sulopenem. Notably, probenecid inhibits tubular secretion and may increase plasma concentrations of other drugs (eg methotrexate, piperacillin, and sulfonamides) [49].

TMP-SMX, as a 3-day course, is effective for ESBL-E uUTI when the infecting isolate is susceptible, with clinical efficacy comparable to 3-day courses of fluoroquinolones [50]. However, susceptibility among ESBL-E urinary isolates in the United States is limited to approximately 50% [51], necessitating close clinical monitoring when used empirically. Short-course therapy is generally well tolerated, though nausea, emesis, and hypersensitivity reactions may occur [52].

Oral fosfomycin as a single dose is an alternative option for uUTI caused by ESBL-producing *E. coli*. Although *E. coli* susceptibility to fosfomycin is not routinely assessed in clinical microbiology laboratories, resistance remains uncommon in the United States [53]. Oral fosfomycin is not suggested for gram-negative organisms other than *E. coli* because of the high prevalence of *fosA* genes, including *K. pneumoniae*, which may enzymatically inactivate fosfomycin and compromise clinical efficacy [54–56]. Admittedly, clinical outcome studies investigating the clinical impact of *fosA* are not available. In a randomized trial, single-dose oral fosfomycin was associated with lower clinical success than a 5-day course of nitrofurantoin among *E. coli* uUTI (58% [139/241] vs 70% [171/244]) [39]. Mild diarrhea is the most common adverse event associated with oral fosfomycin [57]. The clinical benefit of multidose oral fosfomycin for uUTI remains uncertain but may be a reasonable consideration, extrapolating from data in cUTI [58] (Question 1.2). The role of IV fosfomycin for cUTI is also discussed in Question 1.2.

Amoxicillin–clavulanic acid is not suggested for the treatment of ESBL-E uUTI. A randomized clinical trial compared 3-day regimens of amoxicillin–clavulanic acid (500 mg/125 mg twice daily) to ciprofloxacin (250 mg twice daily) for 370 women with *E. coli* uUTI [59]. Clinical cure was observed in 58% (93/160) and 77% (124/162) of the women randomized to the amoxicillin–clavulanic acid and ciprofloxacin arms, respectively. The higher failure with amoxicillin–clavulanic acid may be associated with persistent vaginal bacterial colonization, which occurred in 45% (68/151) and 10% (16/153) of patients in the amoxicillin–clavulanic acid and ciprofloxacin arms, respectively. The proportion of women in the trial infected with ESBL-E isolates is not available. Of note, both agents were administered at dosages lower than suggested (Table 1). Even though clavulanic acid inhibits ESBL enzymes in vitro [60, 61], this may not translate to clinical efficacy, particularly when extrapolating from studies evaluating the efficacy of amoxicillin–clavulanic acid for uUTI, independent of resistance phenotype [59, 62].

The panel suggests avoiding doxycycline for ESBL-E uUTI given its relatively low urinary concentrations, with only 35%–60% excreted unchanged in urine [63]. Limited clinical data supporting its use for UTI derive from small studies that primarily involved *P. aeruginosa*, an organism resistant to oral tetracyclines, thereby limiting their interpretability [64, 65].

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Although cefepime–enmetazobactam, fluoroquinolones (eg, ciprofloxacin and levofloxacin)—if susceptibility has been demonstrated, and carbapenems (eg, ertapenem, meropenem, and imipenem–cilastatin; henceforth, imipenem) are typically effective against ESBL-E uUTI (Question 1.2), the panel suggests reserving these agents for cUTI and other invasive ESBL-E infections, given the availability of multiple effective alternatives for uUTI. Traditional β-lactams (eg, piperacillin–tazobactam, cefepime, and cephamycins) are likewise not designated as preferred options for ESBL-E uUTI; considerations regarding continuation of these agents when empirically initiated for uUTI are addressed in Question 1.4, Question 1.5, and Question 1.6, respectively.

Question 1.2: What are Preferred Antibiotics for the Treatment of Complicated Urinary Tract Infection Caused by Extended-Spectrum β-Lactamase–Producing Enterobacterales?

Suggested Approach

TMP-SMX, ciprofloxacin, or levofloxacin are preferred agents for the treatment of cUTI caused by ESBL-E, when in vitro susceptibility is demonstrated. Cefepime–enmetazobactam or carbapenems (ertapenem, meropenem, or imipenem) are preferred when resistance, intolerance, or toxicities preclude use of TMP-SMX or fluoroquinolones. Intravenous fosfomycin, aminoglycosides, and piperacillin–tazobactam are alternative options for ESBL-E cUTI.

Rationale

TMP-SMX, ciprofloxacin, and levofloxacin are preferred treatment options for ESBL-E cUTI when susceptibility is confirmed, based on their ability to achieve sustained urinary concentrations, supportive clinical trial data, and extensive clinical experience [66–68]. When IV therapy is initiated and susceptibility to TMP-SMX, ciprofloxacin, or levofloxacin is demonstrated, transition to one of these oral agents is preferred.

Cefepime–enmetazobactam or carbapenems are preferred when oral TMP-SMX or fluoroquinolones cannot be administered (Question 1.3). Surveillance data indicate that enmetazobactam restores cefepime's activity against almost all ESBL-E isolates [69]. In a randomized trial of cUTI, treatment success was observed in 74% (56/76) of patients receiving cefepime–enmetazobactam for infections due to ESBL-E, compared with 52% (34/66) for those receiving piperacillin–tazobactam [70]. In the absence of direct comparisons between cefepime–enmetazobactam and carbapenems for the treatment of ESBL-E cUTI, the panel considers both to be equally effective options.

Intravenous fosfomycin, approved by the FDA in October 2025, is an alternative treatment for ESBL-E cUTIs. In a randomized clinical trial, among 111 patients with ESBL-E cUTI, 7 days of IV fosfomycin and piperacillin–tazobactam were associated with clinical cure in 93% (52/56) and 93% (51/55) of patients, respectively [71]. In a randomized open-label clinical trial that included 81 patients with presumed ESBL-*E. coli* cUTI and bacteremia, clinical and microbiological cure occurred in 59% (23/39) of patients receiving IV fosfomycin and 71% (30/42) of those receiving meropenem [72]. Although this difference was not statistically significant, fosfomycin did not achieve noninferiority to comparator agents in the overall trial. Additional clinical data are needed before considering IV fosfomycin as a preferred therapy. The sodium content of IV fosfomycin exceeds recommended daily intake and warrants caution in older adults and patients at risk for heart failure [73]. The panel suggests IV fosfomycin as an alternative treatment option for cUTIs caused by *E. coli*. The FDA susceptibility criteria exist for IV fosfomycin against *K. pneumoniae*, in contrast to oral fosfomycin (Table 2). However, *K. pneumoniae* frequently harbors *fosA*, which may inactivate fosfomycin [54–56]. Reassuringly, in a subgroup analysis of a clinical trial including patients with *K. pneumoniae* cUTI, clinical cure was achieved in 93% (25/27) and 100% (25/25) of patients randomized to 7 days of IV fosfomycin or piperacillin–tazobactam, respectively [71]. Pending additional supportive clinical data, the panel advises careful consideration when prescribing IV fosfomycin for ESBL-producing *K. pneumoniae* infections.

Aminoglycosides are alternative options for ESBL-E cUTI. Although expected to be effective as they achieve high renal parenchymal concentrations [74], duration-dependent nephrotoxicity should be carefully considered [75, 76]. In a randomized clinical trial of 609 patients with cUTI (28% caused by ESBL-E), clinical cure occurred in 89% (170/191) and 90% (178/197) of patients receiving plazomicin and meropenem, respectively [77]. Acute kidney injury occurred in 7% (21/300) of patients receiving plazomicin compared with 4% (12/297) receiving meropenem. Among patients with ESBL-E cUTI, microbiological eradication was achieved in 82% (42/51) and 75% (45/60) of patients, respectively. Other aminoglycosides are likely equally effective if susceptibility is demonstrated [78–80]. A randomized clinical trial of 61 patients with cUTI (including 40 with ESBL-E cUTI) demonstrated similar clinical cure between those receiving 3 doses of amikacin 1 g every 48 hours versus meropenem for 7 days [78]. Aminoglycosides may be reasonable to consider for completing treatment courses (eg, transitioning from another agent for terminal doses) given their prolonged duration of activity in the renal cortex and the convenience of once-daily dosing [79, 81]. The role of piperacillin–tazobactam as an alternative treatment for cUTI is summarized in Question 1.4.

In a randomized trial investigating sulopenem for the treatment of cUTI, both arms received 5 initial days of IV therapy (ie, sulopenem or ertapenem) prior to transitioning to oral therapy [82]. Sulopenem did not achieve noninferiority for the primary endpoint of combined clinical and microbiologic response at day 21 (68% [301/444] vs 74% [325/440]). Clinical success, however, was similar (89% [397/444] vs 88%

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[389/440]), as was overall favorable response among patients with ESBL-positive infections (72% [79/110] vs 68% [85/125]). Given the inclusion of 5 initial days of IV sulopenem in this trial, oral sulopenem is suggested as an alternative option for ESBL-E cUTI only after initial clinical improvement has been observed. Notably, IV sulopenem is not currently FDA-approved for clinical use.

Although oral fosfomycin is not suggested as initial therapy for cUTI because of limited renal parenchymal concentrations, it may be considered for transition therapy in select circumstances for ESBL *E. coli* cUTI when other oral options are unavailable. Clinical trials evaluating multidose oral fosfomycin following short courses of IV therapy demonstrated similar clinical cure compared with fluoroquinolones; however, ESBL-E representation was limited, and IV treatment durations prior to initiation of oral fosfomycin often approximated the total duration typically required for cUTI [58, 83]. Further data are needed before oral fosfomycin can be considered a preferred option for transition therapy for cUTIs.

Question 1.3: What are Preferred Antibiotics for the Treatment of Infections Outside of the Urinary Tract Caused by Extended-Spectrum β-Lactamase–Producing Enterobacterales?

Suggested Approach

Ertapenem, imipenem, and meropenem are preferred agents for the treatment of ESBL-E infections outside of the urinary tract. Imipenem or meropenem are preferred for patients who are critically ill or those with hypoalbuminemia. Cefepime–enmetazobactam is an alternative option for ESBL-E infections outside of the urinary tract. Once an appropriate clinical response has been achieved, transition to oral ciprofloxacin, levofloxacin, or TMP-SMX may be considered, provided in vitro susceptibility is demonstrated.

Rationale

Carbapenems are a preferred treatment for invasive ESBL-E infections based on an open-label clinical trial that enrolled 391 patients with ESBL-E bloodstream infections (86% with confirmed ESBL genes) [84]. Patients were randomized to receive either piperacillin–tazobactam 4.5 g IV every 6 hours or meropenem 1 g IV every 8 hours, administered as 30-minute infusions. Thirty-day survival occurred in 88% (164/187) of patients in the piperacillin–tazobactam arm compared with 96% (184/191) in the meropenem arm. These trial data were subsequently reanalyzed to include only patients with clinical isolates for which piperacillin–tazobactam MICs were ≤16 μg/mL as determined by broth microdilution, the reference standard for AST [24]. Among the 320 patients (82%) with isolates available for retesting, 30-day survival occurred in 91% (134/147) of patients receiving piperacillin–tazobactam and 96% (149/155) of those receiving meropenem. Although the absolute risk difference was attenuated and no longer statistically significant in the reanalysis (95% confidence interval, −1% to 11%) [24], the panel continues to favor carbapenem therapy as the preferred treatment for ESBL-producing bloodstream infections given the consistent direction of the mortality difference overall and across individual subgroup analyses. Additional limitations of piperacillin–tazobactam that suggest caution with its administration for invasive infections due to ESBL-E are discussed in Question 1.4. The panel further suggests extrapolating evidence from ESBL-E bloodstream infections to other common sites of infection, including intra-abdominal infections, skin and soft tissue infections, and pneumonia.

For patients who are critically ill and/or have hypoalbuminemia, imipenem or meropenem is preferred over ertapenem. In contrast to imipenem and meropenem, ertapenem is highly protein bound, resulting in a prolonged serum half-life [85]. In patients with hypoalbuminemia, the unbound fraction of ertapenem increases, leading to enhanced clearance and a shortened serum half-life [85–88]. In an observational study of 279 patients with Enterobacterales infections, hypoalbuminemia (defined as serum albumin <2.5 g/dL) was associated with a higher odds of 30-day mortality among patients treated with ertapenem compared with those receiving imipenem or meropenem [89].

Clinical data comparing ertapenem with other carbapenems in critically ill patients are limited and conflicting [87, 90]. However, given concerns regarding variable and unpredictable interpatient pharmacokinetics (PK) [87, 91], the panel suggests meropenem or imipenem rather than ertapenem as initial therapy for critically ill patients with ESBL-E infections. Although higher doses of ertapenem (eg, 1.5 g daily) or more frequent dosing (eg, every 12 hours) may mitigate some probability-of-target-attainment concerns in obese or critically ill patients and those with hypoalbuminemia, data demonstrating improved outcomes with these alternative dosing strategies are limited [91–93].

Cefepime–enmetazobactam represents an alternative for the treatment of ESBL-E infections outside the urinary tract. This suggestion is supported by extensive clinical experience with cefepime for invasive infections; particularly when administered at high doses via extended infusion, combined with the inhibitory activity of enmetazobactam against ESBLs [69], and PK/pharmacodynamic (PD) modeling demonstrating adequate intrapulmonary penetration and a high probability of target attainment for both agents against Enterobacterales isolates with MICs ≤8/8 μg/mL [94]. Compared with tazobactam, enmetazobactam contains a methyl substitution that confers a net neutral charge and zwitterionic structure, enhancing bacterial cell penetration and increasing periplasmic concentrations, thereby complementing cefepime's zwitterionic oxyiminocephalosporin activity [95]. Clinical data evaluating cefepime–enmetazobactam for ESBL-E infections outside the

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urinary tract remain limited. Although its pharmacologic properties suggest expected efficacy for invasive ESBL-E infections, additional evidence is needed before it can be suggested as a preferred therapy.

Given the high oral bioavailability and sustained serum exposures achieved with oral TMP-SMX and fluoroquinolones, transition to these agents should be considered for patients with ESBL-E infections when all of the following criteria are met: (1) in vitro susceptibility to the selected agent is confirmed, (2) the patient is hemodynamically stable, and (3) there are no concerns regarding impaired gastrointestinal absorption. In contrast, oral step-down therapy with nitrofurantoin, fosfomycin, amoxicillin–clavulanate, omadacycline, or doxycycline is not suggested for ESBL-E bloodstream infections due to concerns with achieving adequate serum concentrations and/or limited clinical data supporting their efficacy for ESBL-E bacteremia.

Until more PK/PD analyses and clinical data are available, the use of oral sulopenem as transition therapy for ESBL-E bloodstream infections is not suggested. In a subgroup analysis of 77 patients with cUTI and associated bacteremia enrolled in a randomized clinical trial, treatment success occurred in 57% (25/44) and 65% (28/43) of those transitioned to oral sulopenem (from IV sulopenem) and oral ciprofloxacin or amoxicillin–clavulanate (from IV ertapenem), respectively [82]. In a randomized clinical trial of patients with intra-abdominal infections, clinical success was 82% (204/249) for sulopenem-treated patients (ie, 5 days of IV sulopenem followed by oral sulopenem) versus 88% (233/265) in the alternative arm (ie, 5 days of IV ertapenem followed by oral ciprofloxacin/metronidazole or amoxicillin–clavulanate) [96].

Question 1.4: Is There a Role for Piperacillin–Tazobactam in the Treatment of Infections Caused by Extended-Spectrum β-Lactamase–Producing Enterobacterales?

Suggested Approach

If piperacillin–tazobactam is initiated as empiric therapy for an uUTI caused by an organism subsequently identified as an ESBL-E and the patient demonstrates clinical improvement, neither a change in therapy nor an extension of treatment duration is necessary. The panel suggests piperacillin–tazobactam as an alternative treatment option for cUTI caused by ESBL-E in patients who are not critically ill and who do not have concomitant bacteremia. In contrast, piperacillin–tazobactam is not suggested for the treatment of ESBL-E infections at sites outside of the urinary tract, even when in vitro susceptibility to piperacillin–tazobactam is demonstrated.

Rationale

Piperacillin–tazobactam often demonstrates in vitro activity against ESBL-E [97]. However, there are several concerns regarding tazobactam's ability to function as an effective β-lactamase inhibitor for ESBL-E. First, piperacillin–tazobactam MIC testing may be inaccurate and/or poorly reproducible when ESBL enzymes are present, or in the presence of other β-lactamase enzymes such as OXA-1, making it unclear if an isolate that tests susceptible to this agent is reliably susceptible [24, 98–100]. Second, preclinical data indicate that with increased bacterial inoculum which may be present in certain clinical infections (eg, abscesses), regrowth of ESBL-E isolates is more likely with piperacillin–tazobactam than with meropenem, although the clinical implications of these findings are unclear [101–103]. Third, the effectiveness of tazobactam may be diminished for organisms with increased expression of ESBL enzymes or by the presence of multiple ESBL or other β-lactamases (eg, AmpC enzymes) [95]. This may in part be due to the low concentration of tazobactam relative to the amount of piperacillin [104]. More specifically, in a 4.5 g dose of piperacillin–tazobactam there is an 8:1 ratio of piperacillin to tazobactam (ie, 4 g of piperacillin and 0.5 g of tazobactam). In contrast, in a 3 g dose of ceftolozane–tazobactam there is a 2:1 ratio of ceftolozane to tazobactam. Finally, the piperacillin–tazobactam breakpoint for Enterobacterales is primarily based on PK/PD considerations of piperacillin dosing strategies and not on whether a fixed concentration of 4 μg/mL of tazobactam in testing wells is reflective of the restorative ability of commonly prescribed tazobactam dosages to reestablish the activity of piperacillin in infections with ESBL production.

If piperacillin–tazobactam was initiated as empiric therapy for uUTI caused by an organism later identified as an ESBL-E and clinical improvement occurs, no change or extension of antibiotic therapy is necessary, given the high urinary concentration of piperacillin–tazobactam [105].

Piperacillin–tazobactam is an alternative treatment for cUTI caused by ESBL-E. A systematic review of 1 randomized clinical trial [106] and 6 observational studies including 1156 patients treated with either piperacillin–tazobactam or carbapenems for ESBL-E UTIs found comparable treatment success between the regimens [107]. The included clinical trial of 66 patients with ESBL-E cUTI demonstrated similar treatment success with piperacillin–tazobactam and ertapenem, at 94% (31/33) and 97% (32/33), respectively [106]. In a separate randomized clinical trial involving 111 patients with ESBL-E cUTI, IV fosfomycin and piperacillin–tazobactam were associated with 93% (52/56) and 93% (51/55) clinical cure [71]. However, among the subgroup of 231 patients with ESBL-E bloodstream infections originating from the urinary tract enrolled in a clinical trial comparing piperacillin–tazobactam with meropenem, survival was numerically lower in the piperacillin–tazobactam arm (93% [95/102] vs 97% [124/128]) [84]. The panel does not suggest piperacillin–tazobactam for the treatment of ESBL-E cUTI in patients who are critically ill or with concomitant bacteremia.

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Piperacillin–tazobactam is also not suggested for the treatment of invasive ESBL-E infections based on the results of the aforementioned clinical trial which indicated inferior results with piperacillin–tazobactam compared with carbapenem therapy [84]. A second trial investigating the role of piperacillin–tazobactam for the treatment of ESBL-E bloodstream infections is ongoing [108]. A 2023 meta-analysis including 2786 patients across the clinical trial [84] and 25 observational studies found no difference in survival between β-lactam/β-lactamase inhibitors (not limited to piperacillin–tazobactam) and carbapenems [109]. Two subsequent observational studies, one including 644 patients [110] and another including 264 patients [111] had conflicting findings with one suggesting no difference in survival (92% [283/309] vs 92% [308/335]) [110] and the second indicating lower survival in the empiric piperacillin–tazobactam group (86% [172/200] vs 96% [61/64]) [111]. Given the concerns with preclinical data for piperacillin–tazobactam against higher burden infections and clinical trial results indicating suboptimal outcomes with piperacillin–tazobactam for ESBL-E bloodstream infections, piperacillin–tazobactam is not suggested as a preferred or alternative treatment for ESBL-E invasive infections.

Question 1.5: Is There a Role for Cefepime in the Treatment of Infections Caused by Extended-Spectrum β-Lactamase–Producing Enterobacterales?

Suggested Approach

If cefepime was initiated as empiric therapy for uUTI caused by an organism later identified as an ESBL-E and clinical improvement occurs, no change or extension of antibiotic therapy is necessary. The panel suggests avoiding cefepime for the treatment of cUTI or infections outside of the urinary tract caused by ESBL-E, even if susceptibility to cefepime is demonstrated.

Rationale

ESBLs commonly hydrolyze cefepime [95, 112, 113] in the absence of protection with a β-lactamase inhibitor (eg, enmetazobactam). Furthermore, even if ESBL-producing isolates test susceptible to cefepime, cefepime MIC testing may be inaccurate and/or poorly reproducible with commercial AST methods [114]. Clinical trials comparing the outcomes of patients with ESBL-E bloodstream infections treated with cefepime or carbapenems have not been conducted.

If cefepime was initiated as empiric therapy for uUTI caused by an organism later identified as an ESBL-E and clinical improvement occurs, no change or extension of antibiotic therapy is necessary, given the high urinary concentration of cefepime [115].

Although some observational studies evaluating the role of cefepime versus carbapenems for ESBL-E cUTI suggest no difference in clinical outcomes [116, 117], in a clinical trial of ESBL-E cUTI the cefepime arm was terminated early because of a clinical failure signal with cefepime (2 g IV every 12 hours), despite all isolates having cefepime MICs of 1–2 μg/mL; with clinical success in 33% (2/6) and 97% (32/33) of patients receiving cefepime and ertapenem, respectively [106]. The panel suggests avoiding cefepime for the treatment of ESBL-E cUTI.

Observational studies comparing cefepime and carbapenems for the treatment of invasive ESBL-E infections demonstrated either no difference in outcomes or poorer outcomes with cefepime [118–122]. Clinical trials comparing cefepime to carbapenems for ESBL-E bloodstream infections have not been conducted. However, a subgroup analysis of patients with ESBL-E pneumonia enrolled in a clinical trial identified clinical success in 69% (9/13) versus 100% (10/10) of patients randomized to cefepime (2 g IV every 8 hours) and imipenem (500 mg IV every 6 hours), respectively [123]. The panel suggests avoiding cefepime for the treatment of invasive ESBL-E infections.

Question 1.6: Is There a Role for the Cephamycins in the Treatment of Infections Caused by Extended-Spectrum β-Lactamase–Producing Enterobacterales?

Suggested Approach

Cephamycins are not suggested for the treatment of ESBL-E infections until more clinical outcomes data using cefoxitin or cefotetan are available and optimal dosing has been defined.

Rationale

Cephamycins are a subclass of cephalosporins that are generally stable against hydrolysis by ESBL enzymes [124, 125]. In the United States, cefoxitin and cefotetan are the only available cephamycins; both are administered IV. At least 10 observational studies have compared clinical outcomes of patients with ESBL-E infections treated with cephamycins versus carbapenems, most commonly UTIs or bloodstream infections from urinary sources [126–132]. Eight of these studies reported no differences in clinical outcomes, whereas 2 demonstrated poorer outcomes with cephamycins [127, 132]—including the largest study which identified 30-day survival of 71% (94/132) versus 87% (115/132) for cephamycins and carbapenems, respectively, among patients with ESBL-E bacteremia [132]. Interpretation of these studies is limited by the observational design of all studies, heterogeneity in infection sources, substantial selection bias, and variability in cephamycin agents, dosing strategies, and durations of therapy.

Although not available for clinical use in the United States, the most encouraging data supporting the effectiveness of cephamycins against ESBL-E are for cefmetazole [130, 131, 133, 134], which is being compared with meropenem in a clinical trial of patients with ESBL-E bacteremia [135].

Many cephamycins evaluated in published studies are not available in the United States, and relatively few patients have received cefoxitin or cefotetan in clinical investigations [129, 136, 137]. Additional clinical data, including optimized dosing

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and administration strategies, are needed before these agents can be endorsed for the treatment of ESBL-E infections. While limited data suggest improved outcomes with high-dose, continuous-infusion cefoxitin (eg, 6 g daily) [136, 137], the IV-only formulation and relatively short half-lives of both agents, particularly cefoxitin, offer no clear feasibility or clinical advantage over preferred agents for the treatment of ESBL-E infections. Based on available data, the panel does not suggest cephamycins for the treatment of ESBL-E infections.

Question 1.7: What is the Role of β-Lactam Agents With Activity Against Carbapenem-Resistant Organisms for the Treatment of Infections Caused by Extended-Spectrum β-Lactamase–Producing Enterobacterales?

Suggested Approach

While effective against ESBL-E infections, the panel suggests that aztreonam–avibactam, ceftazidime–avibactam, meropenem–vaborbactam, imipenem–relebactam, ceftolozane–tazobactam, and cefiderocol be preferentially reserved for treating infections caused by organisms exhibiting carbapenem resistance.

Rationale

Aztreonam–avibactam, ceftazidime–avibactam, meropenem–vaborbactam, imipenem–cilastatin–relebactam (henceforth, imipenem–relebactam), ceftolozane–tazobactam, and cefiderocol exhibit activity against ESBL-E [47, 138]. Avibactam can successfully protect aztreonam and ceftazidime against hydrolysis by ESBL enzymes [18, 139]. Subgroup analysis of clinical trial data support ceftazidime–avibactam's effectiveness against ESBL-E infections [140–144]. The carbapenem component of meropenem–vaborbactam and imipenem–relebactam provide sufficient activity against ESBL-E, even without the addition of a β-lactamase inhibitor.

Ceftolozane–tazobactam is more potent against ESBL-E than piperacillin–tazobactam, with the addition of tazobactam leading to ceftolozane MICs that are several dilutions lower than piperacillin MICs [145–150]. Moreover, ceftolozane appears to have greater stability to hydrolysis by common ESBL enzymes (eg, CTX-M-15) compared with piperacillin, making ceftolozane less reliant than piperacillin on tazobactam's inhibitory properties [151, 152]. Additionally, the ratio of β-lactam to tazobactam present in ceftolozane–tazobactam (2:1) results in greater concentration of tazobactam compared with piperacillin–tazobactam (8:1). In a subgroup analysis of patients with ESBL-E intra-abdominal infections and UTIs from randomized clinical trials, ceftolozane–tazobactam was associated with similar clinical cure as meropenem (97% [76/78] vs 89% [23/26]) [153]. In a subgroup analysis of a clinical trial comparing ceftolozane–tazobactam versus meropenem for ESBL-E pneumonia, 28-day survival was similar between patients receiving ceftolozane–tazobactam (79% [66/84]) and meropenem (71% [52/73]) [154, 155].

Although aztreonam–avibactam, ceftazidime–avibactam, meropenem–vaborbactam, imipenem–relebactam, ceftolozane–tazobactam, and cefiderocol are expected to be effective against ESBL-E infections, the panel suggests that these agents be preferentially reserved for treating carbapenem-resistant organisms or polymicrobial infections with organisms exhibiting carbapenem resistance (eg, ceftolozane–tazobactam for coinfection with DTR *P. aeruginosa* and ESBL-E).

SECTION 2: AMPC β-LACTAMASE–PRODUCING ENTEROBACTERALES

AmpC β-lactamases (herein, referred to as AmpC for brevity) are produced at basal levels by several Enterobacterales species, where they play a physiological role in cell-wall recycling [156]. These enzymes hydrolyze a range of β-lactams—rendering these agents ineffective—either at basal expression levels (eg, cefazolin) or, more notably, when AmpC production is increased above basal levels (eg, ceftriaxone) [157–159]. In Enterobacterales, increased AmpC production arises via 3 main mechanisms:

1. Inducible chromosomal expression of *ampC*
2. Constitutive chromosomal *ampC* hyperexpression due to mutations in regulatory genes
3. Plasmid-mediated *ampC* genes—which may be integrated into the bacterial chromosome in some species

Inducible expression is triggered by exposure to specific β-lactam agents, leading to significant periplasmic AmpC accumulation sufficient to elevate MICs to agents such as aztreonam, cefotaxime, ceftazidime, ceftriaxone, and piperacillin–tazobactam. Clinically, this can result in an isolate that initially tests ceftriaxone-susceptible becoming resistant during therapy, sometimes after only a few doses of the β-lactam [160]. In this guidance, such organisms are classified as having a moderate risk for clinically significant AmpC production.

In contrast, with constitutive chromosomal *ampC* hyperexpression or plasmid-mediated *ampC*, AmpC production remains persistently elevated; isolates typically demonstrate nonsusceptibility to cefotaxime, ceftazidime, and ceftriaxone at baseline. Constitutive hyperexpression may arise in certain *E. coli* or *Shigella* species through mutations in *ampC* promoters or attenuators, or in *ampC* regulatory genes such as *ampD*, *ampR*, or *ampG*. Given the complexity of the *ampC* regulatory network, the prevalence and functional impact of specific mutations on AmpC overproduction is not well defined [161, 162].

Plasmid-mediated *ampC* genes (eg, bla_{ACT}, bla_{CMY}, bla_{DHA}, bla_{FOX}, and bla_{MIR}) may be carried on plasmids (eg, bla_{CMY} in *E. coli*) or integrated into the chromosome (eg, bla_{CMY} in *Citrobacter freundii*) [163]. Their prevalence varies geographically. In US cohorts, plasmid-mediated *ampC* genes are

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detected in <10% of ceftriaxone–nonsusceptible isolates and are largely confined to *E. coli* and *K. pneumoniae* [162, 164]. When present, CMY-type enzymes, particularly CMY-2, predominate in *E. coli*, whereas both CMY- and DHA-type enzymes are produced by *K. pneumoniae* [162, 164]. This guidance focuses on the treatment of infections caused by Enterobacterales species with a moderate likelihood of inducible *ampC* expression, corresponding to the first of the 3 mechanisms outlined above.

Question 2.1: Which Commonly Identified Enterobacterales Species Should be Considered at Moderate Risk for Clinically Significant Inducible *ampC*?

Suggested Approach

Enterobacter cloacae complex, *Klebsiella aerogenes*, and *C. freundii* are the most frequently encountered Enterobacterales at moderate risk for clinically significant inducible *ampC*. Although clinical experience is limited, in vitro data suggest *Hafnia alvei* exhibits similar inducible *ampC* potential and may be considered in the same risk category.

Rationale

The optimal method to define the likelihood of clinically significant *ampC* induction across bacterial species would be to systematically identify organisms that initially test susceptible to select β-lactams (eg, ceftriaxone) and, following β-lactam exposure, subsequently test resistant—confirmed through genotyping and expression analyses to verify recovery of the same strain and demonstrate increased AmpC production. Such comprehensive studies, however, are not available.

Commonly used mnemonics for AmpC-producing organisms (eg, SPACE, SPICE, and ESCPM) oversimplify the wide spectrum of *ampC* induction potential among gram-negative bacteria and obscure important interspecies differences. For example, *C. freundii* carries a chromosomal *ampC* gene, whereas *Citrobacter koseri* does not [165, 166]. As a result, these acronyms may both underestimate and overestimate the true risk of clinically significant AmpC production. Similarly, "indole-positive *Proteus* species" have been historically included in such acronyms; however, *Proteus vulgaris*, a currently recognized indole-positive species, generally lacks a chromosomal *ampC*. Common mnemonics previously encompassed *Proteus rettgeri* and *Proteus morganii* (now *Providencia rettgeri* and *Morganella morganii*, respectively) [167], making their continued inclusion in acronyms inaccurate.

The emergence of clinically significant *ampC* expression during treatment is most frequently documented for *En. cloacae* complex (hereafter, referred to as *En. cloacae*), *K. aerogenes* (formerly *Enterobacter aerogenes*), and *C. freundii*. Clinical reports suggest that resistance emergence after exposure to agents such as ceftriaxone may occur in approximately 20% of infections caused by these organisms [160, 168–170]. These observations are consistent with in vitro data, which also indicate that *H. alvei* is capable of AmpC overproduction [171, 172]. Accordingly, when *En. cloacae*, *K. aerogenes*, *C. freundii*, or *H. alvei* are recovered in clinical cultures, the panel suggests generally avoiding cefotaxime, ceftazidime, and ceftriaxone, even if initial AST results indicate susceptibility (Question 2.2). For nonsevere infections with adequate source control caused by these species, if one of these agents has been initiated and the patient demonstrates appropriate clinical improvement, completing the treatment course may be reasonable. Regardless of inducibility, basal AmpC production in these organisms confers intrinsic resistance to ampicillin, amoxicillin–clavulanate, ampicillin–sulbactam, first-generation cephalosporins, and cephamycins [8].

In contrast, organisms historically considered at moderate risk for clinically significant *ampC* derepression—such as *M. morganii*, *Providencia* spp., and *Serratia marcescens*—appear less prone to *ampC* overexpression, based on both in vitro analyses [171, 173] and clinical observations [160, 168]. Notably, one observational study reported poorer outcomes with ceftriaxone therapy compared with cefepime or carbapenem therapy among patients infected with organisms at lower risk of *ampC* induction, primarily *M. morganii* and *S. marcescens* [174]. However, this association was not replicated in a subsequent observational study [175]. Overall, available evidence suggests that clinically significant AmpC production occurs in fewer than 5% of infections due to these species. Accordingly, when *M. morganii*, *Providencia* spp., and *S. marcescens* are isolated, the panel suggests selecting antibiotic therapy according to AST results, given the lack of data suggesting harm with this approach.

Several less common clinical pathogens (eg, *Citrobacter youngae* and *Yersinia enterocolitica*) harbor inducible chromosomal *ampC* genes but remain insufficiently studied [171, 176–178]. Consequently, data regarding their propensity for clinically significant AmpC overproduction are limited. For these organisms, it is reasonable to base therapeutic decisions on AST results (eg, ceftriaxone can be administered if susceptible). However, in the context of infections caused by these less frequently encountered organisms—or by *M. morganii*, *Providencia* spp., or *S. marcescens*—that are characterized by a high bacterial burden and limited source control (eg, endocarditis and central nervous system infections), consideration of cefepime as an alternative to ceftriaxone is reasonable, even when ceftriaxone susceptibility is demonstrated. As with all infections, if an adequate clinical response is not achieved despite appropriate antibiotic dosing and necessary source-control measures, clinicians should maintain a high index of suspicion for the emergence of resistance to the initially prescribed agent.

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Question 2.2: What Features Should be Considered in Selecting Antibiotics for Infections Caused by Organisms at Moderate Risk of Clinically Significant AmpC Production Due to an Inducible *ampC* Gene?

Suggested Approach

Several β-lactam antibiotics are at moderate risk for inducing *ampC* genes. Both the ability to induce *ampC* genes and the relative stability of the β-lactam agent against hydrolysis by AmpC-E should inform antibiotic decision-making.

Rationale

β-Lactam antibiotics fall within a spectrum of potential for inducing *ampC* genes. Aminopenicillins (ie, amoxicillin and ampicillin), narrow-spectrum (ie, first-generation) cephalosporins, and cephamycins are potent *ampC* inducers [179]. However, both organisms at low risk (eg, *S. marcescens*) and at moderate risk (eg, *En. cloacae*) for clinically significant *ampC* induction hydrolyze these antibiotics even at basal *ampC* expression levels. Therefore, AmpC-E isolates, regardless of *ampC* induction status, will generally test as resistant to these drugs.

Imipenem is also a potent *ampC* inducer but it generally remains stable to AmpC-E hydrolysis because of the formation of stable acyl enzyme complexes [180]. The induction potential of ertapenem and meropenem has not been formally investigated but, like imipenem, they are generally stable to AmpC hydrolysis [181, 182].

Aztreonam, cefotaxime, ceftazidime, ceftriaxone, and piperacillin–tazobactam are relatively weak *ampC* inducers [179, 183, 184]. Despite their limited ability to induce *ampC*, the susceptibility of these agents to hydrolysis makes them less likely to be effective for the treatment of infections by organisms at moderate risk for clinically significant AmpC production [182, 185–188]. They remain, however, reasonable treatment options for Enterobacterales at lower risk for clinically significant AmpC production (eg, *S. marcescens*).

Cefepime has the advantage of both being a weak inducer of *ampC* and of withstanding hydrolysis by AmpC β-lactamases because of the formation of stable acyl enzyme complexes [189–191]. Therefore, cefepime is generally an effective agent for the treatment of AmpC-E infections [192, 193]. TMP-SMX, fluoroquinolones, aminoglycosides, tetracyclines, and other non-β-lactam antibiotics do not induce *ampC* and are also not substrates for AmpC hydrolysis (Question 2.7).

Question 2.3: What is the Role of Cefepime for the Treatment of Infections Caused by Enterobacterales at Moderate Risk of Clinically Significant AmpC Production due to an Inducible *ampC* Gene?

Suggested Approach

Cefepime is suggested for the treatment of infections caused by organisms at moderate risk of significant AmpC production (ie, *En. cloacae* complex, *K. aerogenes*, *C. freundii*, and *H. alvei*), if cefepime MICs are in the susceptible or susceptible dose-dependent range (ie, ≤8 μg/mL).

Rationale

Cefepime, an oxyimino-cephalosporin, demonstrates relative stability against AmpC β-lactamases and exhibits a low propensity for *ampC* induction [189–191, 194]. Clinical trials comparing cefepime with carbapenems for the treatment of AmpC-E infections have not been conducted. A 2025 meta-analysis encompassing 7 studies and 1099 patients with bacteremia caused by *En. cloacae*, *K. aerogenes*, *C. freundii*, *M. morganii*, *Providencia* spp., or *S. marcescens* found no differences in mortality, clinical cure, or infection relapse between cefepime and carbapenem recipients [193]. Notably, this analysis was limited by inter-study heterogeneity, potential confounding due to preferential carbapenem use in more acutely ill patients, and species-level variability in the risk of clinically significant AmpC production. Nonetheless, in view of cefepime's favorable stability against AmpC hydrolysis, the high probability of PK/PD target attainment with appropriate dosing, and the absence of compelling evidence of inferiority in clinical outcomes studies, the panel suggests cefepime as a preferred treatment option for infections caused by organisms at moderate risk of clinically significant AmpC production (Table 1). Because cefepime is generally active against AmpC-E, cefepime–enmetazobactam is also anticipated to retain activity; however, its use is preferentially reserved for infections caused by ESBL-E or organisms co-producing AmpC and ESBL enzymes.

Although cefepime may be effective for the treatment of AmpC-E infections, it is suboptimal against infections caused by ESBL-E, which is a consideration if both enzymes are produced by Enterobacterales (Question 1.5). Data are incomplete on the frequency of ESBL production by Enterobacterales at moderate risk of clinically significant AmpC production in the United States. An evaluation of consecutive *En. cloacae* isolates from 66 US hospitals from 2019 to 2020 indicated that 3% (6/211) contained a $bla_{\text{CTX-M}}$ gene [17]. A study from Pennsylvania found that 33% (15/45) of *En. cloacae* bloodstream isolates collected between 2003 and 2005 produced SHV-type ESBLs [195]. However, there was no association between ESBL production and the cefepime MIC. A study from Maryland found that ESBL genes were identified in 22% of *K. aerogenes* (4/18), in 14% of *En. cloacae* (7/51), and in no *C. freundii* (0/8) bloodstream isolates collected between 2018 and 2021 [196]. There was no correlation between the presence of an ESBL gene and the cefepime MIC; none of the ESBL-producing isolates had cefepime MICs of 4–8 μg/mL. Contemporary data specific to the United States are needed to better understand how frequently ESBLs are produced by Enterobacterales at moderate risk of clinically significant AmpC production. Available evidence does not suggest an association between cefepime susceptible dose-dependent MICs (ie, MICs 4–8 μg/mL) and ESBL production. Accordingly, the panel suggests cefepime as a preferred treatment option for

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AmpC-E infections when the cefepime MIC is ≤8 μg/mL, provided an ESBL or its gene have not been identified.

Question 2.4: What is the Role of Ceftriaxone for the Treatment of Infections Caused by Enterobacterales at Moderate Risk of Clinically Significant AmpC Production due to an Inducible *ampC* Gene?

Suggested Approach

Ceftriaxone (or cefotaxime or ceftazidime) is not suggested for the treatment of invasive infections caused by organisms at moderate risk of clinically significant AmpC production (ie, *En. cloacae* complex, *K. aerogenes*, *C. freundii*, and *H. alvei*). In cases of nonsevere infection in which ceftriaxone was initiated empirically, and the patient demonstrates clinical improvement with adequate source control, continuation of ceftriaxone to complete the treatment course may be reasonable.

Rationale

Clinical reports vary in their estimates of how often resistance to ceftriaxone arises during treatment of infections caused by Enterobacterales considered to be at moderate risk for clinically significant AmpC induction. Interpreting the available evidence is complicated by several methodological and biological challenges. First, no CLSI-endorsed methods or FDA-cleared devices exist for the detection of *ampC* genes or AmpC enzymes in clinical isolates, precluding definitive confirmation of AmpC production. Second, ceftriaxone resistance may result from alternative mechanisms (eg, ESBL production); yet this has been minimally investigated outside of *E. coli*, *K. pneumoniae*, and *K. oxytoca*. Third, many studies aggregate data from species at low risk for AmpC-mediated ceftriaxone resistance (eg, *S. marcescens*) with those at higher risk (eg, *En. cloacae*), thereby obscuring species-specific estimates of the likelihood of ceftriaxone resistance emergence. Fourth, studies assessing resistance following ceftriaxone exposure often omit molecular confirmation of relatedness between index and subsequent isolates. Moreover, several investigations employed pre-2010 CLSI ceftriaxone breakpoints (MIC ≤8 μg/mL), complicating the extrapolation of findings to current standards (MIC ≤1 μg/mL) [8]. Finally, substantial heterogeneity across studies, including variation in infection sources, illness severity, comorbidities, concomitant antibiotic therapy, and ceftriaxone dosing and duration, further limits the interpretability of clinical data.

These limitations notwithstanding, combining available data, the emergence of resistance after ceftriaxone exposure occurs in approximately 20% of infections caused by *En. cloacae*, *K. aerogenes*, or *C. freundii* [160, 168–170, 197]. Similar clinical data are lacking for *H. alvei*. An individual patient's risk may differ based on host factors (eg, higher with a persistent source of infection or prolonged durations of therapy).

No clinical trials have compared the outcomes of patients with presumed AmpC-E infections treated with ceftriaxone compared with alternate agents (eg, cefepime). Several observational studies compared the clinical outcomes of patients with infections caused by *En. cloacae*, *K. aerogenes*, and *C. freundii* treated with ceftriaxone compared with either cefepime or carbapenems [174, 198–201]; 2 of the 5 cohorts demonstrated poorer clinical outcomes with the use of ceftriaxone [174, 201, 202]. Based on the available data, the panel suggests generally avoiding third-generation cephalosporins when treating infections caused by organisms at moderate risk of clinically significant AmpC production. However, for uncomplicated infections it may be reasonable to transition to ceftriaxone after clear clinical improvement has been achieved and if there are no concerns for ongoing sources of infection (eg, abscesses and indwelling catheters), weighing the convenience of once-daily ceftriaxone dosing with the potentially increased risk of emergence of resistance and possible treatment failure.

Question 2.5: What is the Role of Piperacillin–Tazobactam for the Treatment of Infections Caused by Enterobacterales at Moderate Risk of Clinically Significant AmpC Production due to an Inducible *ampC* Gene?

Suggested Approach

Piperacillin–tazobactam is not suggested for the treatment of invasive infections caused by Enterobacterales at moderate risk of clinically significant inducible AmpC production.

Rationale

Tazobactam is less effective at protecting β-lactams from AmpC hydrolysis than newer β-lactamase inhibitors, such as avibactam and relebactam [138]. As an example, in an in vitro study of *En. cloacae* isolates, tazobactam (paired with ceftolozane) was able to inhibit 100% (32/32) of wild type, 67% (6/9) of ESBL, and only 19% (10/53) of AmpC-producing *En. cloacae* isolates [203]. In another study, increasing levels of *ampC* expression were associated with dramatic increases in piperacillin–tazobactam MICs [182]. Regarding clinical data, a 2019 meta-analysis summarized the findings of 12 observational studies, building upon a previous meta-analysis [192], and did not identify a difference in mortality between patients treated with piperacillin–tazobactam versus carbapenems for bacteremia caused by *Enterobacter* spp., *Citrobacter* spp., *Serratia* spp., Providencia spp., or *Morganella* spp. [204]. However, moderate heterogeneity across studies and confounding by indication likely existed (ie, ill appearing patients were more likely to be prescribed carbapenems). At least 3 observational studies published after this meta-analysis, including 759 patients, suggest poorer outcomes for patients receiving piperacillin–tazobactam compared with cefepime and/or carbapenems [174, 205, 206].

A clinical trial compared the outcomes of 72 patients with bloodstream infections caused by *Enterobacter* spp., *K. aerogenes*, *C. freundii*, *M. morganii*, *Providencia* spp., or *S. marcescens* randomized to piperacillin–tazobactam or meropenem

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[207]. There were no significant differences in the primary outcome (a composite outcome including 30-day mortality, clinical failure, microbiological failure, or microbiological relapse) between study arms. However, some notable and seemingly conflicting findings were observed for individual components of this composite outcome: mortality (0% [0/38] vs 6% [2/34]); clinical failure (21% [8/38] vs 12% [4/34]); microbiological failure (13% [5/38] vs 0% [0/34]); and microbiological relapse (0% [0/38] vs 9% [3/34]) for the piperacillin–tazobactam and meropenem arms, respectively. The findings of this trial are challenging to interpret and a larger trial is needed to determine the role of piperacillin–tazobactam for the treatment of organisms at moderate risk for clinically significant *ampC* induction.

In light of the limited ability of tazobactam to protect piperacillin from AmpC hydrolysis in vitro and at least a few observational studies suggesting poorer clinical outcomes with the use of piperacillin–tazobactam compared with cefepime/carbapenems, the panel suggests caution with prescribing piperacillin–tazobactam for serious infections caused by AmpC-E.

Question 2.6: What is the Role of β-Lactam Agents With Activity Against Carbapenem-Resistant Organisms for the Treatment of Infections Caused by Enterobacterales at Moderate Risk of Clinically Significant AmpC Production?

Suggested Approach

Although aztreonam–avibactam, ceftazidime–avibactam, imipenem–relebactam, meropenem–vaborbactam, and cefiderocol are expected to be active against AmpC-E, their use should generally be reserved for infections caused by carbapenem-resistant organisms. Ceftolozane–tazobactam is not suggested for the treatment of AmpC-E infections.

Rationale

Avibactam, vaborbactam, and relebactam have been shown to inhibit AmpC enzymes [139, 208–210]. Likewise, cefiderocol generally retains activity against AmpC-E [211]. Nonetheless, resistance to ceftazidime–avibactam among AmpC-E has been reported, most often driven by amino acid substitutions that expand the AmpC substrate-binding pocket, thereby improving accommodation of cephalosporins, such as ceftazidime and cefiderocol [212–215]. The frequency with which AmpC-E acquire resistance to ceftazidime–avibactam remains undefined.

Carbapenems are typically stable to AmpC-mediated hydrolysis; consequently, imipenem–relebactam and meropenem–vaborbactam are expected to retain activity against AmpC-E. Although aztreonam–avibactam, ceftazidime–avibactam, imipenem–relebactam, meropenem–vaborbactam, and cefiderocol are likely to be effective therapeutic options, the panel advises reserving these agents for infections caused by carbapenem-resistant organisms, where they are most needed.

Ceftolozane was engineered to confer greater stability to hydrolysis by pseudomonal AmpC cephalosporinases (ie, *Pseudomonas*-derived cephalosporinases [PDCs]) compared with earlier cephalosporins such as ceftazidime [216]. However, considerably less is known regarding ceftolozane–tazobactam's activity against AmpC-E. Tazobactam provides relatively weak protection against AmpC-mediated hydrolysis compared with newer β-lactamase inhibitors, such as avibactam and relebactam (Question 2.5). Limited in vitro data suggest that ceftolozane–tazobactam may retain some activity against AmpC-E [217]; however, in at least one investigation, the agent was active against only 19% (10/53) of *En. cloacae* isolates producing moderate levels of AmpC enzymes [203]. Based on the available evidence, the panel does not suggest the use of ceftolozane–tazobactam for the treatment of AmpC-E infections. In the context of polymicrobial infections where both DTR *P. aeruginosa* and AmpC-E are isolated, ceftolozane–tazobactam may be considered as a therapeutic option to minimize exposure to multiple agents and their associated toxicities. If this approach is pursued, close clinical monitoring is suggested to ensure an appropriate response.

Question 2.7: What is the Role of Non-β-Lactam Agents for the Treatment of Infections Caused by Enterobacterales at Moderate Risk of Clinically Significant AmpC Production due to an Inducible *ampC* Gene?

Suggested Approach

Non-β-lactam agents are not substrates for AmpC enzymes and, therefore, remain effective treatment options against AmpC-E when bacterial isolates test susceptible, if appropriate based on the site of infection, severity of illness, and agent-specific toxicities.

Rationale

For AmpC-E uUTI, preferred treatment options include nitrofurantoin or TMP-SMX (Question 1.1). Alternative options include single-dose aminoglycosides, ciprofloxacin, levofloxacin, pivmecillinam, and sulopenem (Question 1.1). Mecillinam (the active moiety of pivmecillinam) and sulopenem demonstrate in vitro activity against >90% of AmpC-E isolates [41, 218–223]; however, clinical outcomes data are limited. In contrast, susceptibility data for gepotidacin against AmpC-E are comparatively sparse, and available evidence suggests elevated MICs among common AmpC-E isolates [31]. In the absence of confirmed susceptibility testing, gepotidacin is not suggested for the treatment of AmpC-E uUTI.

For cUTI, ciprofloxacin, levofloxacin, or TMP-SMX are preferred agents (Question 1.2), with aminoglycosides and sulopenem as alternative options (Question 1.2). Because TMP-SMX and fluoroquinolones are not substrates for AmpC-mediated hydrolysis, oral transition therapy with ciprofloxacin, levofloxacin, or TMP-SMX may be considered for invasive AmpC-E

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infections once susceptibility has been confirmed and clinical stability occurs (Question 1.3).

SECTION 3: CARBAPENEM-RESISTANT ENTEROBACTERALES

CRE are members of the Enterobacterales order that are resistant to at least one carbapenem (ie, ertapenem, imipenem, and meropenem) or that produce a carbapenemase enzyme [224]. Certain Enterobacterales, including *Morganella* spp., *Proteus* spp., and *Providencia* spp., exhibit decreased susceptibility to imipenem due to structural characteristics of their penicillin-binding proteins (PBPs) that reduce the antibiotic's binding affinity [225, 226]. Consequently, for these species, resistance to at least one carbapenem other than imipenem is required for classification as CRE.

CRE represent a heterogeneous group that can be broadly categorized as non-carbapenemase producing or carbapenemase producing. Non-carbapenemase–producing CRE typically arise from amplification of non-carbapenemase β-lactamase genes (eg, $bla_{\text{CTX-M}}$, *ampC*) in combination with outer membrane porin loss (eg, OmpC/OmpF in *E. coli* or *En. cloacae*; OmpK35/OmpK36 in *K. pneumoniae*) [227, 228].

Carbapenemase-producing isolates account for 35%–83% of CRE cases in the United States, with higher proportions observed when CRE definitions require resistance to meropenem or imipenem [229–232]. The epidemiology of carbapenemase-producing Enterobacterales in the United States continues to evolve. Although bla_{KPC} has historically most commonly been identified in *K. pneumoniae* isolates, bla_{KPC} is increasingly detected in other species, including *En. cloacae* and *E. coli* [232]. Since 2021, a marked increase in New Delhi metallo-β-lactamase–producing Enterobacterales (NDM-E) has been observed, along with a more gradual increase in OXA-48-like producers (OXA-48-E) [231, 233]. KPC and OXA-48-like enzymes are serine carbapenemases, whereas NDM, Verona integron-encoded metallo-β-lactamases (VIM), and imipenem-hydrolyzing metallo-β-lactamases (IMP) are collectively classified as metallo-β-lactamase (MBL) carbapenemases. These enzyme classes differ fundamentally in catalytic mechanism: serine β-lactamases use an active-site serine to hydrolyze β-lactams, while MBLs depend on 1 or 2 zinc ions at the catalytic site, which coordinate and activate a water molecule for nucleophilic attack [234].

Carbapenemase families, and sometimes specific variants within a carbapenemase family, differ in β-lactam substrate profiles and susceptibility to inhibition by β-lactamase inhibitors. Consequently, identification of the specific carbapenemase family (eg, KPC) produced by an organism causing infection optimizes therapeutic decision-making to ultimately improve patient outcomes [235–238]. Treatment suggestions for CRE infections assume that in vitro activity of preferred or alternative agents has been demonstrated.

Question 3.1: What are Preferred Antibiotics for the Treatment of Uncomplicated Urinary Tract Infections Caused by Carbapenem-Resistant Enterobacterales?

Suggested Approach

Aminoglycosides (as a single dose), ciprofloxacin, levofloxacin, nitrofurantoin, or TMP-SMX are preferred treatment options for uUTIs caused by CRE, acknowledging the likelihood of CRE susceptibility to several of these agents may be low. Colistin, oral fosfomycin (for *E. coli* only), gepotidacin, or pivmecillinam are alternative options; there is also a low likelihood of CRE susceptibility to several of these agents. Several IV agents (eg, aztreonam–avibactam, cefiderocol, ceftazidime–avibactam, fosfomycin [preferentially for *E. coli*], imipenem–relebactam, and meropenem–vaborbactam) are likely to be effective, but it is suggested that their use be preserved for situations where resistance, availability, or intolerance preclude the use of other options.

Rationale

Because treatment courses for uUTI are relatively short and susceptibility data, particularly for newer oral uUTI agents, may not be available, therapy for CRE uUTI is frequently initiated and completed before carbapenem resistance is recognized. Review of prior urine culture results is therefore essential when evaluating patients with suspected uUTI to determine whether empiric CRE coverage is warranted [239]. Although several oral agents may retain activity against CRE, susceptibility rates are generally low, necessitating close clinical monitoring when these agents are used empirically. More comprehensive discussions of mechanisms of action, resistance patterns, and clinical efficacy are provided in Question 1.1.

Single-dose aminoglycosides are preferred agents given their renal elimination and high urinary concentrations, which support clinical efficacy in uUTI [28, 29]. United States surveillance data suggest CRE have variable susceptibility to aminoglycosides: amikacin (59%), gentamicin (47%), plazomicin (97%), and tobramycin (27%) [240]. Even in the setting of elevated MICs, urinary concentrations may theoretically exceed resistance thresholds and translate to clinical success [241].

Ciprofloxacin, levofloxacin, nitrofurantoin, and TMP-SMX are also preferred, based on favorable urinary PK, clinical trial data, and extensive clinical experience [39, 66–68]. However, available data suggest that fewer than 20% of CRE isolates are susceptible to fluoroquinolones or TMP-SMX, and approximately 30% to nitrofurantoin [240, 242–245].

Colistin is converted from its inactive prodrug, colistimethate sodium, to its active form within the urinary tract, thereby making colistin, but not polymyxin B which is primarily cleared through nonrenal routes [246], an alternative option for uUTI [247, 248]. Notably, CLSI does not define a susceptible breakpoint for colistin against Enterobacterales; MICs $\leq$2 μg/mL

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are categorized as intermediate [8]. Given the high urinary concentrations achieved, colistin is nevertheless anticipated to retain activity for infections confined to the urinary tract. In a US collection of 117 CRE isolates, approximately 91% had colistin MICs within the intermediate range [240]. Despite this potential activity, colistin is suggested only as an alternative rather than a preferred option for CRE uUTI due to its toxicity profile. Even with relatively short treatment courses, the risk of nephrotoxicity is approximately 30% [249–252].

Oral fosfomycin is an alternative option for uUTI caused by carbapenem-resistant *E. coli*; it is not considered a preferred agent because clinical trial data demonstrate inferior outcomes compared with nitrofurantoin [39]. Susceptibility testing for fosfomycin is not routinely performed in many clinical microbiology laboratories, and contemporary US data specifically evaluating activity against carbapenem-resistant *E. coli* are not available. Using the CLSI oral fosfomycin breakpoint (which differs from the IV breakpoint; Table 2), approximately 80% of carbapenem-resistant *E. coli* isolates are expected to be susceptible to oral fosfomycin [245, 253].

As discussed in Question 1.1, gepotidacin and pivmecillinam received approval for the treatment of uUTI in 2025. Data specifically evaluating gepotidacin activity against CRE isolates are extremely limited. However, given its novel mechanism of action targeting bacterial DNA gyrase and topoisomerase IV at sites distinct from fluoroquinolones, it is anticipated to retain activity against at least a subset of CRE isolates that are not susceptible to fluoroquinolones. In a surveillance study including approximately 600 AMR uropathogens, only a few of which were carbapenem resistant, resistance to gepotidacin was observed in <1% of isolates [254]. Clinical outcomes data evaluating gepotidacin for uUTI caused by CRE are not available. Accordingly, gepotidacin is suggested as an alternative option.

Pivmecillinam is distinct from other oral agents suggested for CRE uUTI in that it is a β-lactam antibiotic, administered as the prodrug of the microbiologically active compound, mecillinam. In a collection of CRE isolates, mecillinam susceptibility was 0% among 174 KPC producers, 71% among 1094 NDM-E, and 84% among 4042 OXA-48-E; susceptibility among 2349 non-carbapenemase–producing CRE isolates was 70% [255]. A separate collection of 105 molecularly characterized CRE isolates demonstrated limited activity (<50%) against NDM-E and OXA-48-E and no activity against KPC-producing Enterobacterales (KPC-E) [256]. A murine UTI model suggested promising in vivo activity against NDM-E [257]. Given the apparent ability of KPC enzymes to hydrolyze mecillinam, knowledge of the likely carbapenemase mechanism involved should inform consideration of pivmecillinam. Clinical outcomes data against CRE are lacking. It is suggested as an alternative agent for uUTI caused by CRE, if the likelihood of KPC production is low.

Although several IV agents (eg, aztreonam–avibactam, cefiderocol, ceftazidime–avibactam, fosfomycin [particularly for *E. coli*], imipenem–relebactam, and meropenem–vaborbactam) have demonstrated in vitro activity and clinical efficacy against CRE that cause uUTI [71, 72, 140, 258–264], the panel suggests preferentially reserving their use for invasive CRE infections.

Question 3.2: What are the Preferred Antibiotics for the Treatment of Complicated Urinary Tract Infection Caused by Carbapenem-Resistant Enterobacterales?

Suggested Approach

Ciprofloxacin, levofloxacin, and TMP-SMX, are preferred treatment options for cUTIs caused by CRE, acknowledging the likelihood of CRE susceptibility to these agents is low. Several IV agents (eg, aztreonam–avibactam, cefiderocol, ceftazidime–avibactam, imipenem–relebactam, and meropenem–vaborbactam) are also preferred options, particularly when clinical instability is present. Intravenous fosfomycin (preferentially for *E. coli*) and once-daily aminoglycosides (ie, amikacin, gentamicin, plazomicin, or tobramycin) are alternative agents for the treatment of cUTI caused by CRE.

Rationale

Ciprofloxacin, levofloxacin, or TMP-SMX are preferred treatment options for cUTI caused by CRE only after susceptibility is demonstrated, given the overall low likelihood of activity (Question 3.1). Intravenous agents, including aztreonam–avibactam, cefiderocol, ceftazidime–avibactam, imipenem–relebactam, and meropenem–vaborbactam, are also preferred options, particularly in the setting of clinical instability, supported by in vitro susceptibility data and clinical trial evidence demonstrating noninferiority to alternative regimens for the treatment of UTIs [140, 258–264]. Notably, the proportion of CRE isolates included across trials is low. Selection among these agents should be guided primarily by the specific carbapenemase mechanism identified, in addition to AST results (Questions 3.4–3.6).

Intravenous fosfomycin is an alternative agent for the treatment of cUTI (preferably those caused by *E. coli*) exhibiting carbapenem resistance (Question 1.2), only after susceptibility is confirmed, as only about 50% of isolates are likely to be susceptible when applying IV fosfomycin breakpoints (Table 2) [253, 265]. Once-daily aminoglycosides (ie, amikacin, gentamicin, plazomicin, and tobramycin) are also alternative agents for the treatment of CRE cUTI (Question 1.2); agent-specific differences exist in their likelihood of activity against CRE isolates (Question 3.1). Although expected to be effective as they achieve high renal parenchymal concentrations [74], duration-dependent nephrotoxicity should be carefully considered [75, 76]. Aminoglycosides can be useful for completing treatment courses, given their prolonged renal cortical

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concentrations and the logistical advantages of once-daily dosing [79, 81] (Table 1, Supplementary Material).

Question 3.3: What are the Preferred Antibiotics for the Treatment of Invasive Infections Caused by Carbapenem-Resistant Enterobacterales that are not Carbapenemase Producing?

Suggested Approach

For invasive infections due to Enterobacterales isolates that do not produce carbapenemases and demonstrate susceptibility to meropenem and imipenem (MIC ≤1 μg/mL) but are not susceptible to ertapenem (MIC ≥1 μg/mL), administration of extended-infusion meropenem or imipenem is suggested. In cases where isolates are not susceptible to ertapenem and retain susceptibility to either meropenem or imipenem, but not both, use of the active carbapenem via extended infusion may be considered, provided the patient is not critically ill and adequate source control has been achieved. For infections caused by non-carbapenemase–producing Enterobacterales that lack susceptibility to all carbapenems, preferred therapeutic options include aztreonam–avibactam, ceftazidime–avibactam, imipenem–relebactam, and meropenem–vaborbactam. Cefiderocol, eravacycline, and tigecycline are alternative options. Notably, tetracycline derivatives exhibit rapid tissue distribution with low serum and urinary concentrations, limiting their utility in bacteremia and UTIs and potentially contributing to poorer outcomes in these settings [266, 267].

Rationale

For infections caused by non-carbapenemase–producing Enterobacterales isolates that are susceptible to imipenem and meropenem (MIC ≤1 μg/mL) but resistant to ertapenem (MIC ≥1 μg/mL), extended-infusion imipenem or meropenem is suggested. Surveillance data from a US CRE isolate collection demonstrated that fewer than 3% of 1249 isolates exhibiting ertapenem resistance while retaining susceptibility to meropenem and imipenem harbored a carbapenemase gene [268]. Instead, such isolates frequently exhibit amplification of non-carbapenemase β-lactamase genes (eg, ESBLs or narrow-spectrum β-lactamases), typically in the presence of porin disruption, which have already attenuated ertapenem activity. Increasing β-lactamase production has the theoretical potential to compromise imipenem or meropenem efficacy over the course of therapy [227, 269]. Accordingly, when extended-infusion carbapenem therapy is employed in this setting, close clinical monitoring is warranted. For uUTI caused by isolates with this phenotype, standard-infusion imipenem or meropenem may be a reasonable alternative.

In situations where isolates are non-carbapenemase–producing and susceptible to one carbapenem but not another (eg, susceptible to imipenem but resistant to meropenem or vice versa), evidence to guide optimal management is limited. In such cases, treatment decisions should be individualized based on patient-specific factors. In critically ill patients or in the presence of ongoing infection sources, consideration may be given to newer β-lactam/β-lactamase inhibitor combinations, as the inhibitor component may mitigate carbapenem hydrolysis secondary to significant β-lactamase (eg, ESBL, CMY, and OXA-1) expression [270].

For isolates that are non-carbapenemase producing yet not susceptible to any carbapenem, newer β-lactam agents are preferred, including aztreonam–avibactam, ceftazidime–avibactam, imipenem–relebactam, and meropenem–vaborbactam, with agent selection guided by AST results. Among these, there is modest panel preference for ceftazidime–avibactam and imipenem–relebactam, as in vitro data suggest that avibactam and relebactam may demonstrate somewhat greater inhibitory activity in this context compared with vaborbactam [271, 272]. Cefiderocol may also demonstrate activity against such isolates; however, its lack of a companion β-lactamase inhibitor raises theoretical concerns of its effectiveness in the setting of high-level β-lactamase expression. Consequently, cefiderocol is considered an alternative option in these scenarios.

Eravacycline and tigecycline may serve as alternative options for CRE infections that do not involve the bloodstream or urinary tract (Question 3.7). Their antibiotic activity is independent of carbapenemase production status.

Question 3.4: What are the Preferred Antibiotics for the Treatment of Invasive Infections Caused by KPC-Producing Enterobacterales?

Suggested Approach

Ceftazidime–avibactam, imipenem–relebactam, and meropenem–vaborbactam are preferred treatment options for KPC-E infections. Among these, resistance emergence during therapy has been reported in approximately 10% of cases with ceftazidime–avibactam, compared with <3% for imipenem–relebactam and meropenem–vaborbactam. Aztreonam–avibactam, cefiderocol, eravacycline, and tigecycline are alternative options, with eravacycline and tigecycline not suggested for infections involving the bloodstream or urinary tract.

Rationale

Preferred agents for KPC-E infections include ceftazidime–avibactam, imipenem–relebactam, and meropenem–vaborbactam. Although no clinical trials directly comparing these agents are available, they have consistently been associated with improved clinical outcomes and significantly less toxicity compared with alternative regimens, which were commonly polymyxin or aminoglycoside based [235, 273–277].

United States surveillance data indicate that ceftazidime–avibactam, imipenem–relebactam, and meropenem–vaborbactam demonstrate >95% activity against KPC-E [278]. While all 3 are preferred options, the panel slightly favors meropenem–vaborbactam, followed by ceftazidime–avibactam, and then imipenem–relebactam. This ranking reflects composite

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observational data suggesting numerically more favorable clinical outcomes with meropenem–vaborbactam compared with ceftazidime–avibactam, along with a lower likelihood of resistance emergence, although these differences were generally not statistically significant [279–281].

In one observational study of patients with CRE infections (approximately 73% KPC-E), 30-day survival was 81% (85/105) in the ceftazidime–avibactam group and 88% (23/26) in the meropenem–vaborbactam group; clinical cure occurred in 62% (65/105) and 69% (18/26), respectively [281]. Among patients with recurrent infections, resistance emerged in 20% (3/15) of those receiving ceftazidime–avibactam and in none (0/3) receiving meropenem–vaborbactam. In a second observational study, 30-day survival was 73% (38/52) with ceftazidime–avibactam versus 81% (29/36) with meropenem–vaborbactam; patients receiving meropenem–vaborbactam had approximately twice the odds of a favorable clinical response at 72 hours [279]. Estimates of resistance emergence were not reported. In a third observational study of 73 patients with invasive KPC-E infections, overall survival did not differ between groups; however, among patients with recurrent infection within 90 days, resistance developed in 12% (2/17) of those treated with ceftazidime–avibactam and in none (0/7) treated with meropenem–vaborbactam [280]. Clinical cure was not reported.

Comparative clinical data evaluating imipenem–relebactam against either ceftazidime–avibactam or meropenem–vaborbactam for KPC-E infections are lacking. In an uncontrolled observational study including 6 patients with invasive KPC-E infections, all patients survived and achieved clinical cure with imipenem–relebactam [282]. However, in a clinical trial of imipenem–nonsusceptible Enterobacterales infections, favorable clinical response occurred in only 40% (2/5) of patients treated with imipenem–relebactam [276]. Although the sample size precludes definitive conclusions, these trial results warrant caution. Nevertheless, the demonstrated in vitro activity of imipenem–relebactam against KPC-E isolates, extensive clinical experience with imipenem, and the inhibitory potential of relebactam [283] collectively support the expectation of clinical efficacy of imipenem–relebactam for KPC-E infections.

A shared vulnerability of ceftazidime–avibactam, imipenem–relebactam, and meropenem–vaborbactam is amplification of bla_{KPC}, which increases enzyme expression and hydrolytic capacity, thereby diminishing β-lactam activity [284]. Resistance emergence during ceftazidime–avibactam therapy has been reported in approximately 10% of cases [284–288]. The predominant mechanism involves substitutions within or adjacent to the KPC Ω-loop, and less commonly alterations in AmpC enzymes [284]. These changes remodel the active site, enhancing ceftazidime hydrolysis while reducing avibactam binding. Notably, ceftazidime–avibactam–resistant KPC variants frequently exhibit collateral reductions in carbapenem MICs, occasionally restoring in vitro carbapenem susceptibility [289–293]; the clinical significance of this phenotypic reversion remains unclear. In contrast, resistance during imipenem–relebactam or meropenem–vaborbactam therapy is reported in <3% of cases and is primarily mediated by reduced outer membrane permeability (eg, disruptions of the ompK35 and ompK36 porin genes in *K. pneumoniae*) [284, 294–300].

Aztreonam–avibactam and cefiderocol are alternative agents for KPC-E infections. Both demonstrate >95% in vitro activity against KPC-E isolates [301–303]. Clinical data supporting aztreonam–avibactam for KPC-E infections are very limited [304]. In a subgroup analysis of a clinical trial comparing cefiderocol with alternative regimens (primarily polymyxin-based therapy), survival in the KPC-E subgroup was 78% (7/9) with cefiderocol versus 79% (11/14) with alternative therapy [261].

Eravacycline and tigecycline are additional alternative options for KPC-E infections that do not involve the bloodstream or urinary tract, although clinical data are limited (Question 3.7). Their activity is independent of the presence or type of carbapenemase.

Question 3.5: What are the Preferred Antibiotics for the Treatment of Invasive Infections Caused by New Delhi Metallo-β-Lactamase–Producing Enterobacterales?

Suggested Approach

Aztreonam–avibactam and cefiderocol are preferred treatment options for NDM-E infections. If aztreonam–avibactam is not available, ceftazidime–avibactam plus aztreonam is an alternative. Eravacycline and tigecycline are alternative options for infections not involving the bloodstream or urinary tract.

Rationale

Aztreonam–avibactam and cefiderocol are the preferred agents for infections caused by NDM-E. New Delhi metallo-β-lactamase enzymes belong to the larger category of MBL enzymes and hydrolyze all traditional β-lactam antibiotics except aztreonam [305]. Despite its stability to hydrolysis by NDM, aztreonam remains susceptible to hydrolysis by serine β-lactamases frequently co-produced by NDM-E isolates (eg, ESBLs, AmpCs, and OXA-type enzymes) [305]. Avibactam inhibits these serine β-lactamases, thereby protecting aztreonam from enzymatic degradation and permitting it to reach its primary target, PBP3. Notably, the preferred agents for KPC-E are ineffective against NDM-E because neither avibactam, relebactam, nor vaborbactam inhibit NDM.

Using the CLSI susceptibility breakpoint of ≤4/4 μg/mL (Table 2), surveillance data indicate that >95% of NDM-E isolates are susceptible to aztreonam–avibactam [230, 306–308]. The suggested loading and subsequent maintenance dosing of aztreonam–avibactam is supported by population PK/PD modeling incorporating joint PD targets (≥60% fT > MIC for

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aztreonam and ≥50% fT >2.5 μg/mL for avibactam), as well as animal model data [309–311] (Table 1).

Clinical outcome data for aztreonam–avibactam remain limited and derive primarily from subgroup analyses of 2 trials [304, 312]. In one subgroup analysis of 6 patients with NDM-E infections, clinical cure occurred in 33% (2/6) of patients treated with aztreonam–avibactam compared with 0% (0/1) in the meropenem arm; 28-day survival was 83% (5/6) and 100% (1/1), respectively [304]. In a second clinical trial that included 11 patients with NDM-E infections, clinical cure was observed in 44% (4/9) of patients receiving aztreonam–avibactam versus 0% (0/2) receiving alternative therapy (polymyxin- or aminoglycoside-based regimens); 28-day survival was 89% (8/9) and 50% (1/2), respectively [312].

When aztreonam–avibactam is not available, ceftazidime–avibactam plus aztreonam is a reasonable alternative [305]. Although no direct clinical comparisons exist, available data do not clearly favor one regimen over the other. It has been hypothesized that any residual ceftazidime activity—via affinity for additional PBPs (PBP1a, PBP1b, and PBP2)—beyond ceftazidime's primary affinity for PBP3—may enhance bactericidal effects beyond aztreonam alone, though this remains unproven [313]. In a study of 102 adults with MBL-producing bloodstream infections (82 NDM producing), 30-day survival was 81% (42/52) with ceftazidime–avibactam plus aztreonam versus 56% (28/50) with polymyxin- or tigecycline-based therapy [314]. In another cohort including 328 NDM-E isolates, 30-day survival was 78% (167/215) with ceftazidime–avibactam, compared with 67% (22/33) for cefiderocol and 50% (13/26) for colistin [315]. A validated MIC testing method for ceftazidime–avibactam plus aztreonam is not available; however, the CLSI-endorsed broth disk elution method can be used to determine susceptibility [8, 316]. Strategies for administering ceftazidime–avibactam plus aztreonam [317–319] are summarized in Table 1 and Supplementary Material.

In a study of healthy volunteers given ceftazidime–avibactam ± aztreonam, asymptomatic transaminase elevations occurred in 40% of participants and resolved with drug cessation [319]. Elevations were dependent on aztreonam dosing, occurring primarily in groups receiving aztreonam 8 g per day either as a continuous infusion or every 6 hours; transaminitis was not increased with the co-administration of ceftazidime–avibactam. Asymptomatic transaminase elevations occurred in approximately 6% of patients in aztreonam–avibactam trials where the total daily dose of aztreonam was 6 g per day [304, 312].

Despite comparable in vitro activity between ceftazidime–avibactam plus aztreonam and aztreonam–avibactam, the fixed 3:1 aztreonam–avibactam formulation ensures synchronized β-lactam/β-lactamase inhibitor exposure and improves probability of joint target attainment [311]. In addition, single-agent administration enhances feasibility. For these reasons, the panel preferentially suggests aztreonam–avibactam when available. Notably, although in vitro data suggest that combining aztreonam with meropenem–vaborbactam or imipenem–relebactam may provide activity against NDM-E [320, 321], clinical experience with these combinations remains limited [322].

Cefiderocol is a catechol-substituted siderophore cephalosporin that co-opts bacterial iron acquisition pathways to facilitate periplasmic entry via active ferric iron transport systems [323]. Once internalized, cefiderocol demonstrates high-affinity binding predominantly to PBP3, thereby inhibiting peptidoglycan cross-linking, impairing cellwall biosynthesis, and ultimately inducing bacterial cell lysis. United States surveillance data indicate that cefiderocol retains activity against approximately 85% of NDM-E isolates [324, 325].

Clinical data evaluating cefiderocol in infections due to NDM-E are limited. A subgroup of patients enrolled in 2 clinical trials with NDM-E infections demonstrated higher proportions of clinical cure with cefiderocol compared with alternative therapy (primarily polymyxin-based regimens), 60% (6/10) versus 20% (1/5), respectively, as well as improved 28-day survival, 90% (9/10) versus 40% (2/5) [326]. Conversely, in a separate trial evaluating a subgroup of patients with NDM-E bloodstream infections, 30-day survival was 50% (8/16) in the cefiderocol arm compared with 82% (9/11) in the alternative therapy arm, again largely polymyxin based [327].

Direct comparative data evaluating aztreonam–avibactam versus cefiderocol for the treatment of infections caused by NDM-E are not available. Both agents are mechanistically rational options in this setting; however, clinical data supporting their use are limited and derived primarily from small subgroup analyses, with variable and, in some cases, suboptimal outcomes. In the absence of evidence demonstrating superiority of one regimen over the other, both aztreonam–avibactam and cefiderocol are considered preferred therapeutic options.

For both aztreonam–avibactam and cefiderocol, resistance in NDM-producing *E. coli* isolates is strongly associated with specific 4–amino acid insertions in PBP3 (eg, YRIN, YRIK), particularly when combined with CMY-type AmpC variants [328–332]. These insertions reduce affinity for aztreonam, ceftazidime, and cefiderocol, compromising the activity of these agents. Although first recognized as a major concern in Asia, these mutants are emerging in the United States and have been described in approximately 30% of NDM-producing *E. coli* isolates in the mid-Atlantic region [333]. When NDM-producing *E. coli* with resistance to both aztreonam–avibactam and cefiderocol are encountered, clinical data to inform optimal therapeutic management remain limited; in such circumstances, individualized combination regimens incorporating agents such as IV fosfomycin, polymyxin B, and/or tigecycline warrant consideration.

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In NDM-producing *K. pneumoniae*, resistance to aztreonam–avibactam is less common than in *E. coli*, but reported mechanisms include permeability changes (eg, porin loss and increased efflux) and mutations in plasmid-encoded AmpC β-lactamases, particularly CMY variants [334, 335]. As aztreonam–avibactam has only recently entered clinical use, further data on resistance mechanisms and frequency are anticipated.

Beyond PBP3 insertions with CMY variants, cefiderocol resistance has been associated with mutations in the TonB-dependent iron transport system [336–339] and increased expression of NDM enzymes [340, 341]. Data suggest that rates of cefiderocol resistance development may vary by MBL family and even by specific NDM allele [234].

Eravacycline and tigecycline represent alternative therapeutic options for NDM-E infections that do not involve the bloodstream or urinary tract, although clinical data are limited (Question 3.7).

Question 3.6: What are the Preferred Antibiotics for the Treatment of Invasive Infections Caused by Carbapenem-Resistant Enterobacterales if OXA-48-like Production is Present?

Suggested Approach

Ceftazidime–avibactam is the preferred treatment option for OXA-48-E infections. Aztreonam–avibactam, cefiderocol, eravacycline, and tigecycline are alternative treatment options, with eravacycline and tigecycline not suggested for infections involving the bloodstream or urinary tract.

Rationale

Ceftazidime–avibactam is the preferred treatment for infections caused by OXA-48-E. In vitro data demonstrate that >95% of OXA-48-E isolates are susceptible to ceftazidime–avibactam [342, 343]. Clinical outcomes data are limited to observational studies.

In a single-arm observational cohort of 171 patients with OXA-48-E infections treated with ceftazidime–avibactam, 79% (135/171) achieved clinical cure, and 78% (134/171) were alive at day 30 [344]. In a second observational study of 76 patients with OXA-48-E bloodstream infections, clinical success was observed in 91% (30/33) of patients treated with ceftazidime–avibactam compared with 58% (25/43) of those receiving alternative regimens (predominantly polymyxin-based therapies). Thirty-day survival was 88% (29/33) in the ceftazidime–avibactam group and 74% (32/43) in the comparator group [345]. Unlike avibactam, both vaborbactam and relebactam demonstrate limited inhibitory activity against OXA-48-like enzymes [346–349]. In a neutropenic thigh infection model evaluating 51 OXA-48-E isolates, the in vivo activity of ceftazidime–avibactam was consistently greater than that of meropenem–vaborbactam or imipenem–relebactam, regardless of MIC values [347]. Accordingly, meropenem–vaborbactam and imipenem–relebactam are not suggested treatment options for OXA-48-E isolates, even when in vitro susceptibility is reported.

Published clinical reports describing treatment-emergent resistance during ceftazidime–avibactam therapy for OXA-48-E infections remain exceedingly sparse. However, in vitro-selected resistant mutants suggest that resistance may arise through alterations in efflux pump systems, modifications or loss of outer membrane porins, structural changes in PBP3, and/or amino acid substitutions within OXA-48-like β-lactamases; the clinical significance of these observations is unknown [350, 351].

Aztreonam–avibactam and cefiderocol represent alternative therapeutic options for OXA-48-E infections. Both agents are active against over 95% of OXA-48-E isolates [47, 301, 303]. Clinical outcomes data for aztreonam–avibactam in OXA-48-E infections are lacking and clinical data for cefiderocol are limited. In a subgroup analysis of 10 patients with OXA-48-E infections enrolled in 2 clinical trials and treated with cefiderocol, 70% (7/10) achieved clinical cure, and 100% (10/10) were alive at day 28 [352].

Eravacycline and tigecycline are also considered alternative options for OXA-48-E infections that do not involve the bloodstream or urinary tract (Question 3.7).

Question 3.7: What is the Role of Tetracycline Derivatives for the Treatment of Infections Caused by Carbapenem-Resistant Enterobacterales?

Suggested Approach

Although β-lactam agents remain preferred treatment options for invasive CRE infections, eravacycline and tigecycline are alternative options when β-lactam agents are either not active or unable to be tolerated. Tetracycline derivatives are not suggested for the treatment of CRE urinary tract or bloodstream infections.

Rationale

Tetracycline derivatives retain activity against CRE isolates independent of carbapenemase production [242]. Resistance is commonly mediated by efflux pump overexpression (eg, AcrAB-TolC) [353]. Tetracycline derivatives exhibit rapid tissue distribution with low serum and urinary concentrations, limiting their utility in bacteremia and UTIs and potentially contributing to poorer outcomes in these settings [266, 267]. Eravacycline and tigecycline may be considered for CRE infections in which tissue penetration is advantageous, such as intra-abdominal infections, skin and soft tissue infections, osteomyelitis, and pneumonia.

Tigecycline has substantially more clinical data for treating CRE infections than eravacycline [354–356]. Although direct comparisons with newer β-lactams are limited, observational data generally favor the newer β-lactams, although tigecycline was not consistently the comparator in these analyses [357].

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Pharmacodynamic modeling and clinical data suggest that tigecycline MICs ≥0.5 μg/mL are associated with poor outcomes in CRE infections, despite remaining within the FDA susceptible range (≤2 μg/mL) [267, 358, 359] (Table 2). Thus, isolates reported as susceptible may still be associated with suboptimal clinical response. CLSI breakpoints are not established for tigecycline or eravacycline.

Clinical trial data evaluating eravacycline for CRE infections are limited, with fewer than 5 CRE cases included across studies [360, 361], and postmarketing data remain sparse [362]. Gastrointestinal intolerance is more frequent with tigecycline than with eravacycline; nausea occurs in approximately 20%–25% of patients receiving tigecycline compared with about 5% with eravacycline [363].

Clinical data evaluating minocycline for CRE infections are also limited, and susceptibility proportions appear lower than for tigecycline or eravacycline [242, 364–366]. Although omadacycline demonstrates greater in vitro activity than minocycline [353, 367, 368], clinical outcome data for CRE infections are lacking. The panel suggests caution when considering minocycline or omadacycline for the treatment of CRE infections.

Question 3.8: What is the Role of Combination Antibiotic Therapy for the Treatment of Infections Caused by Carbapenem-Resistant Enterobacterales?

Suggested Approach

Combination antibiotic therapy (ie, the use of a β-lactam agent in combination with an aminoglycoside, fluoroquinolone, IV fosfomycin, tetracycline, or polymyxin) is not routinely suggested for the treatment of infections caused by CRE.

Rationale

While empiric combination therapy may increase the likelihood that at least one active agent is administered in patients at risk for CRE infection, available evidence does not support the continuation of combination therapy once a β-lactam agent with confirmed in vitro activity has been identified. Continued use of a second agent has not been shown to improve clinical outcomes and instead increases the risk of antibiotic-associated adverse events [369]. Furthermore, clinical data demonstrating that combination therapy prevents the emergence of resistance are lacking.

No clinical trials have compared novel β-lactam agents as monotherapy versus as components of combination regimens (eg, ceftazidime–avibactam alone vs ceftazidime–avibactam plus tobramycin). The limited available observational data have not demonstrated improved outcomes with combination therapy [344, 370, 371]. The largest observational study evaluated 577 patients with KPC-E infections treated with either ceftazidime–avibactam monotherapy or ceftazidime–avibactam in combination with a second agent [371]. Thirty-day survival was comparable between groups—approximately 74% (122/165) in the monotherapy arm and 75% (309/412) in the combination arm—indicating no apparent survival advantage with combination therapy.

Intravenous fosfomycin has been used for several decades in many regions, and observational data evaluating its use as part of combination therapy for CRE infections have yielded variable results [372–375]. Substantial heterogeneity across studies precludes definitive conclusions regarding its role. Although routine use of IV fosfomycin in combination regimens is not suggested, it may be considered in select cases of carbapenem-resistant *E. coli* infection when no active β-lactam agents are available (eg, NDM-E with PBP3 insertions and CMY enzyme production resistant to both aztreonam–avibactam and cefiderocol). In such circumstances, IV fosfomycin is suggested only in combination with another active agent (eg, tigecycline and polymyxin B), recognizing that supporting clinical data are limited.

SECTION 4: *PSEUDOMONAS AERUGINOSA* WITH DIFFICULT-TO-TREAT RESISTANCE

Multidrug-resistant (MDR) *P. aeruginosa* is defined as *P. aeruginosa* that is not susceptible to at least one antibiotic in at least 3 antibiotic classes for which susceptibility is generally expected, including penicillins, cephalosporins, carbapenems, fluoroquinolones, and aminoglycosides [376]. Difficult-to-treat resistance is defined as isolates exhibiting nonsusceptibility to all the following agents: aztreonam, cefepime, ceftazidime, ciprofloxacin, imipenem, levofloxacin, meropenem, and piperacillin–tazobactam [377].

Multidrug-resistant and DTR *P. aeruginosa* generally arise through the interplay of multiple resistance mechanisms. These include decreased expression of outer membrane porins (eg, OprD), increased production of, or amino acid substitutions within, PDC enzymes (commonly referred to as pseudomonal AmpC enzymes), upregulation of efflux pumps (eg, MexAB-OprM), mutations in PBP targets, the presence of ESBLs (eg, $bla_{\text{OXA-10}}$), and occasionally, although rarely in the United States, carbapenemase enzymes (eg, VIM) [378].

Currently, 4 β-lactam agents with potential activity against DTR *P. aeruginosa* are available in the United States. These include cefiderocol, ceftazidime–avibactam, ceftolozane–tazobactam, and imipenem–relebactam. The panel encourages clinical microbiology laboratories to perform AST for all 4 agents when DTR *P. aeruginosa* isolates are identified in clinical cultures given the variable activity of the 4 agents against DTR *P. aeruginosa*. Treatment suggestions for DTR *P. aeruginosa* infections assume that in vitro activity of preferred or alternative agents has been demonstrated.

Question 4.1: What are the Preferred Antibiotics for the Treatment of Infections Caused by Multidrug-Resistant *Pseudomonas aeruginosa*?

Suggested Approach

When *P. aeruginosa* isolates test susceptible to both traditional non-carbapenem β-lactam agents (ie, aztreonam, cefepime,

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ceftazidime, and piperacillin–tazobactam) and to carbapenems, the former are preferred over carbapenem therapy. For infections caused by *P. aeruginosa* isolates not susceptible to any carbapenem but susceptible to traditional β-lactams, the administration of a traditional non-carbapenem β-lactam is also suggested, using high dosages and prolonged infusions. For critically ill patients or those with poor source control with *P. aeruginosa* isolates resistant to carbapenems but susceptible to traditional β-lactams, use of newer β-lactam agents to which *P. aeruginosa* test susceptible (eg, cefiderocol, ceftazidime–avibactam, ceftolozane–tazobactam, and imipenem–relebactam) is also a reasonable treatment approach.

Rationale

In general, when a *P. aeruginosa* isolate demonstrates in vitro susceptibility to traditional antipseudomonal β-lactam agents (eg, aztreonam, cefepime, ceftazidime, and piperacillin–tazobactam) or fluoroquinolones (eg, ciprofloxacin and levofloxacin), the panel favors selection of an agent from these classes rather than carbapenem therapy, with the goal of preserving carbapenem activity for future infections caused by increasingly drug-resistant organisms. Notably, carbapenem use for *P. aeruginosa* infections has been associated with a higher likelihood of resistance emergence during therapy compared with other traditional antipseudomonal β-lactams [379, 380], without corresponding improvements in clinical outcomes [379].

Minimum inhibitory concentration distributions for traditional β-lactam agents are generally higher for *P. aeruginosa* than for Enterobacterales [325]. For instance, the MIC distribution of wild-type *P. aeruginosa* isolates extends to 8 μg/mL for both cefepime and ceftazidime and to 16/4 μg/mL for piperacillin–tazobactam [381]. In contrast, wild-type *E. coli* isolates exhibit substantially lower MIC values, extending to 0.125, 1, and 8/4 μg/mL for cefepime, ceftazidime, and piperacillin–tazobactam, respectively [381].

Given relatively elevated β-lactam MICs, even among wild-type *P. aeruginosa* isolates, prolonged infusions may be required to maximize the proportion of the dosing interval during which free drug concentrations exceed the MIC, a key determinant of β-lactam efficacy and clinical success [382–385]. Consistent with these PK/PD principles, clinical data suggest improved patient outcomes with prolonged β-lactam infusions (eg, 3–4 hours) compared with standard infusions (eg, 30 minutes) for the treatment of *P. aeruginosa* infections [386, 387].

Pseudomonas aeruginosa isolates that are not susceptible to carbapenems (eg, imipenem or meropenem MICs ≥4 μg/mL) yet susceptible to other traditional antipseudomonal β-lactam agents account for at least 40% of carbapenem-resistant *P. aeruginosa* isolates [388, 389]. This resistance phenotype is most commonly attributable to reduced expression or alterations of the OprD porin, which mediates carbapenem entry across the outer membrane of *P. aeruginosa* into the periplasmic space [390, 391].

To date, comparative effectiveness data to inform optimal treatment strategies for infections caused by carbapenem-resistant *P. aeruginosa* isolates that retain susceptibility to traditional β-lactams (eg, cefepime) are limited [392–394] and the optimal approach is unclear. In such cases, the panel's preferred strategy is administration of a traditional β-lactam to which the organism is susceptible using a high-dose, extended-infusion regimen (eg, cefepime 2 g IV every 8 hours infused over ≥3 hours) to optimize drug exposure (Table 1).

An alternative therapeutic strategy is the use of newer β-lactam agents (eg, cefiderocol, ceftazidime–avibactam, ceftolozane–tazobactam, and imipenem–relebactam). While the panel favors preserving their activity for future AMR infections, it is reasonable to consider their use for critically ill patients or in those with ongoing, uncontrolled sources of infection (eg, large intra-abdominal abscess).

Regardless of the antibiotic selected, patients with *P. aeruginosa* infections should be closely monitored for clinical response, given this organism's remarkable capacity to develop resistance through mutational and adaptive changes in gene expression during antibiotic exposure. In an evaluation of 767 episodes of *P. aeruginosa* bacteremia, emergence of resistance within 30 days occurred at the following frequencies for each agent: ceftazidime (12%), imipenem (27%), meropenem (15%), and piperacillin–tazobactam (8%) [380]. Accordingly, clinicians are encouraged to obtain repeat AST for subsequent *P. aeruginosa* isolates recovered from the same patient to guide ongoing therapeutic decision-making.

Question 4.2: What are Preferred Antibiotics for the Treatment of Complicated Urinary Tract Infections Caused by Difficult-to-Treat Resistance *Pseudomonas aeruginosa*?

Suggested Approach

Cefiderocol, ceftazidime–avibactam, ceftolozane–tazobactam, and imipenem–relebactam (listed alphabetically, no order of preference) are the preferred treatment options for cUTI caused by DTR *P. aeruginosa*. Once-daily amikacin or tobramycin are alternative agents for the treatment of DTR *P. aeruginosa* cUTI.

Rationale

Cefiderocol, ceftazidime–avibactam, ceftolozane–tazobactam, and imipenem–relebactam are preferred therapies for cUTI caused by DTR *P. aeruginosa*, supported by clinical trials demonstrating noninferiority compared with standard comparator regimens [140, 260, 261, 264, 395]. Available data are insufficient to preferentially suggest one agent over another. However, given cefiderocol's unique activity compared with the other newer antipseudomonal β-lactams against NDM-E and nonfermenting gram-negative organisms beyond *P.*

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aeruginosa, consideration should be given to reserve its use for these other AMR infections. Treatment considerations for cUTI caused by *P. aeruginosa* isolates producing a carbapenemase are addressed in Question 4.4.

Once-daily amikacin or tobramycin are alternative agents for the treatment of DTR *P. aeruginosa* cUTI [247]. Although expected to be effective as they achieve high renal parenchymal concentrations [74], duration-dependent nephrotoxicity should be carefully considered [75, 76]. These agents are useful for completing treatment courses, given their prolonged renal cortical concentrations and the logistical advantages of once-daily dosing [79, 81] (Table 1, Supplementary Material). Notably, plazomicin's in vitro activity against *P. aeruginosa* is comparable to or slightly inferior to amikacin; it does not offer improved activity over traditional aminoglycosides against this species [396]. Neither CLSI nor FDA breakpoints for plazomicin against *P. aeruginosa* are available (Table 2).

Uncomplicated urinary tract infection due to DTR *P. aeruginosa* is exceedingly uncommon. If identified, potential treatment options include a single dose of amikacin or tobramycin (Question 1.1), cefiderocol, ceftazidime–avibactam, ceftolozane–tazobactam, imipenem–relebactam, or colistin, with the same caveats regarding cefiderocol as described for cUTI. Importantly, colistin, but not polymyxin B, is converted from its inactive prodrug, colistimethate sodium, to its active form within the urinary tract, thereby making colistin, but not polymyxin B, a potential therapeutic option for uUTI [247, 397].

Question 4.3: What are Preferred Antibiotics for the Treatment of Infections Outside of the Urinary Tract Caused by Difficult-to-Treat Resistance *Pseudomonas aeruginosa*?

Suggested Approach

Ceftazidime–avibactam, ceftolozane–tazobactam, and imipenem–relebactam are the preferred agents for the treatment of infections outside of the urinary tract caused by DTR *P. aeruginosa*. Among these, ceftolozane–tazobactam is the preferred option for pneumonia caused by DTR *P. aeruginosa*. Cefiderocol is an alternative option for infections outside of the urinary tract caused by DTR *P. aeruginosa*.

Rationale

Ceftazidime–avibactam, ceftolozane–tazobactam, and imipenem–relebactam are preferred therapies for DTR *P. aeruginosa*, whereas cefiderocol is suggested as an alternative option. Clinical trials directly comparing these agents are not available. However, clinical trials have identified ceftazidime–avibactam, ceftolozane–tazobactam, and imipenem–relebactam to be associated with favorable clinical outcomes for pneumonia, where *P. aeruginosa* is a common pathogen [144, 154, 398]. Although these trials enrolled few patients with DTR *P. aeruginosa*, observational studies have consistently demonstrated improved patient outcomes with these newer agents for infections due to MDR or DTR *P. aeruginosa* compared with alternative regimens, which typically consist of combinations of polymyxins, aminoglycosides, and/or carbapenems [399–403].

In contrast, cefiderocol has not been found to be superior to alternative agents for infections due to AMR *P. aeruginosa* [261]. In a subgroup analysis of a randomized clinical trial that included 22 patients with carbapenem-resistant *P. aeruginosa* infections, survival was 82% in both the cefiderocol and alternative (primarily polymyxin-based) therapy arms [261]. Similarly, in another randomized trial comparing cefiderocol with traditional therapy, a subgroup of 16 patients with carbapenem-resistant *P. aeruginosa* bacteremia experienced identical 30-day survival of 75% in each treatment arm, with the alternative arm consisting of mostly polymyxin-based combinations [327]. Observational data assessing cefiderocol for the treatment of DTR *P. aeruginosa* infections are emerging; however, these studies are generally limited by small sample sizes and the absence of comparator groups, restricting their ability to inform relative effectiveness [404–406]. Accordingly, the panel suggests cefiderocol as an alternative option when inactivity or intolerance precludes the use of preferred β-lactam agents for infections outside of the urinary tract caused by DTR *P. aeruginosa*.

Several observational studies have compared outcomes among patients treated with ceftazidime–avibactam versus ceftolozane–tazobactam for DTR *P. aeruginosa* infections [407–410]. None of these studies identified differences in 30-day survival. However, some differences in clinical outcomes have been observed. The largest study which included patients with MDR *P. aeruginosa* pneumonia or bacteremia reported lower clinical success among the patients with pneumonia treated with ceftazidime–avibactam (51% [89/175]) versus ceftolozane–tazobactam (63% [110/175]) [409]. Across the overall cohort, ceftazidime–avibactam was also associated with a higher 30-day risk of recurrent infection (21% [44/210] vs 15% [31/210]). Another observational study reported higher rates of recurrent pneumonia with ceftazidime–avibactam 18% [21/117] versus ceftolozane–tazobactam (8% [6/80] [410]. Although these studies were observational and "clinical success" is a more subjective endpoint than mortality, particularly when it comes to nonfermenting gram-negatives and pneumonia, PK/PD differences between ceftazidime–avibactam and ceftolozane–tazobactam may explain these findings. Ceftolozane achieves epithelial lining fluid concentrations that are 50% of plasma concentrations and exceed 4 μg/mL (ie, the ceftolozane susceptibility breakpoint) for 100% of the dosing interval in patients with pneumonia [411]. In contrast, both ceftazidime and avibactam achieve epithelial lining fluid concentrations of approximately 30% of plasma levels, which

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may be suboptimal given that ceftazidime–avibactam relies on adequate exposure of both components to achieve activity against DTR *P. aeruginosa* [412].

Difficult-to-treat resistance *P. aeruginosa* isolates may be less likely to test susceptible to imipenem–relebactam compared with ceftolozane–tazobactam and ceftazidime–avibactam [413, 414]. Moreover, less clinical outcomes data are available for imipenem–relebactam. In an observational study of 63 patients with MDR *P. aeruginosa* infections, 30-day survival was approximately 80% [415]. Although no comparator arm was included, survival is similar to that observed with ceftazidime–avibactam and ceftolozane–tazobactam in cohort studies [407–410]. Based on available clinical outcomes data, ceftazidime–avibactam, ceftolozane–tazobactam, and imipenem–relebactam are all considered preferred agents for the treatment of DTR *P. aeruginosa* infections. However, the panelists prefer ceftolozane–tazobactam when treating DTR *P. aeruginosa* pneumonia.

Question 4.4: How Does Identification of Carbapenemases Produced by *P. aeruginosa* Influence Treatment Selection?

Suggested Approach

Although carbapenemase production is currently uncommon among *P. aeruginosa* clinical isolates in the United States, carbapenemase identification has important therapeutic implications. For infections caused by KPC-producing *P. aeruginosa*, cefiderocol, ceftazidime–avibactam, and imipenem–relebactam are preferred treatment options. Cefiderocol is the preferred agent for infections caused by *P. aeruginosa* producing NDM, VIM, or IMP carbapenemases (collectively known as MBLs).

Rationale

Carbapenemase production is an infrequent mechanism of carbapenem resistance among *P. aeruginosa* isolates in the United States [389, 416]. In contrast, carbapenemases (most commonly KPC or VIM carbapenemases) are identified in a substantial proportion of carbapenem-resistant *P. aeruginosa* isolates in other regions of the world, including Latin America (approximately 69%), Asia (57%), and Southern Europe (50%) [389, 417–420]. With increasing globalization and international travel, the prevalence of carbapenemase-producing *P. aeruginosa* in the United States is likely to rise. The clinical relevance of this threat was underscored by a 2022–2023 outbreak of *P. aeruginosa*, producing both VIM and GES (ie, Guiana extended-spectrum β-lactamase enzymes), linked to contaminated artificial tears, which resulted in 81 infections across the United States, permanent vision loss in 17% of cases and death in 5% [421–423].

In addition to carbapenemases, other β-lactamase enzymes—such as GES, Vietnamese extended-spectrum β-lactamase (VEB), and *Pseudomonas* extended resistance (PER) enzymes—remain rare in the United States but may confer elevated MICs to multiple β-lactam agents, including newer β-lactams [22, 424]. Notably, some GES alleles (eg, GES-5) have carbapenemase activity whereas other alleles (eg, GES-1) do not [425]. Although emerging molecular technologies can detect bla_{GES}, bla_{VEB}, and bla_{PER} genes in clinical specimens, these assays are infrequently used in routine clinical microbiology practice. In contrast, several FDA-cleared multiplex assays are available for detection of KPC, VIM, IMP, NDM, and OXA-48 carbapenemases, including from blood cultures and respiratory specimens. Consequently, carbapenemase gene results are often available before AST results for newer β-lactam agents, highlighting the importance of understanding how carbapenemase identification should inform antibiotic decision-making.

When bla_{KPC} genes are identified in *P. aeruginosa* isolates, ceftolozane–tazobactam is not suggested, as tazobactam does not inhibit KPC enzymes. While the in vitro potency of cefiderocol, ceftazidime–avibactam, and imipenem–relebactam against KPC-producing Enterobacterales is well established, data characterizing the activity of these agents against KPC-producing *P. aeruginosa* isolates are limited. Although avibactam and relebactam inhibit KPC in vitro, a study that included 44 KPC-producing *P. aeruginosa* isolates found that 21 (48%) were resistant to ceftazidime–avibactam and 32 (75%) were resistant to imipenem–relebactam [414]. The decision between cefiderocol, ceftazidime–avibactam, and imipenem–relebactam for KPC-producing *P. aeruginosa* infections should be made based on AST results.

When NDM, VIM, or IMP are identified (ie, MBLs), ceftazidime–avibactam, ceftolozane–tazobactam, and imipenem–relebactam are not suggested, as none of these β-lactamase inhibitors have activity against MBLs. When carbapenemase testing results are not available for DTR *P. aeruginosa* isolates, concurrent resistance to ceftazidime–avibactam, ceftolozane–tazobactam, and imipenem–relebactam should raise suspicion for MBL production, with VIM being the most common carbapenemase produced by *P. aeruginosa* in the United States [389]. Cefiderocol is the preferred treatment for MBL-producing *P. aeruginosa* [302, 325, 426], although clinical outcomes data remain limited [326, 427]. Of concern, cefiderocol-resistant MBL-producing *P. aeruginosa* isolates are being reported [428, 429], leaving no remaining active β-lactam options. Importantly, identification of carbapenemase genes does not obviate the need for susceptibility testing, and AST for preferred agents should still be pursued.

Although aztreonam–avibactam generally is active against MBL-producing Enterobacterales, this combination has limited activity against MBL-producing *P. aeruginosa*; neither CLSI nor FDA susceptibility breakpoints are available [430, 431]. While avibactam may reduce aztreonam hydrolysis by PDC enzymes, active efflux (eg, MexAB-OprM) limit aztreonam from achieving adequate intracellular concentrations at

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its PBP3 target [432]. The addition of ceftazidime to aztreonam–avibactam increases the in vitro activity of this combination against MBL-producing *P. aeruginosa* [433, 434], but clinical data supporting ceftazidime–avibactam plus aztreonam remain limited to small case series [435–437]. The panel does not suggest aztreonam–avibactam for MBL-producing *P. aeruginosa*; ceftazidime–avibactam plus aztreonam is suggested only when resistance precludes the use of cefiderocol.

Question 4.5: What is the Likelihood of Emergence of Resistance of Difficult-to-Treat Resistance *Pseudomonas aeruginosa* to Newer β-Lactam Agents When Used to Treat Difficult-to-Treat Resistance *Pseudomonas aeruginosa* Infections?

Suggested Approach

Emergence of resistance during therapy is a concern with all β-lactam agents used to treat *P. aeruginosa* infections. Available data suggest that treatment-emergent resistance occurs in approximately 20% of isolates treated with newer β-lactams.

Rationale

The likelihood of activity of the 4 β-lactam agents currently available in the United States with activity against DTR *P. aeruginosa* (ie, cefiderocol, ceftazidime–avibactam, ceftolozane–tazobactam, and imipenem–relebactam) varies by agent. Among 167 DTR *P. aeruginosa* isolates collected from 2019 to 2021 across the United States, susceptibility proportions were 50% for ceftazidime–avibactam, 74% for ceftolozane–tazobactam, and 37% for imipenem–relebactam; cefiderocol was not evaluated [413]. In a separate study of 154 DTR isolates collected from 2020 to 2023, susceptibility was 99%, 61%, 71%, and 62% for cefiderocol, ceftazidime–avibactam, ceftolozane–tazobactam, and imipenem–relebactam, respectively [438]. Across surveillance studies, cefiderocol generally has the greatest likelihood of in vitro activity against MDR and DTR *P. aeruginosa* [426, 438, 439].

As with all antibiotics used for *P. aeruginosa* infections, treatment with newer β-lactams increases the likelihood that subsequent isolates will no longer be susceptible. The emergence of resistance has been most extensively evaluated for ceftazidime–avibactam and ceftolozane–tazobactam, reflecting their more frequent use in clinical practice relative to cefiderocol and imipenem–relebactam [440].

For ceftazidime–avibactam, resistance most commonly arises through amino acid substitutions in chromosomal PDCs, particularly within or adjacent to the omega loop [441–445]; reduced affinity of PBP3 for ceftazidime has also been occasionally identified [446]. Mechanisms of resistance emerging during ceftolozane–tazobactam therapy closely mirror those observed with ceftazidime–avibactam, consistent with their shared cephalosporin core structure. Although ceftolozane contains R2 side-chain modifications that enhance stability against PDCs and improve PBP binding [216], cross-resistance between ceftazidime–avibactam and ceftolozane–tazobactam exceeds 50% [408, 444, 445]. Emergence of nonsusceptibility during ceftazidime–avibactam or ceftolozane–tazobactam therapy has been reported in approximately 24% of DTR *P. aeruginosa* infections [408, 409, 447].

Cefiderocol is a siderophore–cephalosporin that exploits iron acquisition systems of *P. aeruginosa* by mimicking iron-binding siderophores, allowing it to enter bacterial cells through TonB-dependent receptors rather than traditional porins [336]. Once in the periplasmic space, cefiderocol inhibits PBPs, particularly PBP3, disrupting cell-wall synthesis. Consequently, mutations of components of this pathway can confer cefiderocol resistance as can modifications to PBP3 [448, 449]. Moreover, the absence of a β-lactamase inhibitor renders cefiderocol vulnerable to resistance mechanisms mediated by PDC and other β-lactamase overexpression or structural modifications [448, 450, 451]. Given shared structural features among cephalosporins, increases in cefiderocol MICs have been observed in approximately 15% of *P. aeruginosa* isolates following exposure to ceftazidime–avibactam or ceftolozane–tazobactam [450, 452]. In contrast, limited data suggest that fewer than 10% of *P. aeruginosa* isolates develop cefiderocol resistance during treatment [261].

Resistance to imipenem–relebactam most commonly develops through loss or modification of the OprD porin and/or overexpression of efflux pumps such as MexAB-OprM or MexEF-OprN [453, 454]. OprD alterations reduce imipenem uptake, while relebactam—although able to protect imipenem against PDC enzymes—remains susceptible to efflux [453, 454]. Across 2 clinical trials including 50 patients with *P. aeruginosa* infections treated with imipenem–relebactam, no emergence of imipenem–relebactam resistance was observed [276, 398]. However, in 2 observational studies including 46 patients, treatment-emergent imipenem–relebactam nonsusceptibility developed in approximately 24% of patients [415, 455].

Although estimates will continue to evolve, emergence of resistance during therapy remains a concern across all 4 newer β-lactam agents, supporting repeat AST for isolates recovered in culture while on therapy or for those from patients who re-present with suspected DTR *P. aeruginosa* infection.

Question 4.6: What is the Role of Combination Antibiotic Therapy for the Treatment of Infections Caused by Difficult-to-Treat Resistance *Pseudomonas aeruginosa*?

Suggested Approach

Combination antibiotic therapy is not suggested for infections caused by DTR *P. aeruginosa* when in vitro susceptibility to cefiderocol, ceftazidime–avibactam, ceftolozane–tazobactam, or imipenem–relebactam has been confirmed.

Rationale

Although empiric combination therapy (eg, addition of tobramycin to a newer β-lactam) may be reasonable to increase the

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likelihood of initial coverage in patients at risk for DTR *P. aeruginosa* infection, available data do not support continued combination therapy once susceptibility to a β-lactam agent has been confirmed. Specifically, combination therapy has not been shown to improve clinical outcomes compared with β-lactam monotherapy and is associated with a higher risk of antibiotic-related adverse events [369]. Moreover, clinical data demonstrating that combination therapy prevents the emergence of resistance are lacking.

Clinical trials comparing monotherapy versus combination therapy for the newer β-lactam agents are not available. Observational studies have not identified a survival benefit associated with combination therapy [456, 457]. Considering the known toxicities of aminoglycosides and polymyxins, and the absence of demonstrated clinical benefit, the panel does not suggest routine use of combination therapy for DTR *P. aeruginosa* infections when an active β-lactam agent is available.

When no β-lactam agent demonstrates in vitro activity, combination therapy with tobramycin (if susceptible) and a newer β-lactam may be considered, preferentially selecting the β-lactam with an MIC closest to its susceptibility breakpoint. For example, if ceftazidime–avibactam and ceftolozane–tazobactam MICs are >128/4 μg/mL and the imipenem–relebactam MIC is 4/4 μg/mL (intermediate), imipenem–relebactam in combination with tobramycin is favored. Data supporting improved outcomes with this strategy are lacking. Notably, aminoglycoside monotherapy is not suggested for DTR *P. aeruginosa* infections outside of the urinary tract due to suboptimal probabilities of achieving PD exposures associated with bactericidal activity [458, 459].

If tobramycin susceptibility is not demonstrated, polymyxin B may be considered in combination with a newer β-lactam. For infections outside the urinary tract, polymyxin B is preferred over colistin because it is administered in its active form, achieves more reliable plasma concentrations, and may be associated with a lower risk of nephrotoxicity [460–462].

Question 4.7: What is the Role of Nebulized Antibiotics for the Treatment of Difficult-to-Treat Resistance *Pseudomonas aeruginosa* Pneumonia?

Suggested Approach

The panel does not suggest the routine use of nebulized antibiotics for the treatment of pneumonia caused by DTR *P. aeruginosa* when in vitro susceptibility to an active systemic β-lactam agent has been demonstrated.

Rationale

Three clinical trials have evaluated adjunctive nebulized antibiotic therapy versus placebo in adults with gram-negative pneumonia, with all participants permitted to receive concomitant systemic antibiotics. These studies included nebulized colistin in 100 patients (34% with *P. aeruginosa*) [463]; nebulized amikacin/fosfomycin in 142 patients (22% with *P. aeruginosa*) [464]; and nebulized amikacin in 508 patients (32% with *P. aeruginosa*) [465]. None of these trials demonstrated improvements in clinically meaningful outcomes, including survival, either in the overall study populations or within subgroups infected with drug-resistant pathogens.

Multiple systematic reviews and meta-analyses have further analyzed evidence from clinical trials and observational studies evaluating adjunctive inhaled antibiotic therapy for pneumonia, including infections caused by MDR *P. aeruginosa*, but not DTR *P. aeruginosa* [466–469]. None have identified a survival benefit with adjunctive inhaled antibiotics; some have identified modest improvements in clinical response [466–468].

Although PK/PD modeling suggests that aerosolized antibiotics can achieve high epithelial lining fluid concentrations [470], several factors may limit the clinical effectiveness of nebulized antibiotics. These include inadequate penetration into consolidated lung parenchyma and heterogeneous drug distribution within infected airways, which may preclude sustained bactericidal activity [471]. Such limitations likely reflect, in part, the off-label use of parenteral formulations and delivery via nebulization devices (eg, jet nebulizers) not specifically optimized for pulmonary drug delivery [472, 473]. Accordingly, recommendations from professional societies regarding the use of adjunctive inhaled antibiotics remain inconsistent [7, 474–476].

Given the absence of a survival benefit, the panel does not support the routine use of adjunctive nebulized antibiotics for pneumonia caused by DTR *P. aeruginosa*. However, selective adjunctive use (eg tobramycin, amikacin, colistin methanesulfonate) is reasonable in patients for whom no newer β-lactam agents exhibit in vitro activity against *P. aeruginosa* and in whom clinical response to systemic therapy is suboptimal. In such circumstances, adverse events such as bronchoconstriction, laryngeal injury, and hypersensitivity pneumonitis should be considered [477, 478].

SECTION 5: CARBAPENEM-RESISTANT *ACINETOBACTER BAUMANNII* COMPLEX

CRAB is a formidable nosocomial pathogen with a remarkable ability to persist in healthcare environments, evade host immune defenses, and acquire diverse AMR mechanisms [479]. For simplicity, this guidance uses the term "CRAB," recognizing that many clinical microbiology laboratories cannot reliably distinguish *A. baumannii* from other members of the *A. baumannii–calcoaceticus* complex [480].

CRAB is most frequently isolated from respiratory tract and wound cultures, where distinguishing colonization from true infection, particularly in mechanically ventilated patients or those with extensive burns, can be challenging. Invasive CRAB infections occur predominantly in critically ill patients

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with substantial comorbidities, and poor clinical outcomes are often influenced by host factors and severity of illness, complicating attribution to antibiotic therapy alone.

A defining feature of *A. baumannii* is its extraordinary capacity to accumulate resistance mechanisms, which rapidly eliminates most antibiotic options once carbapenem resistance emerges [481]. Resistance is mediated through multiple, often coexisting mechanisms. These include production of enzymes such as *A. baumannii*–derived cephalosporinases (ADCs), which inactivate many traditional β-lactams, and OXA-type carbapenemases (eg, OXA-23, OXA-24/40, and OXA-58), which confer resistance to traditional β-lactam antibiotics, including carbapenems [482–485]. Aminoglycoside activity is frequently compromised by aminoglycoside-modifying enzymes or 16S rRNA methyltransferases, often resulting in class-wide resistance [486]. In addition to enzymatic mechanisms, CRAB commonly exhibits reduced outer membrane permeability due to loss or alteration of porins limiting carbapenem entry (eg, CarO and OmpA) [487]. Moreover, the increased hydrophobicity of the outer membrane contributes to decreased entry of β-lactams and aminoglycosides [488]. Overexpression of intrinsic efflux pumps (eg, AdeABC and AdeIJK) further contributes to resistance across multiple antibiotic classes, including β-lactams, fluoroquinolones, and tetracycline derivatives [489]. Target site modifications also play an important role; alterations in PBPs (eg, PBP1a/1b and PBP3) reduce sulbactam activity [490, 491], while mutations in DNA gyrase and topoisomerase IV decrease fluoroquinolone susceptibility [492]. The convergence of extensive multidrug resistance, diagnostic uncertainty, and limited effective therapies contribute to the challenges of managing CRAB infections. This guidance document focuses on the treatment of invasive CRAB infections.

Question 5.1: What is the Role of Sulbactam–Durlobactam for the Treatment of Invasive Carbapenem-Resistant *Acinetobacter baumannii* Infections?

Suggested Approach

Sulbactam–durlobactam, in combination with imipenem or meropenem, is the preferred treatment for invasive CRAB infections.

Rationale

Sulbactam shares the bicyclic β-lactam core of penicillins, with a distinguishing sulfone moiety, rendering it susceptible to hydrolysis by class A (eg, TEM-1), class C (eg, ADC-30), and class D (eg, OXA-23) β-lactamases commonly produced by CRAB isolates [493]. Durlobactam, a diazabicyclooctane β-lactamase inhibitor structurally related to avibactam, inhibits class A, C, and D enzymes but lacks activity against class B enzymes (ie, MBLs) such as NDMs. By preventing β-lactamase-mediated hydrolysis, durlobactam enables sulbactam to effectively engage PBP1a/1b and PBP3 [494], with surveillance data showing activity against over 95% of CRAB isolates in the United States [495, 496]. Although NDM-producing CRAB isolates remain uncommon in the United States [481, 497], they are increasingly reported in South Asia and the Middle East [498–501].

In a randomized trial including 125 patients with CRAB pneumonia or bloodstream infections, 28-day survival was higher with sulbactam–durlobactam plus imipenem than with colistin plus imipenem (81% [51/63] vs 68% [42/62]) as was clinical cure (62% [39/63] vs 40% [25/62]) [502]. Notably, the comparator regimen (ie, colistin plus imipenem) is not a preferred or alternative therapy for CRAB. Nonetheless, given the higher survival in the sulbactam–durlobactam arm and given that survival in other CRAB trials frequently is <60% [503], these findings support sulbactam–durlobactam as a preferred option for invasive CRAB infections.

Evidence further indicates that adding a carbapenem (imipenem or meropenem) [504] enhances sulbactam–durlobactam activity through complementary PBP binding, with sulbactam targeting PBP1a/1b and PBP3 and imipenem preferentially binding PBP2, while both agents are protected from β-lactamase hydrolysis by durlobactam [505–507]. In vitro, combination therapy with a carbapenem reduces sulbactam MICs by 1- to 2-fold and achieves >2-log CFU/mL reductions in time-kill assays, activity not observed with sulbactam–durlobactam alone [505, 507, 508]. Although confirmation of the clinical benefit of these in vitro observations is needed, the panelists suggest adding imipenem or meropenem to sulbactam–durlobactam for the treatment of invasive CRAB infections. For patients requiring prolonged durations of therapy (eg, CRAB osteomyelitis), although data are not available regarding the benefit of continuing dual therapy, it may be reasonable to discontinue the carbapenem component after sustained clinical improvement has been achieved.

Mechanisms of resistance to sulbactam–durlobactam during therapy are incompletely characterized. Available data suggest resistance may arise from combinations of PBP1a/b or PBP3 mutations and/or overexpression of efflux pumps (eg, AdeIJK) [509–511]. While confirmatory susceptibility testing is suggested, the high likelihood of susceptibility among US CRAB isolates supports empiric initiation of sulbactam–durlobactam for invasive CRAB infections while awaiting AST results. Clinical data to guide management of sulbactam–durlobactam–resistant CRAB infections are lacking. If resistance to sulbactam–durlobactam (MIC ≥16/4 μg/mL) or an MBL gene (eg, bla_{NDM}) is identified, the panel prefers non-sulbactam–based regimens (eg, combinations of cefiderocol, minocycline, polymyxin B, or tigecycline), as sulbactam-based therapy is unlikely to provide meaningful additional benefit,

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although limited in vitro data suggest sulbactam–durlobactam may enhance cefiderocol activity in these settings [512].

Question 5.2: What is the Role of Ampicillin–Sulbactam for the Treatment of Invasive Carbapenem-Resistant *Acinetobacter baumannii* Infections?

Suggested Approach

If sulbactam–durlobactam is not immediately available, high-dose ampicillin–sulbactam (total daily dose of 9 g of the sulbactam component) in combination with at least one additional agent (eg, cefiderocol, minocycline, or polymyxin B) may be used as a temporary bridging therapy for invasive CRAB infections only until sulbactam–durlobactam (in combination with a carbapenem) can be initiated.

Rationale

As described in Question 5.1, sulbactam is a competitive, irreversible β-lactamase inhibitor that, at high exposures, binds and saturates PBP1a/1b and PBP3 of *A. baumannii* [490, 513]. Sulbactam activity against *A. baumannii* has been demonstrated across multiple PD investigations [514–516], animal models [517–519], and clinical outcomes studies [503, 520, 521], including infections caused by isolates categorized as nonsusceptible by standard susceptibility testing.

Ampicillin–sulbactam is formulated in a fixed 2:1 ratio; for example, a 3 g dose contains 2 g of ampicillin and 1 g of sulbactam. Antimicrobial susceptibility testing for ampicillin–sulbactam is likewise reported in a fixed ratio, permitting estimation of sulbactam MICs. For instance, an ampicillin–sulbactam MIC of 8/4 μg/mL corresponds to a sulbactam MIC of 4 μg/mL, which aligns with the CLSI susceptibility breakpoint for sulbactam (Table 2) [8]. Surveillance data suggest that only approximately 10% of CRAB isolates are susceptible to sulbactam [522, 523]. Unlike sulbactam–durlobactam, ampicillin–sulbactam lacks a β-lactamase inhibitor capable of protecting sulbactam from hydrolysis by the β-lactamases commonly produced by CRAB.

Several studies suggest that high-dose ampicillin–sulbactam (eg, 9 g IV every 8 hours administered as a prolonged infusion, corresponding to a total daily sulbactam dose of 9 g) may partially overcome elevated sulbactam MIC distributions in CRAB [518, 524, 525]. Neutropenic murine lung infection models suggest a total daily sulbactam dose of 4 g achieves >90% probability of target attainment for isolates with sulbactam MICs ≤4 μg/L (ie, susceptible), whereas a total daily dose of 9 g extended coverage to isolates with MICs up to 8 μg/mL (ie, intermediate) [518, 519].

Standard doses of ampicillin–sulbactam appear adequate for infections caused by sulbactam-susceptible CRAB isolates. However, inaccuracies associated with commonly used AST methods for ampicillin–sulbactam in CRAB—particularly when methods other than reference broth microdilution are employed raise concern that isolates categorized as "susceptible" may in fact be nonsusceptible [526, 527]. Given the high mortality associated with invasive CRAB infections, misclassification of susceptibility may lead to suboptimal therapy.

Two meta-analyses encompassing more than 20 observational studies or clinical trials and over 2000 patients with CRAB infections suggest that sulbactam-containing regimens are associated with improved clinical outcomes compared with alternative therapies, which most often consisted of polymyxin- or tetracycline-based regimens [520, 521]. Many patients included in these analyses were infected with sulbactam-resistant isolates. Both meta-analyses were limited by the predominantly observational nature of included studies, small sample sizes, heterogeneous antibiotic regimens and dosing strategies, and limited differentiation between colonization and infection. Among the 5 clinical trials evaluating survival in invasive CRAB infections that included sulbactam in one treatment arm [502, 503, 528–530], only one demonstrated a statistically significant survival benefit [531]; however, all trials reported numerically higher survival in the sulbactam-containing arm. Importantly, safety signals have not been identified with high-dose sulbactam across clinical studies.

Taken together, available preclinical and clinical data suggest that standard-dose ampicillin–sulbactam (3 g IV every 6 hours infused over 30 minutes) may be sufficient for infections caused by sulbactam-susceptible CRAB isolates. Nonetheless, due to potential AST misclassification, the absence of significant toxicity with higher doses, and the substantial mortality associated with inadequately treated invasive CRAB infections, the panel favors the use of high-dose ampicillin–sulbactam administered as a prolonged infusion even for isolates reported as sulbactam susceptible (Table 1).

Although clinical evidence demonstrating superior outcomes with combination therapy remains limited, hydrolysis of sulbactam by CRAB isolates is anticipated in the absence of a protective β-lactamase inhibitor. Accordingly, the panel favors combining high-dose ampicillin–sulbactam with a second active agent (eg, cefiderocol, minocycline, or polymyxin B). Importantly, high-dose ampicillin–sulbactam combined with a second agent should only be used as a bridging option while awaiting initiation of sulbactam–durlobactam in combination with a carbapenem, assuming sulbactam–durlobactam is not immediately available. Sulbactam–durlobactam plus a carbapenem remains preferred for invasive CRAB infections.

Question 5.3: What is the Role of Cefiderocol Therapy for the Treatment of Invasive Carbapenem-Resistant *Acinetobacter baumannii* Infections?

Suggested Approach

Cefiderocol in combination with at least one other agent (ie, high-dose ampicillin–sulbactam, minocycline, or polymyxin B) is an alternative treatment for invasive CRAB infections.

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Rationale

Surveillance data indicate that >90% of CRAB isolates demonstrate in vitro susceptibility to cefiderocol [302, 324, 439, 495]. However, although rarely encountered in the United States, NDM-producing CRAB isolates exhibit substantially reduced susceptibility, with only approximately 60% remaining susceptible to cefiderocol [324]. Although the incidence has not been well quantified, the emergence of cefiderocol resistance among CRAB isolates during therapy remains a concern and is most commonly associated with alterations in iron uptake pathways—such as disruption or downregulation of the *pirA* and *piuA* genes—as well as overexpression of β-lactamases (eg, ADCs) [532–535].

Compared with other gram-negative pathogens, *A. baumannii* generally requires higher cefiderocol PK/PD targets to achieve optimal bacterial killing and animal models of *A. baumannii* infection have demonstrated variable bactericidal activity with cefiderocol [536–539].

Clinical outcome data for cefiderocol in CRAB infections remain inconclusive. Notably, no clinical trials or observational studies have compared outcomes among patients treated with cefiderocol versus sulbactam–durlobactam. In a randomized clinical trial that included a subgroup of 54 patients with CRAB infections, 28-day survival was 51% (20/39) among patients treated with cefiderocol compared with 82% (14/17) among those receiving alternative regimens, which were predominantly polymyxin based [261]. In a second randomized clinical trial enrolling 47 patients with CRAB pneumonia, 14-day survival occurred in 78% (18/23) of patients receiving cefiderocol and 83% (20/24) receiving high-dose, extended-infusion meropenem [540]; although limited by small numbers, these results are concerning given that cefiderocol demonstrated outcomes comparable to an agent lacking meaningful activity against CRAB isolates. A third randomized clinical trial included 25 patients with CRAB bacteremia and reported 30-day survival of 55% (6/11) versus 50% (7/14) for cefiderocol and alternative therapy (ie, mostly combinations of ampicillin–sulbactam and polymyxins), respectively [327].

In contrast, observational data have suggested more favorable outcomes with cefiderocol therapy for CRAB infections. A meta-analysis including 1 clinical trial and 7 observational studies reported higher pooled 30-day survival with cefiderocol-based therapy of 58% (200/345) versus 40% (182/455) with alternative regimens [541]. A second meta-analysis comprising 4 observational studies similarly identified higher 30-day survival among patients treated with cefiderocol (62% [104/169]) compared with alternative therapies (37% [95/257]) [542]. No patients in the alternative therapy arms across included observational studies received sulbactam–durlobactam. Additionally, there was significant heterogeneity across studies. Given the ongoing uncertainty regarding the effectiveness of cefiderocol in clinical trials, the panel suggests reserving cefiderocol as an alternative option for the treatment of invasive CRAB infections, in combination with a second agent, only when resistance precludes the use of sulbactam–durlobactam or as a bridging therapy until sulbactam–durlobactam availability.

Question 5.4: What is the Role of Minocycline for the Treatment of Invasive Carbapenem-Resistant *Acinetobacter baumannii* Infections?

Suggested Approach

Minocycline in combination with at least one other agent (ie, high-dose ampicillin–sulbactam, cefiderocol, or polymyxin B) is an alternative treatment for invasive CRAB infections.

Rationale

Minocycline is a tetracycline derivative available in both IV and oral formulations. Applying the revised 2025 CLSI susceptibility breakpoint (≤1 μg/mL), minocycline demonstrates in vitro activity against fewer than 50% of CRAB isolates (Table 2) [543–545]. The frequency of resistance emergence during minocycline therapy is not well described; however, resistance is associated with overexpression of efflux systems such as AdeABC and TetB [546, 547]. Population PK modeling suggests that minocycline dosed at 200 mg every 12 hours provides a high probability of attaining targets associated with bacterial stasis for isolates with MICs up to the susceptible breakpoint of ≤1 μg/mL, but does not reliably attain targets associated with 1-log killing [548]. Minocycline has not been evaluated in clinical trials for CRAB infections. Available evidence is limited to observational studies, which are constrained by small sample sizes, absence of comparator groups, challenges distinguishing colonization from infection, and heterogeneous patient populations [366, 549–551]. Nonetheless, several reports describe favorable clinical outcomes with minocycline-based therapy for CRAB infections.

Tigecycline, another tetracycline derivative, is available only as an IV formulation. Neither CLSI nor FDA susceptibility breakpoints exist for tigecycline against *A. baumannii*. Clinical outcomes data suggest that "high-dose" tigecycline (200 mg IV loading dose followed by 100 mg IV every 12 hours) yields outcomes comparable to alternative agents for CRAB infections [355, 552, 553] (Table 1). Hollow-fiber infection models further demonstrate dose-dependent activity against CRAB isolates [554]. However, as with minocycline, PK data from critically ill patients suggest reduced efficacy when MICs exceed 1 μg/mL [555–557]. Direct comparative studies between minocycline and tigecycline are lacking; however, given the absence of established susceptibility criteria for tigecycline, minocycline is preferred when a tetracycline derivative is selected.

Omadacycline is available in both IV and oral formulations, but neither CLSI nor FDA susceptibility criteria exist for *A. baumannii* isolates. A PK/PD analysis suggests activity of

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omadacycline against CRAB isolates is limited to stasis and requires higher exposures compared with other pathogens [558]; however, a neutropenic murine thigh infection model suggests similar efficacy as tigecycline for CRAB infections [559]. Clinical data are limited [560, 561]. An observational study evaluating 40 patients with CRAB pneumonia suggested similar 28-day survival of 60% (12/20) and 60% (12/20), respectively, when comparing omadacycline and tigecycline [561]. Given limited PK/PD and clinical data, the panel does not suggest omadacycline for the treatment of CRAB infections unless neither minocycline nor tigecycline are available.

Comparative effectiveness data between tetracycline derivatives and sulbactam–durlobactam are not available. Among tetracyclines, the panel suggests minocycline, in combination with a second active agent, as an alternative option for invasive CRAB infections only in settings of resistance to sulbactam–durlobactam or an interim therapy while awaiting access to sulbactam–durlobactam. This suggestion is based on the availability of minocycline susceptibility criteria, PK/PD-informed dosing considerations, and limited but supportive clinical outcomes data. As with other tetracycline derivatives, minocycline exhibits rapid tissue distribution, resulting in relatively low serum and urinary concentrations [63]. Additionally, gastrointestinal intolerance, particularly nausea, occurs in approximately 20%–25% of patients, similar to tigecycline [363, 562, 563].

Question 5.5: What is the Role of Polymyxin B for the Treatment of Invasive Carbapenem-Resistant *Acinetobacter baumannii* Infections?

Suggested Approach

Polymyxin B in combination with at least one other agent (ie, high-dose ampicillin–sulbactam, cefiderocol, or minocycline) is an alternative treatment for invasive CRAB infections.

Rationale

Approximately 85% of CRAB isolates have polymyxin MICs of ≤2 μg/mL [564, 565]. There is no CLSI susceptible category for the polymyxins against *A. baumannii*; rather, *A. baumannii* isolates with polymyxin MICs of ≤2 μg/mL fall into the intermediate category [8] (Table 2). The benefit of polymyxins is diminished when polymyxin MICs are >2 μg/mL [566]. Notably, polymyxin B has a more favorable PK profile than colistin [7, 246, 566].

As previously stated, in a randomized clinical trial including 125 patients with CRAB pneumonia or bloodstream infections, 28-day survival was lower with colistin plus imipenem (68% [42/62]) versus sulbactam–durlobactam plus imipenem (81% [51/63]) as was clinical cure (40% [25/62]) versus 62% [39/63]) [502]. There are several concerns with polymyxins. First, serum concentrations of polymyxins achieved are highly variable and may be inadequate for effective bactericidal activity [246]. Second, the activity of IV polymyxins in pulmonary epithelial lining fluid is suboptimal and generally does not result in adequate bacterial killing in the lungs [567–569]. Third, survival with colistin monotherapy across 6 clinical trials enrolling approximately 500 participants was relatively low, ranging from 27% to 57% across studies [570–575]. It is unclear if trial outcomes would be improved had polymyxin B been administered instead of colistin. Finally, dosages required to treat systemic CRAB infections approach the threshold for nephrotoxicity, making the therapeutic window extremely narrow (ie, approximately 2 μg/mL may be required to achieve 1-$\log_{10}$ reduction in bacterial growth, which is also the threshold associated with nephrotoxicity [249].

Given the ongoing uncertainty regarding the clinical effectiveness of polymyxin B and its unfavorable toxicity profile, the panel suggests reserving it as an alternative option for the treatment of invasive CRAB infections, in combination with a second agent, only when resistance precludes the use of sulbactam–durlobactam or as interim therapy while awaiting access to sulbactam–durlobactam.

Question 5.6: What is the Role of Nebulized Antibiotics for the Treatment of Carbapenem-Resistant *Acinetobacter baumannii* Pneumonia?

Suggested Approach

The panel does not suggest the routine use of nebulized antibiotics for the treatment of pneumonia caused by CRAB.

Rationale

Three randomized clinical trials have assessed adjunctive nebulized antibiotic therapy vs placebo in adults with gram-negative pneumonia, with all participants permitted to receive concomitant systemic antibiotics. These trials evaluated nebulized colistin in 100 patients (65% with *A. baumannii* pneumonia) [463], nebulized amikacin/fosfomycin in 142 patients (20% with *A. baumannii*) [464], and nebulized amikacin in 508 patients (29% with *A. baumannii*) [465]. None of these studies demonstrated improvements in survival, either in the overall study populations or in subgroups infected with drug-resistant pathogens.

Several systematic reviews and meta-analyses have subsequently synthesized data from clinical trials and observational studies evaluating adjunctive inhaled antibiotic therapy for pneumonia, including infections caused by CRAB [466–469]. Consistent with individual trial findings, these analyses have not identified a survival benefit associated with adjunctive inhaled antibiotics, although some have reported modest improvements in clinical response [466–468].

While PK/PD modeling suggests that aerosolized antibiotics can achieve high epithelial lining fluid concentrations [470], several factors likely limit their clinical effectiveness. These include suboptimal penetration into consolidated lung parenchyma and heterogeneous drug distribution within infected airways, which may prevent sustained bactericidal activity

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[471]. Such limitations may be exacerbated by the off-label use of parenteral formulations and delivery via nebulization devices (eg, jet nebulizers) that are not specifically designed for pulmonary drug delivery [472, 473]. Reflecting these uncertainties, recommendations from professional societies regarding the use of adjunctive inhaled antibiotics for pneumonia are inconsistent [7, 474–476].

In light of the absence of a clear survival benefit, the panel does not support the routine use of adjunctive nebulized antibiotics for pneumonia caused by CRAB. Nonetheless, selective use may be reasonable in patients with documented resistance to sulbactam–durlobactam or when clinical response to systemic therapy is suboptimal. In such cases, the potential for adverse events, including bronchoconstriction, laryngeal injury, and hypersensitivity pneumonitis, should be carefully weighed [477, 478].

SECTION 6: *STENOTROPHOMONAS MALTOPHILIA*

Stenotrophomonas maltophilia is an aerobic, glucose nonfermenting, gram-negative bacillus that is ubiquitous in water environments [576]. Although generally believed to be less pathogenic than many other nosocomial organisms, *S. maltophilia* produces biofilm and virulence factors that enable colonization or infection in vulnerable hosts, typically during prolonged hospitalization or extensive antibiotic exposure, particularly carbapenem exposure [577–579]. Accumulating evidence suggests that early initiation of therapy with agents with activity against *S. maltophilia* is critical for optimizing outcomes in patients with invasive *S. maltophilia* infections [580–582].

Stenotrophomonas maltophilia poses several management challenges. First, although *S. maltophilia* has the potential to cause severe infection, determining whether *S. maltophilia* represents a colonizing organism or a true pathogen is difficult, particularly in patients with underlying pulmonary conditions such as cystic fibrosis or ventilator dependency. Moreover, *S. maltophilia* is often recovered as a component of a polymicrobial infection, further complicating decisions on the necessity of targeted *S. maltophilia* therapy [583]. Importantly, *S. maltophilia* can be a true pathogen that causes considerable morbidity, particularly in patients with hematologic malignancies if it causes hemorrhagic pneumonia or bacteremia [584–588].

Second, treatment selection is hampered by AMR genes carried by *S. maltophilia* isolates [589]. Production of L1 MBLs and L2 cephalosporinases render most conventional β-lactams ineffective [590]. *Stenotrophomonas maltophilia* exhibits resistance to aminoglycosides via chromosomal aminoglycoside acetyl transferase enzymes [591]. Furthermore, *S. maltophilia* can accumulate resistance mechanisms that reduce the activity of antibiotics with expected activity against *S. maltophilia*. These include multidrug efflux pumps that increase the MICs of TMP-SMX, tetracyclines, and fluoroquinolones, as well as chromosomal Sm*qnr* genes that further reduce the effectiveness of fluoroquinolones [592–594].

Third, a "standard of care" antibiotic regimen for *S. maltophilia* infections against which to compare the effectiveness of various treatment regimens is not evident. Clinical trials comparing the effectiveness of commonly used agents for *S. maltophilia* are lacking. Data to prioritize among agents with in vitro activity against *S. maltophilia* and to determine the additive benefit of combination therapy regimens remain incomplete.

Lastly, *S. maltophilia* AST determination is problematic. The CLSI has established breakpoints for 6 agents against *S. maltophilia*: cefiderocol, chloramphenicol, levofloxacin, minocycline, ticarcillin–clavulanate, and TMP-SMX [8]. However, ticarcillin–clavulanate manufacturing has been discontinued and chloramphenicol is rarely used in the United States due to significant toxicities, leaving 4 agents for which interpretable antibiotic MIC values can be provided to clinicians. Confidence in MIC interpretive criteria for several of these agents is further undermined by concerns regarding the reproducibility of commonly employed susceptibility testing methods [595, 596], limited PK/PD data to inform breakpoints for most agents, and insufficient data linking MIC data to clinical outcomes. This guidance document focuses on the treatment of invasive *S. maltophilia* infections.

Question 6.1: What is the Role of Cefiderocol for the Treatment of Invasive *Stenotrophomonas maltophilia* Infections?

Suggested Approach

Cefiderocol, as monotherapy, is the preferred treatment for invasive *S. maltophilia* infections, acknowledging this preference is largely based on data derived from neutropenic animal infection models and not clinical outcomes studies.

Rationale

Surveillance studies consistently demonstrate that *S. maltophilia* susceptibility to cefiderocol approaches 100%, including among isolates resistant to other commonly used agents [47, 324, 597–600]. Reports of cefiderocol resistance (ie, MICs ≥2 μg/mL) remain rare [405, 600, 601]. Multiple AMR markers may contribute to increasing cefiderocol MICs; a prominent mechanism is mutations in the iron transport system (eg, *tonB*) [600].

Multiple neutropenic thigh and lung infection models suggest cefiderocol has potent activity against *S. maltophilia* and indicate concordance between in vitro susceptibility and in vivo efficacy using human-simulated dosing regimens [536, 602–604]. Modeling indicates that patients treated with cefiderocol have a >90% probability of achieving PK/PD exposure targets that correlate with bactericidal activity in animal models

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[605]. As one example, in a neutropenic rabbit pneumonia model, cefiderocol eradicated *S. maltophilia* from lung tissue, whereas TMP-SMX was unable to do so, leaving a residual bacterial burden [604]. Survival was substantially higher among cefiderocol-treated rabbits (88% [7/8]) compared with those receiving TMP-SMX (25% [2/8]); no untreated animals survived.

Clinical data regarding the use of cefiderocol for *S. maltophilia* infections remain limited, with survival reported in approximately 70% of patients in observational cohorts [405, 606–608]. Comparisons between cefiderocol and alternative regimens are challenging because of substantial heterogeneity across studies and the absence of direct comparative data. Notably, a meta-analysis of 14 studies including 663 patients treated with alternative agents for *S. maltophilia* infections (eg, TMP-SMX and fluoroquinolones) similarly reported survival in approximately 70% of patients [609]. Clinical trial data are restricted to small subgroup analyses. In one trial, only 5 patients with *S. maltophilia* infections were assigned to the cefiderocol arm, of whom 1 (20%) survived, precluding meaningful comparisons with alternative therapy [261]. In a second trial, 30-day survival was 60% (3/5) among patients receiving cefiderocol, compared with 100% (3/3) among those receiving alternative therapy, which consisted of levofloxacin-based regimens [327].

Notwithstanding these limited, and in some cases unfavorable, clinical data, the panel suggests cefiderocol as a preferred therapy for *S. maltophilia* infections based on its high likelihood of in vitro activity against *S. maltophilia* isolates and its consistent bactericidal activity demonstrated in animal models. There are insufficient data to indicate improved clinical outcomes with the use of cefiderocol as a component of combination therapy.

Question 6.2: What is the Role of Aztreonam–Avibactam for the Treatment of Invasive *Stenotrophomonas maltophilia* Infections?

Suggested Approach

Aztreonam–avibactam, preferably in combination with a second agent, is an alternative treatment option for invasive *S. maltophilia* infections.

Rationale

The L1 metallo-β-lactamases produced by *S. maltophilia* hydrolyze all β-lactams except aztreonam; whereas, the L2 cephalosporinases hydrolyzes aztreonam, along with other β-lactam agents [590]. Avibactam, however, can inhibit the activity of L2 cephalosporinases [610]. Consequently, the combination of aztreonam–avibactam enables aztreonam to reach and bind to PBP3, its primary target, in *S. maltophilia* [611, 612].

Although CLSI and FDA susceptibility breakpoints for aztreonam–avibactam against *S. maltophilia* have not been established as of 2026, surveillance data indicate that >90% of *S. maltophilia* isolates have MICs ≤4/4 μg/mL, the susceptibility breakpoint used for Enterobacterales (Table 2) [613–616]. Resistance to aztreonam–avibactam in *S. maltophilia* may result from overexpression of the L1 and/or L2 β-lactamase, along with upregulation of the Sme multidrug efflux pumps [610, 617].

Regarding PK/PD data, in a neutropenic murine thigh infection model assessing 27 *S. maltophilia* isolates across a range of aztreonam–avibactam MICs, human-simulated Enterobacterales dosing achieved at least a 1-$\log_{10}$ CFU reduction in 72% mice infected with isolates with MICs ≤4/4 μg/mL [618]. However, the PD target required for aztreonam–avibactam to achieve 1-log killing against *S. maltophilia* has not been defined, complicating interpretation of these animal data and precluding Monte Carlo simulations to estimate the probability of target attainment in humans.

Clinical data evaluating aztreonam–avibactam for *S. maltophilia* infections are sparse. Several case reports describe favorable outcomes [619–622]. Subgroup analyses from 2 clinical trials evaluating aztreonam–avibactam included 4 patients with *S. maltophilia* infections, all of whom were randomized to receive aztreonam–avibactam [261, 312]. Clinical outcomes were favorable in 1 patient, indeterminate in 1 patient, and unfavorable in 2 patients, thereby limiting meaningful interpretation of efficacy in this population. Given the limited PK/PD and clinical data, the panel suggests use of aztreonam–avibactam as part of combination therapy, with transition to monotherapy considered only after clear and sustained clinical improvement and confirmation of in vitro susceptibility.

The combination of ceftazidime–avibactam plus aztreonam provides similar activity as aztreonam–avibactam based on mechanistically-informed PD modeling studies [623], although the administration of a single agent (ie, aztreonam–avibactam) is more convenient [624]. In settings where aztreonam–avibactam is not available, ceftazidime–avibactam plus aztreonam is a reasonable alternative. Practical strategies for administering ceftazidime–avibactam plus aztreonam [317–319] are summarized in Table 1 and Supplementary Material. Notably, the combination of ceftazidime–avibactam and aztreonam has similar limitations in available PK/PD and clinical outcomes data for *S. maltophilia* as aztreonam–avibactam.

Question 6.3: What is the Role of Levofloxacin for the Treatment of Invasive *Stenotrophomonas maltophilia* Infections?

Suggested Approach

Levofloxacin, as a component of combination therapy, is an alternative treatment option for invasive *S. maltophilia* infections.

Rationale

Surveillance data from the United States demonstrate approximately 90% of *S. maltophilia* isolates are susceptible to levofloxacin (ie, MICs ≤2 μg/mL) [613, 625]. Emergence of resistance during levofloxacin therapy appears to occur in

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approximately 20% of cases [626–629]. Resistance to levofloxacin is most commonly mediated by mutations in Sm*qnr* that interfere with fluoroquinolone binding to gyrase and topoisomerase and overexpression of Sme multidrug efflux pumps, both of which may emerge under levofloxacin treatment pressure [592, 630–632].

Pharmacokinetic/PD investigations suggest that levofloxacin frequently does not achieve sustained bacterial suppression or PK/PD targets associated with bacterial killing against *S. maltophilia* [633–636]. Levofloxacin dosed at 750 mg IV every 24 hours had approximately a 50% probability of achieving exposure targets associated with bacterial stasis in a neutropenic murine thigh infection model when the isolate has an MIC value of 2 μg/mL (the CLSI susceptible breakpoint), and an approximately 27% probability of achieving targets associated with a ≥1-log reduction in bacterial load [633]. In a murine pneumonia model, levofloxacin improved survival relative to placebo (50% vs 0%), though eradication of infection was not consistently achieved [637].

Clinical trial data evaluating levofloxacin for *S. maltophilia* infections are lacking. Observational studies are constrained by small sample sizes, heterogeneous infection sites, potential misclassification of colonization versus infection, frequent use of concomitant antibiotics, and inconsistent dosing strategies [581, 609, 627, 628, 638–645]. Across studies, no single agent, levofloxacin, minocycline, or TMP-SMX, has been definitively associated with more favorable patient outcomes. Combination therapy is frequently employed in published cohorts; however, studies have not clearly demonstrated improved outcomes with combinations of levofloxacin, minocycline, or TMP-SMX compared with monotherapy with these agents [581, 646, 647].

Given the variable baseline susceptibility, risk of resistance emergence during therapy, limited PK/PD support for bactericidal activity, and difficult to interpret clinical outcomes data, the panel suggests levofloxacin as a component of combination therapy for invasive *S. maltophilia* infections. Levofloxacin-associated toxicities include tendinopathy, QT prolongation, dysglycemia, increased risk of *Clostridiodes difficile* infections, and central nervous system effects [648]. Oral levofloxacin achieves high bioavailability with systemic exposures comparable to IV administration [649]. Transition to levofloxacin monotherapy should be considered only after clear and sustained clinical improvement has been achieved and susceptibility is confirmed.

Question 6.4: What is the Role of Minocycline for the Treatment of Invasive *Stenotrophomonas maltophilia* Infections?

Suggested Approach

Minocycline, as a component of combination therapy, is an alternative treatment option for invasive *S. maltophilia* infections.

Rationale

Among tetracycline derivatives, minocycline has the most favorable preclinical data supporting its activity for *S. maltophilia* infections. Surveillance data from the United States demonstrate in vitro susceptibility to minocycline in approximately 90% of isolates (MIC ≤1 μg/mL) [613]. *Stenotrophomonas maltophilia* resistance to minocycline primarily involves overexpression of Sme multidrug efflux pumps [650]. The CLSI susceptibility breakpoints are available for minocycline against *S. maltophilia* [8] (Table 2), whereas neither CLSI nor FDA breakpoints exist for other tetracycline derivatives.

The strongest evidence supporting minocycline derives from PK/PD investigations [636, 651, 652]. Minocycline dosed at 200 mg IV every 12 hours has approximately a >90% probability of achieving exposure targets associated with bacterial stasis in a neutropenic murine thigh infection model when the isolate has an MIC value of 1 μg/mL (the CLSI susceptible breakpoint), and approximately a 50% probability of achieving targets associated with a ≥1-log reduction in bacterial burden [652]. Similar investigations have not been conducted for other tetracycline derivatives.

As with levofloxacin (Question 6.3), clinical trials evaluating minocycline for *S. maltophilia* are lacking and observational data have not clearly favored levofloxacin, minocycline, or TMP-SMX over each other [642, 643, 653–655]. Cohort studies have not clearly demonstrated improved outcomes with combinations of levofloxacin, minocycline, or TMP-SMX compared with monotherapy with these agents [581, 646, 647].

The panel suggests minocycline as a component of combination therapy for invasive *S. maltophilia* infections, given the limitations of the PK/PD and clinical data. This suggestion is further informed by minocycline's relatively low sustained serum concentrations [63], which raise concern regarding its reliability as monotherapy for invasive disease involving the bloodstream. Transitioning to minocycline as monotherapy should only be considered after clear and sustained clinical improvement is observed. The panel considers IV and oral minocycline interchangeable, as oral administration achieves approximately 95% bioavailability and comparable serum concentrations, steady-state trough levels, and elimination half-lives relative to IV dosing [563]. The most common adverse events associated with minocycline include gastrointestinal symptoms, vestibular effects, and skin reactions [562, 563].

Given the availability of susceptibility criteria and well-characterized PD targets derived from animal models, minocycline is preferred over tigecycline for *S. maltophilia*. In contrast, data supporting the use of eravacycline or omadacycline for *S. maltophilia* infections are sparse or unfavorable [656, 657], and their use is not suggested.

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Question 6.5: What is the Role of Trimethoprim–Sulfamethoxazole for the Treatment of Invasive *Stenotrophomonas maltophilia* Infections?

Suggested Approach

Trimethoprim–sulfamethoxazole, as a component of combination therapy, is an alternative treatment option for invasive *S. maltophilia* infections.

Rationale

Surveillance data from the United States demonstrate in vitro susceptibility of *S. maltophilia* to TMP-SMX in >90% of isolates (MIC ≤2/38 µg/mL) [613, 658]. When resistance occurs, it is commonly mediated by overexpression of multidrug efflux pumps and acquisition of *sul* and *dfrA* genes [594, 659, 660].

Despite its extensive historical use, PK/PD investigations consistently demonstrate that TMP-SMX lacks bactericidal activity against *S. maltophilia*, even when administered at high doses and against isolates with low TMP MICs [597, 651, 661–663]. Across multiple in vitro and PK/PD modeling studies, TMP-SMX achieves, at best, bacterial stasis, with enhanced activity observed when used in combination regimens [651, 662]. A neutropenic rabbit pneumonia model of *S. maltophilia*, demonstrated that TMP-SMX reduced bacterial burden but failed to eradicate infection and was associated with inferior survival compared with cefiderocol (25% [2/8] vs 88% [7/8], respectively) [604], although the TMP-SMX exposures in these rabbits is unknown.

Clinical trial data evaluating TMP-SMX for *S. maltophilia* infections are lacking. Available evidence is derived from observational studies, which have yielded no clear consensus regarding the relative effectiveness of TMP-SMX compared with alternative agents such as levofloxacin or minocycline [642, 643, 653–655]. These analyses are limited by heterogeneous infection syndromes, small sample sizes, frequent concomitant antibiotic use, and/or challenges distinguishing colonization from true infection. Whether TMP-SMX is associated with improved outcomes when used as combination therapy compared with monotherapy remains unclear [581, 646, 647].

Despite the absence of consistent evidence demonstrating improved clinical outcomes with TMP-SMX-based combination therapy, the panel suggests TMP-SMX only as a component of combination therapy for invasive *S. maltophilia* infections, given the limitations of available PK/PD and clinical outcomes data. TMP-SMX is associated with a well-recognized toxicity profile, including hypersensitivity reactions, hematologic abnormalities, gastrointestinal intolerance, hepatotoxicity, renal dysfunction, and electrolyte disturbances, which warrant close monitoring [664]. Oral TMP-SMX achieves essentially complete bioavailability with systemic exposure equivalent to IV administration [665]. Transition to TMP-SMX monotherapy should be considered only after clear and sustained clinical improvement has been achieved.

Question 6.6: What is the Role of Ceftazidime for the Treatment of Invasive *Stenotrophomonas maltophilia* Infections?

Suggested Approach

Ceftazidime is not a suggested treatment for *S. maltophilia* infections due to the presence of β-lactamase genes intrinsic to *S. maltophilia* that are expected to render ceftazidime inactive.

Rationale

The panel does not suggest ceftazidime for the treatment of *S. maltophilia* infections, as L1 and L2 β-lactamases produced by *S. maltophilia* are expected to render it ineffective. Preclinical models suggest ceftazidime is unable to substantially prevent *S. maltophilia* growth [651]. Comparative effectiveness studies evaluating the role of ceftazidime against *S. maltophilia* infections are virtually nonexistent. As of 2024, neither the CLSI nor the FDA no longer have susceptibility breakpoints for ceftazidime against *S. maltophilia* [8].

CONCLUSIONS

The field of AMR is dynamic and rapidly evolving, and the treatment of AMR infections will continue to challenge clinicians. As newer antibiotics against AMR pathogens are incorporated into clinical practice, we are learning more about their effectiveness and propensity to resistance. This treatment guidance will be updated periodically.

Supplementary Data

Supplementary materials are available at *Clinical Infectious Diseases* online. Consisting of data provided by the authors to benefit the reader, the posted materials are not copyedited and are the sole responsibility of the authors, so questions or comments should be addressed to the corresponding author.

Notes

Acknowledgments. The panel acknowledges selected reviewers for providing constructive feedback on the draft manuscript: Ahmed Abdul Azim, Marisa Holubar, and Priya Nori for their contributions as external reviewers. The panel also acknowledges the contributions of Standards and Practice Guidelines Subcommittee primary reviewers Majdi Al Hasan and Nicolás Cortés-Penfield.

Financial support. This work was unfunded.

Potential conflicts of interest. The following list includes what has been reported to IDSA. To provide thorough transparency, IDSA requires full disclosure of all relationships, regardless of relevancy to the guidance topic. Evaluation of such relationships as potential conflicts of interest is determined by a review process which includes assessment by the Board of Directors liaison to the Standards and Practice Guidelines Committee and, if necessary, the Conflicts of Interest and Ethics Committee. The assessment of disclosed relationships for possible conflicts of interests is based on the

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relative weight of the financial relationship (ie, monetary amount) and the relevance of the relationship (ie, the degree to which an association might reasonably be interpreted by an independent observer as related to the topic or recommendation of consideration). IDSA requests panel members to disclose activities and financial relationships/investments related to consultant/advisory roles, promotional speakers bureau, stocks/bonds, honoraria, expert testimony, ownership interest, research grants, organizational benefits, intellectual property, other remuneration, activities with other organizations, and relevant financial interest of family members. Readers of this guidance should be mindful of this when the list of disclosures is reviewed.

P. D. T. reports no disclosures. E. L. H. has served as a scientific consultant for Lexi-Comp . J. A. J. has served as a scientific advisor for bioMérieux, Gilead Sciences, and Shionogi; has received honoraria from Clinical Care Options and Haymarket Medical Education; and has owned stock in Vaxart. A. J. M. has reported no current industry relationships. She previously has served as a scientific advisor to Cepheid, Day Zero Diagnostics, bioMérieux, Merck, Shionogi, Qpex Biopharma, Accelerate Diagnostics, VenatoRX, and OpGen. M. J. S. has received research funding from Merck, bioMérieux, SNIPRBiome, Hardy Diagnostics, and Melinta Therapeutics; has received consulting fees from Beckman Coulter Diagnostics; has received remuneration from AbbVie for serving on a Data and Safety Monitoring Board; has received consulting fees from Shionogi; and has served on a Data and Safety Monitoring Board for Spero Therapeutics. R. A. B. has served as a clinical scientist investigator for Venatorx Pharmaceuticals; has received research funding from Shionogi, Merck, Entasis Therapeutics, Wockhardt, Allecra Therapeutics, AstraZeneca, Harrington Family Foundation, Tetraphase Pharmaceuticals, Steris, Melinta Therapeutics, and VenatoRx; and has received an honorarium from Unilab.

All authors have submitted the ICMJE Form for Disclosure of Potential Conflicts of Interest. Conflicts that the editors consider relevant to the content of the manuscript have been disclosed.

References

1. Antimicrobial Resistance Collaborators. Global burden of bacterial antimicrobial resistance in 2019: a systematic analysis. Lancet **2022**; 399:629–55.
2. GBD 2021 Antimicrobial Resistance Collaborators. Global burden of bacterial antimicrobial resistance 1990–2021: a systematic analysis with forecasts to 2050. Lancet **2024**; 404:1199–226.
3. de Kraker MEA, Lipsitch M. Burden of antimicrobial resistance: compared to what? Epidemiol Rev **2022**; 43:53–64.
4. Wolford H, McCarthy NL, Baggs J, et al. Antimicrobial-resistant infections in hospitalized patients. JAMA Netw Open **2025**; 8:e2462059.
5. Lockowitz CR, Hsu AJ, Chiotos K, et al. Suggested dosing of select beta-lactam agents for the treatment of antimicrobial-resistant gram-negative infections in children. J Pediatric Infect Dis Soc **2025**; 14:piaf004.
6. Hsu AJ, Chiotos K, Heil EL, Shields RK, Tamma PD. The effectiveness of newer beta-lactams for the treatment of antimicrobial-resistant gram-negative meningitis. Clin Infect Dis **2025**; 80:959–68.
7. Tsuji BT, Pogue JM, Zavascki AP, et al. International Consensus Guidelines for the optimal use of the polymyxins: endorsed by the American College of Clinical Pharmacy (ACCP), European Society of Clinical Microbiology and Infectious Diseases (ESCMID), Infectious Diseases Society of America (IDSA), International Society for Anti-infective Pharmacology (ISAP), Society of Critical Care Medicine (SCCM), and Society of Infectious Diseases Pharmacists (SIDP). Pharmacotherapy **2019**; 39:10–39.
8. Clinical and Laboratory Standards Institute. Performance standards for antimicrobial susceptibility testing, 36th ed. CLSI M100. Wayne, PA: Clinical and Laboratory Standards Institute, **2026**.
9. MacFadden DR, Coburn B, Shah N, et al. Utility of prior cultures in predicting antibiotic resistance of bloodstream infections due to Gram-negative pathogens: a multicentre observational cohort study. Clin Microbiol Infect **2018**; 24:493–9.
10. Bakhit M, Hoffmann T, Scott AM, Beller E, Rathbone J, Del Mar C. Resistance decay in individuals after antibiotic exposure in primary care: a systematic review and meta-analysis. BMC Med **2018**; 16:126.
11. Blagojevic C, Brown KA, Diong C, et al. Long-term risk of infection among patients colonized with antimicrobial-resistant pathogens: a population-wide cohort study. Open Forum Infect Dis **2024**; 11:ofae712.
12. de Nies L, Kobras CM, Stracy M. Antibiotic-induced collateral damage to the microbiota and associated infections. Nat Rev Microbiol **2023**; 21:789–804.
13. Tamma PD, Avdic E, Li DX, Dzintars K, Cosgrove SE. Association of adverse events with antibiotic use in hospitalized patients. JAMA Intern Med **2017**; 177:1308–15.
14. Soto CL, Hsu AJ, Lee JH, et al. Identifying effective durations of antibiotic therapy for the treatment of carbapenem-resistant Enterobacterales bloodstream infections: a multicenter observational study. Clin Infect Dis **2024**; 78:27–30.
15. Trautner BW, Cortés-Penfield NW, Gupta K et al. Clinical Practice Guideline by Infectious Diseases Society of America (IDSA): 2025 Guideline on Management and Treatment of Complicated Urinary Tract Infections: Introduction and Methods. Available at: https://www.idsociety.org/practiceguideline/complicated-urinary-tract-infections/. Accessed 01 March 2026.
16. Heil EL, Bork JT, Abbo LM, et al. Optimizing the management of uncomplicated gram-negative bloodstream infections: consensus guidance using a modified delphi process. Open Forum Infect Dis **2021**; 8:ofab434.
17. Tamma PD, Smith TT, Adebayo A, et al. Prevalence of bla(CTX-M) genes in gram-negative bloodstream isolates across 66 hospitals in the United States. J Clin Microbiol **2021**; 59:e00127–21.
18. Castanheira M, Kimbrough JH, DeVries S, Mendes RE, Sader HS. Trends of β-lactamase occurrence among *Escherichia coli* and *Klebsiella pneumoniae* in United States hospitals during a 5-year period and activity of antimicrobial agents against isolates stratified by β-lactamase type. Open Forum Infect Dis **2023**; 10:ofad038.
19. Pham TM, Zhang Y, Nevers M, et al. Trends in infection incidence and antimicrobial resistance in the US Veterans Affairs Healthcare System: a nationwide retrospective cohort study (2007–22). Lancet Infect Dis **2024**; 24:1333–46.
20. Bush K, Bradford PA. Epidemiology of β-lactamase-producing pathogens. Clin Microbiol Rev **2020**; 33:e00047–19.
21. Castanheira M, Farrell SE, Krause KM, Jones RN, Sader HS. Contemporary diversity of β-lactamases among Enterobacteriaceae in the nine U.S. census regions and ceftazidime-avibactam activity tested against isolates producing the most prevalent β-lactamase groups. Antimicrob Agents Chemother **2014**; 58:833–8.
22. Castanheira M, Simner PJ, Bradford PA. Extended-spectrum β-lactamases: an update on their characteristics, epidemiology and detection. JAC Antimicrob Resist **2021**; 3:dlab092.
23. Hareza DA, Cosgrove SE, Simner PJ, et al. Is carbapenem therapy necessary for the treatment of non-CTX-M extended-spectrum β-lactamase-producing Enterobacterales bloodstream infections? Clin Infect Dis **2024**; 78:1103–10.
24. Henderson A, Paterson DL, Chatfield MD, et al. Association between minimum inhibitory concentration, beta-lactamase genes and mortality for patients treated with piperacillin/tazobactam or meropenem from the MERINO study. Clin Infect Dis **2021**; 73:e3842–e50.
25. Hopkins KL, Batchelor MJ, Liebana E, Deheer-Graham AP, Threlfall EJ. Characterisation of CTX-M and AmpC genes in human isolates of Escherichia coli identified between 1995 and 2003 in England and Wales. Int J Antimicrob Agents **2006**; 28:180–92.
26. Hareza DA, Bergman Y, Jacobs E, et al. Establishing the optimal ceftriaxone MIC for predicting ESBL production in bloodstream infections. Open Forum Infect Dis **2025**; 12:ofaf490.

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27. Talan DA, Takhar SS, Krishnadasan A, et al. Emergence of extended-spectrum β-lactamase urinary tract infections among hospitalized emergency department patients in the United States. Ann Emerg Med **2021**; 77:32–43.
28. Goodlet KJ, Benhalima FZ, Nailor MD. A systematic review of single-dose aminoglycoside therapy for urinary tract infection: is it time to resurrect an old strategy? Antimicrob Agents Chemother **2019**; 63:e02165–18.
29. Jenrette JE, Coronato K, Miller MA, Molina KC, Quinones A, Jacknin G. Prospective evaluation of single-dose aminoglycosides for treatment of complicated cystitis in the emergency department. Acad Emerg Med **2024**; 31:649–55.
30. Watkins RR, Thapaliya D, Lemonovich TL, Bonomo RA. Gepotidacin: a novel, oral, 'first-in-class' triazaacenaphthylene antibiotic for the treatment of uncomplicated urinary tract infections and urogenital gonorrhoea. J Antimicrob Chemother **2023**; 78:1137–42.
31. Hackel MA, Karlowsky JA, Sahm DF, West JM, Scangarella-Oman NE. In vitro activity of gepotidacin against urinary tract infection isolates of Enterobacterales, *Enterococcus faecalis*, and *Staphylococcus saprophyticus*. Antimicrob Agents Chemother **2025**; 69:e0029625.
32. Scangarella-Oman NE, Butler DL, Breton J, Brown D, Kasapidis C, Sheets AJ. Efficacy and in vitro activity of gepotidacin against bacterial uropathogens, including drug-resistant phenotypes, in females with uncomplicated urinary tract infections: results from two global, pivotal, phase 3 trials (EAGLE-2 and EAGLE-3). Antimicrob Agents Chemother **2025**; 69:e0164024.
33. Wagenlehner F, Perry CR, Hooton TM, et al. Oral gepotidacin versus nitrofurantoin in patients with uncomplicated urinary tract infection (EAGLE-2 and EAGLE-3): two randomised, controlled, double-blind, double-dummy, phase 3, non-inferiority trials. Lancet **2024**; 403:741–55.
34. Das A, Alam U, Khan AS, et al. Efficacy and safety of gepotidacin for urogenital infections: a systematic review and meta-analysis of randomized controlled trials. Infection **2026**; 54:1063–75.
35. Ahlstrand J, Maatouk I, Doanh LH, et al. High in vitro activity of the novel antimicrobial gepotidacin against *Neisseria gonorrhoeae* isolates in eight WHO Enhanced Gonococcal Antimicrobial Surveillance Programme countries in three WHO regions, 2021–24. J Antimicrob Chemother **2026**; 81:dkaf452.
36. Sanchez GV, Baird AM, Karlowsky JA, Master RN, Bordon JM. Nitrofurantoin retains antimicrobial activity against multidrug-resistant urinary *Escherichia coli* from US outpatients. J Antimicrob Chemother **2014**; 69:3259–62.
37. Bruyndonckx R, Latour K, Atud GA, et al. Time trend of prevalence and susceptibility to nitrofurantoin of urinary MDR *Escherichia coli* from outpatients. J Antimicrob Chemother **2019**; 74:3264–7.
38. Tutone M, Bjerklund Johansen TE, Cai T, Mushtaq S, Livermore DM. SUsceptibility and Resistance to Fosfomycin and other antimicrobial agents among pathogens causing lower urinary tract infections: findings of the SURF study. Int J Antimicrob Agents **2022**; 59:106574.
39. Huttner A, Kowalczyk A, Turjeman A, et al. Effect of 5-day nitrofurantoin vs single-dose fosfomycin on clinical resolution of uncomplicated lower urinary tract infection in women: a randomized clinical trial. JAMA **2018**; 319:1781–9.
40. Sougakoff W, Jarlier VV. Comparative potency of mecillinam and other beta-lactam antibiotics against *Escherichia coli* strains producing different beta-lactamases. J Antimicrob Chemother **2000**; 46 Suppl A:9–14.
41. Lodise TP, Kaye KS, Santerre Henriksen A, Kahlmeter G. Review of the in vitro microbiological activity of mecillinam against common uropathogens in uncomplicated urinary tract infection: focus on resistant pathogens. Open Forum Infect Dis **2024**; 11:ofae296.
42. Raja NS. Emerging clinical role of pivmecillinam in the treatment of urinary tract infections caused by extended spectrum betaeta-lactamase (ESBL) producing Enterobacteriaceae. Int J Clin Pract **2019**; 73:1–5.
43. Vik I, Bollestad M, Grude N, et al. Ibuprofen versus pivmecillinam for uncomplicated urinary tract infection in women—a double-blind, randomized non-inferiority trial. PLoS Med **2018**; 15:e1002569.
44. Menday AP. Comparison of pivmecillinam and cephalexin in acute uncomplicated urinary tract infection. Int J Antimicrob Agents **2000**; 13:183–7.
45. Bollestad M, Grude N, Solhaug S, et al. Clinical and bacteriological efficacy of pivmecillinam treatment for uncomplicated urinary tract infections caused by ESBL-producing *Escherichia coli*: a prospective, multicentre, observational cohort study. J Antimicrob Chemother **2018**; 73:2503–9.
46. Kaye KS, Santerre Henriksen A, Sommer M, Frimodt-Møller N. Safety and tolerability of pivmecillinam during more than four decades of clinical experience: a systematic review. Clin Infect Dis **2025**; 80:280–99.
47. Walkty AJ, Karlowsky JA, Baxter MR, et al. In vitro activity of sulopenem against 1880 bacterial pathogens isolated from Canadian patients with urinary tract infections (CANWARD, 2014-21). J Antimicrob Chemother **2022**; 77:3414–3420.
48. Dunne MW, Aronin SI, Das AF, et al. Sulopenem or ciprofloxacin for the treatment of uncomplicated urinary tract infections in women: a phase 3, randomized trial. Clin Infect Dis **2023**; 76:66–77.
49. Probenecid. FDA drug label. Available at: https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=f2375d90-4132-4fb6-aee7-c4f9d11e98d5. Accessed 31 December 2025.
50. Zalmanovici Trestioreanu A, Green H, Paul M, Yaphe J, Leibovici L. Antimicrobial agents for treating uncomplicated urinary tract infection in women. Cochrane Database Syst Rev **2010**; 2010:CD007182.
51. Critchley IA, Cotroneo N, Pucci MJ, Mendes R. The burden of antimicrobial resistance among urinary tract isolates of *Escherichia coli* in the United States in 2017. PLoS One **2019**; 14:e0220265.
52. Butler AM, Durkin MJ, Keller MR, Ma Y, Powderly WG, Olsen MA. Association of adverse events with antibiotic treatment for urinary tract infection. Clin Infect Dis **2022**; 74:1408–18.
53. Sastry S, Clarke LG, Alrowais H, Querry AM, Shutt KA, Doi Y. Clinical appraisal of fosfomycin in the era of antimicrobial resistance. Antimicrob Agents Chemother **2015**; 59:7355–61.
54. Ito R, Mustapha MM, Tomich AD, et al. Widespread fosfomycin resistance in gram-negative bacteria attributable to the chromosomal *fosA* gene. mBio **2017**; 8:e00749–17.
55. Elliott ZS, Barry KE, Cox HL, et al. The role of *fosA* in challenges with fosfomycin susceptibility testing of multispecies *Klebsiella pneumoniae* carbapenemase-producing clinical isolates. J Clin Microbiol **2019**; 57:e00634–19.
56. Abbott IJ, Dekker J, van Gorp E, et al. Impact of bacterial species and baseline resistance on fosfomycin efficacy in urinary tract infections. J Antimicrob Chemother **2020**; 75:988–96.
57. Wenzler E, Bleasdale SC, Sikka M, et al. Phase I study to evaluate the pharmacokinetics, safety, and tolerability of two dosing regimens of oral fosfomycin tromethamine in healthy adult participants. Antimicrob Agents Chemother **2018**; 62:e00464–18.
58. Rouphael N, Winokur P, Keefer MC, et al. Daily fosfomycin versus levofloxacin for complicated urinary tract infections. mBio **2023**; 14:e0167723.
59. Hooton TM, Scholes D, Gupta K, Stapleton AE, Roberts PL, Stamm WE. Amoxicillin-clavulanate vs ciprofloxacin for the treatment of uncomplicated cystitis in women: a randomized trial. JAMA **2005**; 293:949–55.
60. Campbell JD, Lewis JS 2nd, McElmeel ML, Fulcher LC, Jorgensen JH. Detection of favorable oral cephalosporin-clavulanate interactions by in vitro disk approximation susceptibility testing of extended-spectrum-beta-lactamase-producing members of the Enterobacteriaceae. J Clin Microbiol **2012**; 50:1023–6.
61. Thomson GK, Ayaz M, Lutes K, Thomson KS. An improved extended-spectrum-beta-lactamase detection test utilizing aztreonam plus clavulanate. J Clin Microbiol **2018**; 56:e01309–17.
62. Estebanez A, Pascual R, Gil V, Ortiz F, Santibáñez M, Perez Barba C. Fosfomycin in a single dose versus a 7-day course of amoxicillin-clavulanate for the treatment of asymptomatic bacteriuria during pregnancy. Eur J Clin Microbiol Infect Dis **2009**; 28:1457–64.
63. Agwuh KN, MacGowan A. Pharmacokinetics and pharmacodynamics of the tetracyclines including glycylcyclines. J Antimicrob Chemother **2006**; 58:256–65.
64. Mukerji AC, Sharma MM, Taneja OP, Saxena SN, Bhatnagar RK, Ghosh-Ray B. A clinical trial of alpha-6-deoxyoxytetracycline (doxycycline) in the treatment of urinary tract infections. Chemotherapy **1969**; 14:77–85.
65. Musher DM, Minuth JN, Thorsteinsson SB, Holmes T. Effectiveness of achievable urinary concentrations of tetracyclines against "tetracycline-resistant" pathogenic bacteria. J Infect Dis **1975**; 131 Suppl:S40–4.
66. Sandberg T, Skoog G, Hermansson AB, et al. Ciprofloxacin for 7 days versus 14 days in women with acute pyelonephritis: a randomised, open-label and double-blind, placebo-controlled, non-inferiority trial. Lancet **2012**; 380:484–90.
67. Ren H, Li X, Ni ZH, et al. Treatment of complicated urinary tract infection and acute pyelonephritis by short-course intravenous levofloxacin (750 mg/day) or conventional intravenous/oral levofloxacin (500 mg/day): prospective, open-label, randomized, controlled, multicenter, non-inferiority clinical trial. Int Urol Nephrol **2017**; 49:499–507.
68. Talan DA, Stamm WE, Hooton TM, et al. Comparison of ciprofloxacin (7 days) and trimethoprim-sulfamethoxazole (14 days) for acute uncomplicated pyelonephritis pyelonephritis in women: a randomized trial. JAMA **2000**; 283:1583–90.
69. Morrissey I, Hawser S, Kothari N, et al. Evaluation of the activity of cefepime/enmetazobactam against Enterobacterales bacteria collected in Europe from 2019 to 2021, including third-generation cephalosporin-resistant isolates. J Glob Antimicrob Resist **2024**; 38:71–82.
70. Kaye KS, Belley A, Barth P, et al. Effect of cefepime/enmetazobactam vs piperacillin/tazobactam on clinical cure and microbiological eradication in patients with complicated urinary tract infection or acute pyelonephritis: a randomized clinical trial. JAMA **2022**; 328:1304–14.
71. Kaye KS, Rice LB, Dane AL, et al. Fosfomycin for injection (ZTI-01) versus piperacillin-tazobactam for the treatment of complicated urinary tract infection

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including acute pyelonephritis: ZEUS, a phase 2/3 randomized trial. Clin Infect Dis **2019**; 69:2045–56.
72. Sojo-Dorado J, López-Hernández I, Rosso-Fernandez C, et al. Effectiveness of fosfomycin for the treatment of multidrug-resistant *Escherichia coli* bacteremic urinary tract infections: a randomized clinical trial. JAMA Netw Open **2022**; 5: e2137277.
73. Tamma PD, Bonomo RA, Stiefel U. The role of intravenous fosfomycin: finding our way out of Dante's forest dark. JAMA Netw Open **2022**; 5:e2138691.
74. Bergeron MG, Trottier S. Influence of single or multiple doses of gentamicin and netilmicin on their cortical, medullary, and papillary distribution. Antimicrob Agents Chemother **1979**; 15:635–41.
75. Bertino JS Jr, Booker LA, Franck PA, Jenkins PL, Franck KR, Nafziger AN. Incidence of and significant risk factors for aminoglycoside-associated nephrotoxicity in patients dosed by using individualized pharmacokinetic monitoring. J Infect Dis **1993**; 167:173–9.
76. Drusano GL, Ambrose PG, Bhavnani SM, Bertino JS, Nafziger AN, Louie A. Back to the future: using aminoglycosides again and how to dose them optimally. Clin Infect Dis **2007**; 45:753–60.
77. Wagenlehner FME, Cloutier DJ, Komirenko AS, et al. Once-daily plazomicin for complicated urinary tract infections. N Engl J Med **2019**; 380:729–40.
78. Mohsenpour B, Ahmadi A, Azizzadeh H, et al. Comparison of three doses of amikacin on alternate days with a daily dose of meropenem during the same period for the treatment of urinary tract infection with *E. coli*: a double-blind clinical trial. BMC Res Notes **2024**; 17:38.
79. Elbaz M, Zadka H, Weiss-Meilik A, Ben-Ami R. Effectiveness and safety of an institutional aminoglycoside-based regimen as empirical treatment of patients with pyelonephritis. J Antimicrob Chemother **2020**; 75:2307–13.
80. Vidal L, Gafter-Gvili A, Borok S, Fraser A, Leibovici L, Paul M. Efficacy and safety of aminoglycoside monotherapy: systematic review and meta-analysis of randomized controlled trials. J Antimicrob Chemother **2007**; 60:247–57.
81. Chou A, Welch E, Hunter A, Trautner BW. Antimicrobial treatment options for difficult-to-treat resistant gram-negative bacteria causing cystitis, pyelonephritis, and prostatitis: a narrative review. Drugs **2022**; 82:407–38.
82. Dunne MW, Aronin SI, Das AF, et al. Sulopenem for the treatment of complicated urinary tract infections including pyelonephritis: a phase 3, randomized trial. Clin Infect Dis **2023**; 76:78–88.
83. Ten Doesschate T, Kuiper S, van Nieuwkoop C, et al. Fosfomycin vs ciprofloxacin as oral step-down treatment for *Escherichia coli* febrile urinary tract infections in women: a randomized, placebo-controlled, double-blind, multicenter trial. Clin Infect Dis **2022**; 75:221–9.
84. Harris PNA, Tambyah PA, Lye DC, et al. Effect of piperacillin-tazobactam vs meropenem on 30-day mortality for patients with *E coli* or *Klebsiella pneumoniae* bloodstream infection and ceftriaxone resistance: a randomized clinical trial. JAMA **2018**; 320:984–94.
85. Liebchen U, Kratzer A, Wicha SG, Kees F, Kloft C, Kees MG. Unbound fraction of ertapenem in intensive care unit patients. J Antimicrob Chemother **2014**; 69: 3108–11.
86. Burkhardt O, Kumar V, Katterwe D, et al. Ertapenem in critically ill patients with early-onset ventilator-associated pneumonia: pharmacokinetics with special consideration of free-drug concentration. J Antimicrob Chemother **2007**; 59: 277–84.
87. Brink AJ, Richards GA, Schillack V, Kiem S, Schentag J. Pharmacokinetics of once-daily dosing of ertapenem in critically ill patients with severe sepsis. Int J Antimicrob Agents **2009**; 33:432–6.
88. Shah S, Clarke LG, Balasubramani GK, Peng L, Smith BJ, Shields RK. Impact of hypoalbuminemia on patients receiving ertapenem combination therapy for methicillin-susceptible *Staphylococcus aureus* bacteraemia. J Antimicrob Chemother **2025**; 80:1823–7.
89. Zusman O, Farbman L, Tredler Z, et al. Association between hypoalbuminemia and mortality among subjects treated with ertapenem versus other carbapenems: prospective cohort study. Clin Microbiol Infect **2015**; 21:54–8.
90. Lee NY, Huang WH, Tsui KC, Hsueh PR, Ko WC. Carbapenem therapy for bacteremia due to extended-spectrum β-lactamase-producing *Escherichia coli* or *Klebsiella pneumoniae*. Diagn Microbiol Infect Dis **2011**; 70:150–3.
91. Chen M, Nafziger AN, Drusano GL, Ma L, Bertino JS Jr. Comparative pharmacokinetics and pharmacodynamic target attainment of ertapenem in normal-weight, obese, and extremely obese adults. Antimicrob Agents Chemother **2006**; 50:1222–7.
92. Goutelle S, Valour F, Gagnieu MC, et al. Population pharmacokinetics and probability of target attainment of ertapenem administered by subcutaneous or intravenous route in patients with bone and joint infection. J Antimicrob Chemother **2018**; 73:987–94.
93. Lass J, Tamme K, Kipper K, Starkopf J. Dosing of ertapenem in an extreme obesity: a case report of 250 kg patient. Case Rep Crit Care **2017**; 2017:5310768.
94. Das S, Fitzgerald R, Ullah A, et al. Intrapulmonary pharmacokinetics of cefepime and enmetazobactam in healthy volunteers: towards new treatments for nosocomial pneumonia. Antimicrob Agents Chemother **2020**; 65:e01468–20.
95. Papp-Wallace KM, Bethel CR, Caillon J, et al. Beyond piperacillin-tazobactam: cefepime and AAI101 as a potent β-lactam-β-lactamase inhibitor combination. Antimicrob Agents Chemother **2019**; 63:e00105–19.
96. Dunne MW, Aronin SI, Das AF, et al. A phase 3 randomized trial of sulopenem vs. ertapenem in patients with complicated intra-abdominal infections. Clin Microbiol Infect **2025**; 31:396–401.
97. Bush K, Macalintal C, Rasmussen BA, Lee VJ, Yang Y. Kinetic interactions of tazobactam with beta-lactamases from all major structural classes. Antimicrob Agents Chemother **1993**; 37:851–8.
98. Livermore DM, Andrews JM, Hawkey PM, et al. Are susceptibility tests enough, or should laboratories still seek ESBLs and carbapenemases directly? J Antimicrob Chemother **2012**; 67:1569–77.
99. Zhou M, Wang Y, Liu C, et al. Comparison of five commonly used automated susceptibility testing methods for accuracy in the China Antimicrobial Resistance Surveillance System (CARSS) hospitals. Infect Drug Resist **2018**; 11:1347–58.
100. Livermore DM, Day M, Cleary P, et al. OXA-1 β-lactamase and non-susceptibility to penicillin/β-lactamase inhibitor combinations among ESBL-producing *Escherichia coli*. J Antimicrob Chemother **2019**; 74:326–33.
101. Burgess DS, Hall RG 2nd. In vitro killing of parenteral beta-lactams against standard and high inocula of extended-spectrum beta-lactamase and non-ESBL producing *Klebsiella pneumoniae*. Diagn Microbiol Infect Dis **2004**; 49:41–6.
102. Harada Y, Morinaga Y, Kaku N, et al. In vitro and in vivo activities of piperacillin-tazobactam and meropenem at different inoculum sizes of ESBL-producing *Klebsiella pneumoniae*. Clin Microbiol Infect **2014**; 20:O831–9.
103. Thomson KS, Moland ES. Cefepime, piperacillin-tazobactam, and the inoculum effect in tests with extended-spectrum beta-lactamase-producing Enterobacteriaceae. Antimicrob Agents Chemother **2001**; 45:3548–54.
104. Monogue ML, Heil EL, Aitken SL, Pogue JM. The role of tazobactam-based combinations for the management of infections due to extended-spectrum beta-lactamase-producing Enterobacterales: insights from the Society of Infectious Diseases Pharmacists. Pharmacotherapy **2021**; 41:864–80.
105. Piperacillin-tazobactam. FDA drug label. Available at: https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=b1cb1314-d3e0-4b44-a1df-de14c32e8cc6. Accessed 7 March 2026.
106. Seo YB, Lee J, Kim YK, et al. Randomized controlled trial of piperacillin-tazobactam, cefepime and ertapenem for the treatment of urinary tract infection caused by extended-spectrum beta-lactamase-producing *Escherichia coli*. BMC Infect Dis **2017**; 17:404.
107. Sutton SP, Reinert JP. Evaluation of the efficacy of piperacillin/tazobactam in extended-spectrum beta-lactamase-producing Enterobacteriaceae urinary tract infections: a systematic review of the literature. Ann Pharmacother **2025**; 59: 648–56.
108. NCT03671967. PipEracillin Tazobactam Versus mERoPENem for treatment of bloodstream infections caused by cephalosporin-resistant Enterobacteriaceae (PETERPEN). Accessed 31 December 2025.
109. Zhang H, Xu J, Xiao Q, et al. Carbapenem-sparing beta-lactam/beta-lactamase inhibitors versus carbapenems for bloodstream infections caused by extended-spectrum beta-lactamase-producing Enterobacteriaceae: a systematic review and meta-analysis. Int J Infect Dis **2023**; 128:194–204.
110. Månsson TS, Askemyr A, Sunnerhagen T, Tham J, Riesbeck K, Mellhammar L. Piperacillin/tazobactam versus carbapenems for 30-day mortality in patients with ESBL-producing Enterobacterales bloodstream infections: a retrospective, multicenter, non-inferiority, cohort study. Infection **2025**; 53:1769–77.
111. Capsoni N, Bettina M, Bellone B, et al. Empirical piperacillin/tazobactam vs carbapenems to treat bloodstream infections due to ESBL Enterobacterales: a propensity score-matched observational study conducted in an emergency department in Northern Italy. Infect Dis (Lond) **2025**; 58:495–506.
112. Rebuck JA, Olsen KM, Fey PD, Bergman KL, Rupp ME. In vitro activities of parenteral beta-lactam antimicrobials against TEM-10-, TEM-26- and SHV-5-derived extended-spectrum beta-lactamases expressed in an isogenic *Escherichia coli* host. J Antimicrob Chemother **2000**; 46:461–4.
113. Sirot D. Extended-spectrum plasmid-mediated beta-lactamases. J Antimicrob Chemother **1995**; 36 Suppl A:19–34.
114. Smith KP, Brennan-Krohn T, Weir S, Kirby JE. Improved accuracy of cefepime susceptibility testing for extended-spectrum-beta-lactamase-producing Enterobacteriaceae with an on-demand digital dispensing method. J Clin Microbiol **2017**; 55:470–8.
115. Cefepime. FDA drug label. Available at: https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=02aa374f-37b4-456a-b5de-cfd3bbb6ce6e. Accessed 31 December 2025.

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116. Kim SA, Altshuler J, Paris D, Fedorenko M. Cefepime versus carbapenems for the treatment of urinary tract infections caused by extended-spectrum β-lactamase-producing enterobacteriaceae. Int J Antimicrob Agents **2018**; 51: 155–8.
117. Branton AC, Vu CH, Venugopalan V, et al. Re-evaluation of cefepime or piperacillin/tazobactam to decrease use of carbapenems in ESBL-producing Enterobacterales urinary tract infections (REDUCE-UTI). JAC Antimicrob Resist **2023**; 5:dlad021.
118. Wang R, Cosgrove SE, Tschudin-Sutter S, et al. Cefepime therapy for cefepime-susceptible extended-spectrum β-lactamase-producing Enterobacteriaceae bacteremia. Open Forum Infect Dis **2016**; 3:ofw132.
119. Lee NY, Lee CC, Li CW, et al. Cefepime therapy for monomicrobial *Enterobacter cloacae* bacteremia: unfavorable outcomes in patients infected by cefepime-susceptible dose-dependent isolates. Antimicrob Agents Chemother **2015**; 59: 7558–63.
120. Chopra T, Marchaim D, Veltman J, et al. Impact of cefepime therapy on mortality among patients with bloodstream infections caused by extended-spectrum-β-lactamase-producing *Klebsiella pneumoniae* and *Escherichia coli*. Antimicrob Agents Chemother **2012**; 56:3936–42.
121. Lee NY, Lee CC, Huang WH, Tsui KC, Hsueh PR, Ko WC. Cefepime therapy for monomicrobial bacteremia caused by cefepime-susceptible extended-spectrum beta-lactamase-producing Enterobacteriaceae: MIC matters. Clin Infect Dis **2013**; 56:488–95.
122. Frescas BE, McCoy CM, Kirby J, Bowden R, Mercuro NJ. Outcomes associated with empiric cefepime for bloodstream infections caused by ceftriaxone-resistant, cefepime-susceptible *Escherichia coli* and *Klebsiella pneumoniae*. Int J Antimicrob Agents **2023**; 61:106762.
123. Zanetti G, Bally F, Greub G, et al. Cefepime versus imipenem-cilastatin for treatment of nosocomial pneumonia in intensive care unit patients: a multicenter, evaluator-blind, prospective, randomized study. Antimicrob Agents Chemother **2003**; 47:3442–7.
124. Stewart AG, Cottrell K, Henderson A, et al. In vitro activity of cefotetan against ESBL-producing *Escherichia coli* and *Klebsiella pneumoniae* bloodstream isolates from the MERINO trial. Microbiol Spectr **2021**; 9:e0022621.
125. Kusumoto M, Kanao Y, Narita H, et al. In vitro efficacy of cephamycins against multiple extended-spectrum β-lactamase-producing *Klebsiella pneumoniae*, *Proteus mirabilis*, and *Enterobacter cloacae* isolates from dogs and cats. J Vet Med Sci **2023**; 85:653–6.
126. Lee CH, Su LH, Tang YF, Liu JW. Treatment of ESBL-producing *Klebsiella pneumoniae* bacteraemia with carbapenems or flomoxef: a retrospective study and laboratory analysis of the isolates. J Antimicrob Chemother **2006**; 58:1074–7.
127. Yang CC, Li SH, Chuang FR, et al. Discrepancy between effects of carbapenems and flomoxef in treating nosocomial hemodialysis access-related bacteremia secondary to extended spectrum beta-lactamase producing *Klebsiella pneumoniae* in patients on maintenance hemodialysis. BMC Infect Dis **2012**; 12:206.
128. Doi A, Shimada T, Harada S, Iwata K, Kamiya T. The efficacy of cefmetazole against pyelonephritis caused by extended-spectrum beta-lactamase-producing Enterobacteriaceae. Int J Infect Dis **2013**; 17:e159–63.
129. Pilmis B, Parize P, Zahar JR, Lortholary O. Alternatives to carbapenems for infections caused by ESBL-producing Enterobacteriaceae. Eur J Clin Microbiol Infect Dis **2014**; 33:1263–5.
130. Matsumura Y, Yamamoto M, Nagao M, et al. Multicenter retrospective study of cefmetazole and flomoxef for treatment of extended-spectrum-beta-lactamase-producing *Escherichia coli* bacteremia. Antimicrob Agents Chemother **2015**; 59:5107–13.
131. Fukuchi T, Iwata K, Kobayashi S, Nakamura T, Ohji G. Cefmetazole for bacteremia caused by ESBL-producing enterobacteriaceae comparing with carbapenems. BMC Infect Dis **2016**; 16:427.
132. Lee CH, Su LH, Chen FJ, et al. Comparative effectiveness of flomoxef versus carbapenems in the treatment of bacteraemia due to extended-spectrum β-lactamase-producing *Escherichia coli* or *Klebsiella pneumoniae* with emphasis on minimum inhibitory concentration of flomoxef: a retrospective study. Int J Antimicrob Agents **2015**; 46:610–5.
133. Hayakawa K, Matsumura Y, Uemura K, et al. Effectiveness of cefmetazole versus meropenem for invasive urinary tract infections caused by extended-spectrum β-lactamase-producing *Escherichia coli*. Antimicrob Agents Chemother **2023**; 67:e0051023.
134. Takemura W, Tashiro S, Hayashi M, et al. Cefmetazole as an alternative to carbapenems against extended-spectrum beta-lactamase-producing *Escherichia coli* infections based on in vitro and in vivo pharmacokinetics/pharmacodynamics experiments. Pharm Res **2021**; 38:1839–46.
135. Randomised Controlled Trial of Meropenem versus Cefmetazole for definitive treatment of bloodstream infections due to ESBL-producing *Escherichia coli*. Available at: https://center6.umin.ac.jp/cgi-open-bin/ctr_e/ctr_view.cgi?recptno=R000055809. Accessed 31 December 2025.
136. Senard O, Lafaurie M, Lesprit P, et al. Efficacy of cefoxitin versus carbapenem in febrile male urinary tract infections caused by extended spectrum beta-lactamase-producing *Escherichia coli*: a multicenter retrospective cohort study with propensity score analysis. Eur J Clin Microbiol Infect Dis **2020**; 39: 121–9.
137. Chabert P, Provoost J, Cohen S, et al. Pharmacokinetics, efficacy and tolerance of cefoxitin in the treatment of cefoxitin-susceptible extended-spectrum beta-lactamase producing Enterobacterales infections in critically ill patients: a retrospective single-center study. Ann Intensive Care **2022**; 12:90.
138. Bush K, Bradford PA. Interplay between β-lactamases and new β-lactamase inhibitors. Nat Rev Microbiol **2019**; 17:295–306.
139. Karlowsky JA, Biedenbach DJ, Kazmierczak KM, Stone GG, Sahm DF. Activity of ceftazidime-avibactam against extended-spectrum- and AmpC β-lactamase-producing Enterobacteriaceae collected in the INFORM global surveillance study from 2012 to 2014. Antimicrob Agents Chemother **2016**; 60: 2849–57.
140. Carmeli Y, Armstrong J, Laud PJ, et al. Ceftazidime-avibactam or best available therapy in patients with ceftazidime-resistant Enterobacteriaceae and *Pseudomonas aeruginosa* complicated urinary tract infections or complicated intra-abdominal infections (REPRISE): a randomised, pathogen-directed, phase 3 study. Lancet Infect Dis **2016**; 16:661–73.
141. Mendes RE, Castanheira M, Woosley LN, Stone GG, Bradford PA, Flamm RK. Molecular β-lactamase characterization of aerobic gram-negative pathogens recovered from patients enrolled in the ceftazidime-avibactam phase 3 trials for complicated intra-abdominal infections, with efficacies analyzed against susceptible and resistant subsets. Antimicrob Agents Chemother **2017**; 61:e02447–16.
142. Mendes RE, Castanheira M, Woosley LN, Stone GG, Bradford PA, Flamm RK. Characterization of β-lactamase content of ceftazidime-resistant pathogens recovered during the pathogen-directed phase 3 REPRISE trial for ceftazidime-avibactam: correlation of efficacy against β-lactamase producers. Antimicrob Agents Chemother **2019**; 63:e02655–18.
143. Isler B, Ezure Y, Romero JLG, Harris P, Stewart AG, Paterson DL. Is ceftazidime/avibactam an option for serious infections due to extended-spectrum-beta-lactamase- and AmpC-producing Enterobacterales? A systematic review and meta-analysis. Antimicrob Agents Chemother **2020**; 65:e01052–20.
144. Torres A, Zhong N, Pachl J, et al. Ceftazidime-avibactam versus meropenem in nosocomial pneumonia, including ventilator-associated pneumonia (REPROVE): a randomised, double-blind, phase 3 non-inferiority trial. Lancet Infect Dis **2018**; 18:285–95.
145. Farrell DJ, Flamm RK, Sader HS, Jones RN. Antimicrobial activity of ceftolozane-tazobactam tested against Enterobacteriaceae and *Pseudomonas aeruginosa* with various resistance patterns isolated in U.S. Hospitals (2011–2012). Antimicrob Agents Chemother **2013**; 57:6305–10.
146. Sutherland CA, Nicolau DP. Susceptibility profile of ceftolozane/tazobactam and other parenteral antimicrobials against *Escherichia coli*, *Klebsiella pneumoniae*, and *Pseudomonas aeruginosa* from US hospitals. Clin Ther **2015**; 37:1564–71.
147. Shortridge D, Pfaller MA, Castanheira M, Flamm RK. Antimicrobial activity of ceftolozane-tazobactam tested against Enterobacteriaceae and *Pseudomonas aeruginosa* collected from patients with bloodstream infections isolated in United States hospitals (2013–2015) as part of the Program to Assess Ceftolozane-Tazobactam Susceptibility (PACTS) surveillance program. Diagn Microbiol Infect Dis **2018**; 92:158–63.
148. Shortridge D, Pfaller MA, Castanheira M, Flamm RK. Antimicrobial activity of ceftolozane-tazobactam tested against Enterobacteriaceae and *Pseudomonas aeruginosa* with various resistance patterns isolated in U.S. hospitals (2013–2016) as part of the surveillance program: program to assess ceftolozane-tazobactam susceptibility. Microb Drug Resist **2018**; 24:563–77.
149. Sader HS, Castanheira M, Streit JM, Flamm RK. Frequency of occurrence and antimicrobial susceptibility of bacteria isolated from patients hospitalized with bloodstream infections in United States medical centers (2015–2017). Diagn Microbiol Infect Dis **2019**; 95:114850.
150. Sader HS, Flamm RK, Carvalhaes CG, Castanheira M. Comparison of ceftazidime-avibactam and ceftolozane-tazobactam in vitro activities when tested against gram-negative bacteria isolated from patients hospitalized with pneumonia in United States medical centers (2017–2018). Diagn Microbiol Infect Dis **2020**; 96:114833.
151. Nicasio AM, VanScoy BD, Mendes RE, et al. Pharmacokinetics-pharmacodynamics of tazobactam in combination with piperacillin in an in vitro infection model. Antimicrob Agents Chemother **2016**; 60:2075–80.

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152. VanScoy B, Mendes RE, Nicasio AM, et al. Pharmacokinetics-pharmacodynamics of tazobactam in combination with ceftolozane in an in vitro infection model. Antimicrob Agents Chemother **2013**; 57:2809–14.
153. Popejoy MW, Paterson DL, Cloutier D, et al. Efficacy of ceftolozane/tazobactam against urinary tract and intra-abdominal infections caused by ESBL-producing *Escherichia coli* and *Klebsiella pneumoniae*: a pooled analysis of Phase 3 clinical trials. J Antimicrob Chemother **2017**; 72:268–72.
154. Kollef MH, Nováček M, Kivistik U, et al. Ceftolozane-tazobactam versus meropenem for treatment of nosocomial pneumonia (ASPECT-NP): a randomised, controlled, double-blind, phase 3, non-inferiority trial. Lancet Infect Dis **2019**; 19:1299–311.
155. Martin-Loeches I, Timsit JF, Kollef MH, et al. Clinical and microbiological outcomes, by causative pathogen, in the ASPECT-NP randomized, controlled, Phase 3 trial comparing ceftolozane/tazobactam and meropenem for treatment of hospital-acquired/ventilator-associated bacterial pneumonia. J Antimicrob Chemother **2022**; 77:1166–77.
156. Johnson JW, Fisher JF, Mobashery S. Bacterial cell-wall recycling. Ann N Y Acad Sci **2013**; 1277:54–75.
157. Tamma PD, Doi Y, Bonomo RA, Johnson JK, Simner PJ; Antibacterial Resistance Leadership Group. A primer on AmpC β-lactamases: necessary knowledge for an increasingly multidrug-resistant world. Clin Infect Dis **2019**; 69:1446–55.
158. Jacoby GA. AmpC beta-lactamases. Clin Microbiol Rev **2009**; 22:161–82, Table of Contents.
159. Philippon A, Arlet G, Labia R, Iorga BI. Class C β-lactamases: molecular characteristics. Clin Microbiol Rev **2022**; 35:e0015021.
160. Jacobson KL, Cohen SH, Inciardi JF, et al. The relationship between antecedent antibiotic use and resistance to extended-spectrum cephalosporins in group I beta-lactamase-producing organisms. Clin Infect Dis **1995**; 21:1107–13.
161. Honoré N, Nicolas MH, Cole ST. Inducible cephalosporinase production in clinical isolates of *Enterobacter cloacae* is controlled by a regulatory gene that has been deleted from *Escherichia coli*. EMBO J **1986**; 5:3709–14.
162. Hareza DA, Bergman Y, Jacobs E, et al. Molecular epidemiology of β-lactamases in ceftriaxone-resistant Enterobacterales bloodstream infections in the mid-Atlantic United States. Antimicrob Agents Chemother **2025**; 69:e0125824.
163. Philippon A, Arlet G, Jacoby GA. Plasmid-determined AmpC-type beta-lactamases. Antimicrob Agents Chemother **2002**; 46:1–11.
164. Alvarez M, Tran JH, Chow N, Jacoby GA. Epidemiology of conjugative plasmid-mediated AmpC beta-lactamases in the United States. Antimicrob Agents Chemother **2004**; 48:533–7.
165. Lindberg F, Westman L, Normark S. Regulatory components in *Citrobacter freundii* ampC beta-lactamase induction. Proc Natl Acad Sci U S A **1985**; 82: 4620–4.
166. Underwood S, Avison MB. *Citrobacter koseri* and *Citrobacter amalonaticus* isolates carry highly divergent beta-lactamase genes despite having high levels of biochemical similarity and 16S rRNA sequence homology. J Antimicrob Chemother **2004**; 53:1076–80.
167. Matsen JM, Blazevic DJ, Ryan JA, Ewing WH. Characterization of indole-positive *Proteus mirabilis*. Appl Microbiol **1972**; 23:592–4.
168. Choi SH, Lee JE, Park SJ, et al. Emergence of antibiotic resistance during therapy for infections caused by Enterobacteriaceae producing AmpC beta-lactamase: implications for antibiotic use. Antimicrob Agents Chemother **2008**; 52: 995–1000.
169. Chow JW, Fine MJ, Shlaes DM, et al. *Enterobacter bacteremia*: clinical features and emergence of antibiotic resistance during therapy. Ann Intern Med **1991**; 115:585–90.
170. Kaye KS, Cosgrove S, Harris A, Eliopoulos GM, Carmeli Y. Risk factors for emergence of resistance to broad-spectrum cephalosporins among *Enterobacter* spp. Antimicrob Agents Chemother **2001**; 45:2628–30.
171. Kohlmann R, Bähr T, Gatermann SG. Species-specific mutation rates for ampC derepression in Enterobacterales with chromosomally encoded inducible AmpC β-lactamase. J Antimicrob Chemother **2018**; 73:1530–6.
172. Girlich D, Naas T, Bellais S, Poirel L, Karim A, Nordmann P. Heterogeneity of AmpC cephalosporinases of *Hafnia alvei* clinical isolates expressing inducible or constitutive ceftazidime resistance phenotypes. Antimicrob Agents Chemother **2000**; 44:3220–3.
173. Lazarus JE, Wang Y, Waldor MK, Hooper DC. Divergent genetic landscapes drive lower levels of AmpC induction and stable de-repression in *Serratia marcescens* compared to *Enterobacter cloacae*. Antimicrob Agents Chemother **2024**; 68:e0119323.
174. Maillard A, Dortet L, Delory T, Lafaurie M, Bleibtreu A; Treatment of AmpC producing Enterobacterales study group. Mutation rate of AmpC β-lactamase-producing Enterobacterales and treatment in clinical practice: a word of caution. Clin Infect Dis **2024**; 79:52–5.
175. Mulbah JL, Kenney RM, Tibbetts RJ, Shallal AB, Veve MP. Ceftriaxone versus cefepime or carbapenems for definitive treatment of low-risk AmpC-harboring Enterobacterales bloodstream infections in hospitalized adults: a retrospective cohort study. Diagn Microbiol Infect Dis **2025**; 111: 116557.
176. Liu C, Wang X, Chen Y, et al. Three *Yersinia enterocolitica* AmpD homologs participate in the multi-step regulation of chromosomal cephalosporinase, AmpC. Front Microbiol **2016**; 7:1282.
177. Seoane A, Francia MV, García Lobo JM. Nucleotide sequence of the ampC-ampR region from the chromosome of *Yersinia enterocolitica*. Antimicrob Agents Chemother **1992**; 36:1049–52.
178. Liu Z, Shen S, Zhang X, et al. Identification of CMY-190, a novel chromosomally encoded AmpC β-lactamase, and plasmid-encoded KPC-2 in a clinical isolate of *Citrobacter youngae*. Front Microbiol **2025**; 16:1526882.
179. Sanders CC, Bradford PA, Ehrhardt AF, et al. Penicillin-binding proteins and induction of AmpC beta-lactamase. Antimicrob Agents Chemother **1997**; 41: 2013–5.
180. Beadle BM, Shoichet BK. Structural basis for imipenem inhibition of class C beta-lactamases. Antimicrob Agents Chemother **2002**; 46:3978–80.
181. Livermore DM, Oakton KJ, Carter MW, Warner M. Activity of ertapenem (MK-0826) versus Enterobacteriaceae with potent beta-lactamases. Antimicrob Agents Chemother **2001**; 45:2831–7.
182. Kurpiel PM, Hanson ND. Point mutations in the inc antisense RNA gene are associated with increased plasmid copy number, expression of blaCMY-2 and resistance to piperacillin/tazobactam in *Escherichia coli*. J Antimicrob Chemother **2012**; 67:339–45.
183. Weber DA, Sanders CC. Diverse potential of beta-lactamase inhibitors to induce class I enzymes. Antimicrob Agents Chemother **1990**; 34:156–8.
184. Akata K, Muratani T, Yatera K, et al. Induction of plasmid-mediated AmpC β-lactamase DHA-1 by piperacillin/tazobactam and other β-lactams in Enterobacteriaceae. PLoS One **2019**; 14:e0218589.
185. Sakurai Y, Yoshida Y, Saitoh K, Nemoto M, Yamaguchi A, Sawai T. Characteristics of aztreonam as a substrate, inhibitor and inducer for beta-lactamases. J Antibiot (Tokyo) **1990**; 43:403–10.
186. Endimiani A, Doi Y, Bethel CR, et al. Enhancing resistance to cephalosporins in class C beta-lactamases: impact of Gly214Glu in CMY-2. Biochemistry **2010**; 49: 1014–23.
187. Dahyot S, Mammeri H. Hydrolysis spectrum extension of CMY-2-like β-lactamases resulting from structural alteration in the Y-X-N loop. Antimicrob Agents Chemother **2012**; 56:1151–6.
188. Custodio MM, Sanchez D, Anderson B, Ryan KL, Walraven C, Mercier RC. Emergence of resistance in *Klebsiella aerogenes* to piperacillin-tazobactam and ceftriaxone. Antimicrob Agents Chemother **2021**; 65:e01038–20.
189. Hancock RE, Bellido F. Antibacterial in vitro activity of fourth generation cephalosporins. J Chemother **1996**; 8 Suppl 2:31–6.
190. Negri MC, Baquero F. In vitro selective concentrations of cefepime and ceftazidime for AmpC beta-lactamase hyperproducer *Enterobacter cloacae* variants. Clin Microbiol Infect **1999**; 5 Suppl 1:S25–8.
191. Fung-Tomc JC, Gradelski E, Huczko E, Dougherty TJ, Kessler RE, Bonner DP. Differences in the resistant variants of *Enterobacter cloacae* selected by extended-spectrum cephalosporins. Antimicrob Agents Chemother **1996**; 40:1289–93.
192. Harris PN, Wei JY, Shen AW, et al. Carbapenems versus alternative antibiotics for the treatment of bloodstream infections caused by *Enterobacter*, *Citrobacter* or *Serratia* species: a systematic review with meta-analysis. J Antimicrob Chemother **2016**; 71:296–306.
193. Cheo HM, Ho DYK, Koh LP, Koh MCY, Ngiam JN, Lye DCB. Cefepime versus carbapenem therapy for the treatment of invasive infections with inducible chromosomal AmpC-producing Enterobacterales: a systematic review and meta-analysis. Open Forum Infect Dis **2025**; 12:ofaf413.
194. Hancock RE, Bellido F. Factors involved in the enhanced efficacy against gram-negative bacteria of fourth generation cephalosporins. J Antimicrob Chemother **1992**; 29 Suppl A:1–6.
195. Szabó D, Bonomo RA, Silveira F, et al. SHV-type extended-spectrum beta-lactamase production is associated with reduced cefepime susceptibility in *Enterobacter cloacae*. J Clin Microbiol **2005**; 43:5058–64.
196. Hareza D, Simner PJ, Bergman Y, Jacobs E, Cosgrove SE, Tamma PD. The frequency of extended-spectrum β-lactamase genes harbored by Enterobacterales isolates at high risk for clinically significant chromosomal *ampC* expression. Open Forum Infect Dis **2023**; 10:ofad175.
197. Tamma PD, Girdwood SC, Gopaul R, et al. The use of cefepime for treating AmpC β-lactamase-producing Enterobacteriaceae. Clin Infect Dis **2013**; 57: 781–8.

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198. Peters DM, Winter JB, Droege CA, Ernst NE, Liao S. Comparison of ceftriaxone and antipseudomonal β-lactam antibiotics utilized for potential AmpC β-lactamase-producing organisms. Hosp Pharm **2021**; 56:560–8.
199. Drozdinsky G, Neuberger A, Rakedzon S, et al. Treatment of bacteremia caused by *Enterobacter* spp.: should the potential for AmpC induction dictate therapy? A retrospective study. Microb Drug Resist **2021**; 27:410–4.
200. Derrick C, Bookstaver PB, Lu ZK, et al. Multicenter, observational cohort study evaluating third-generation cephalosporin therapy for bloodstream infections secondary to *Enterobacter*, *Serratia*, and *Citrobacter* Species. Antibiotics (Basel) **2020**; 9:254.
201. Maillard A, Delory T, Bernier J, et al. Effectiveness of third-generation cephalosporins or piperacillin compared with cefepime or carbapenems for severe infections caused by wild-type AmpC β-lactamase-producing Enterobacterales: a multi-centre retrospective propensity-weighted study. Int J Antimicrob Agents **2023**; 62:106809.
202. Hilty M, Sendi P, Seiffert SN, et al. Characterisation and clinical features of *Enterobacter cloacae* bloodstream infections occurring at a tertiary care university hospital in Switzerland: is cefepime adequate therapy? Int J Antimicrob Agents **2013**; 41:236–49.
203. Robin F, Auzou M, Bonnet R, et al. In vitro activity of ceftolozane-tazobactam against *Enterobacter cloacae* complex clinical isolates with different beta-lactam resistance phenotypes. Antimicrob Agents Chemother **2018**; 62:e00675–18.
204. Cheng MP, Lee RS, Cheng AP, et al. Beta-lactam/beta-lactamase inhibitor therapy for potential AmpC-producing organisms: a systematic review and meta-analysis. Open Forum Infect Dis **2019**; 6:ofz248.
205. Lu B, Wong M, Ha D, et al. Piperacillin/tazobactam versus cefepime or carbapenems for cefoxitin-non-susceptible *Enterobacter cloacae*, *Klebsiella aerogenes*, *Citrobacter freundii*, *Serratia marcescens* and *Morganella morganii* bacteraemia in immunocompromised patients. J Antimicrob Chemother **2023**; 78:1009–14.
206. da Cunha Ferreira T, Martins IS. Risk factors of death in bloodstream infections caused by AmpC β-lactamase-producing *Enterobacterales* in patients with neoplasia. Infect Drug Resist **2021**; 14:3083–97.
207. Stewart AG, Paterson DL, Young B, et al. Meropenem versus piperacillin-tazobactam for definitive treatment of bloodstream infections caused by AmpC β-lactamase-producing *Enterobacter* spp, *Citrobacter freundii*, *Morganella morganii*, *Providencia* spp, or *Serratia marcescens*: a pilot multicenter randomized controlled trial (MERINO-2). Open Forum Infect Dis **2021**; 8: ofab387.
208. Zhanel GG, Lawrence CK, Adam H, et al. Imipenem-relebactam and meropenem-vaborbactam: two novel carbapenem-β-lactamase inhibitor combinations. Drugs **2018**; 78:65–98.
209. Sader HS, Mendes RE, Doyle TB, Davis AP, Castanheira M. Characterization of *Enterobacter cloacae* and *Citrobacter freundii* species complex isolates with decreased susceptibility to cephalosporins from United States hospitals and activity of ceftazidime/avibactam and comparator agents. JAC Antimicrob Resist **2021**; 3:dlab136.
210. Torres A, Wible M, Tawadrous M, et al. Efficacy and safety of ceftazidime/avibactam in patients with infections caused by beta-lactamase-producing Gram-negative pathogens: a pooled analysis from the Phase 3 clinical trial programme. J Antimicrob Chemother **2023**; 78:2672–82.
211. Iregui A, Khan Z, Landman D, Quale J. Activity of cefiderocol against *Enterobacterales*, *Pseudomonas aeruginosa*, and *Acinetobacter baumannii* endemic to medical centers in New York City. Microb Drug Resist **2020**; 26:722–6.
212. Kawai A, McElheny CL, Iovleva A, et al. Structural basis of reduced susceptibility to ceftazidime-avibactam and cefiderocol in *Enterobacter cloacae* due to AmpC R2 loop deletion. Antimicrob Agents Chemother **2020**; 64:e00198–20.
213. Shields RK, Iovleva A, Kline EG, Kawai A, McElheny CL, Doi Y. Clinical evolution of AmpC-mediated ceftazidime-avibactam and cefiderocol resistance in *Enterobacter cloacae* complex following exposure to cefepime. Clin Infect Dis **2020**; 71:2713–6.
214. Shropshire WC, Endres BT, Borjan J, et al. High-level ceftazidime/avibactam resistance in *Escherichia coli* conferred by the novel plasmid-mediated beta-lactamase CMY-185 variant. J Antimicrob Chemother **2023**; 78:2442–50.
215. Rodríguez-Pallares S, Blanco-Martín T, Lence E, et al. In vivo emergence of resistance to ceftazidime/avibactam through modification of chromosomal AmpC beta-lactamase in *Klebsiella aerogenes*. Antimicrob Agents Chemother **2024**; 68: e0130724.
216. Barnes MD, Taracila MA, Rutter JD, et al. Deciphering the evolution of cephalosporin resistance to ceftolozane-tazobactam in *Pseudomonas aeruginosa*. mBio **2018**; 9:e02085–18.
217. Tato M, García-Castillo M, Bofarull AM, Cantón R; CENIT Study Group. In vitro activity of ceftolozane/tazobactam against clinical isolates of *Pseudomonas aeruginosa* and Enterobacteriaceae recovered in Spanish medical centres: results of the CENIT study. Int J Antimicrob Agents **2015**; 46:502–10.
218. Zhanel GG, Pozdirca M, Golden AR, et al. Sulopenem: an intravenous and oral penem for the treatment of urinary tract infections due to multidrug-resistant bacteria. Drugs **2022**; 82:533–57.
219. Aronin SI, Huband MD, Becker H, Fedler KA, Dunne MW. In vitro activity of sulopenem and comparator agents against US Enterobacterales clinical isolates collected during the SENTRY antimicrobial surveillance program in 2023. J Glob Antimicrob Resist **2025**; 44:152–9.
220. Karlowsky JA, Adam HJ, Baxter MR, et al. In vitro activity of sulopenem, an oral penem, against urinary isolates of *Escherichia coli*. Antimicrob Agents Chemother **2019**; 63:e01832–18.
221. Walkty AJ, Karlowsky JA, Baxter MR, Lagace-Wiens PRS, Adam HJ, Zhanel GG. In vitro activity of sulopenem against 1880 bacterial pathogens isolated from Canadian patients with urinary tract infections (CANWARD, 2014–21). J Antimicrob Chemother **2022**; 77:3414–20.
222. Correction to: in vitro activity of sulopenem and comparator agents against Enterobacterales and anaerobic clinical isolates collected during the SENTRY Antimicrobial Surveillance Program. J Antimicrob Chemother **2023**; 78:1559.
223. Hawser S, Morrissey I, Kothari N, Monti F, Henriksen AS. Activity of mecillinam against USA urinary tract clinical isolates from 2017 to 2020 including isolates resistant to comparator antibiotics. J Glob Antimicrob Resist **2024**; 39:137–43.
224. Sabour S, Huang JY, Bhatnagar A, et al. Detection and characterization of targeted carbapenem-resistant health care-associated threats: findings from the Antibiotic Resistance Laboratory Network, 2017 to 2019. Antimicrob Agents Chemother **2021**; 65:e0110521.
225. Zheng X, Zeng W, Liu Y, et al. Clinical characteristics, specific resistance patterns, and molecular mechanisms of carbapenem-resistant *Morganella morganii* isolates. Front Cell Infect Microbiol **2025**; 15:1672736.
226. Neuwirth C, Siébor E, Duez JM, Péchinot A, Kazmierczak A. Imipenem resistance in clinical isolates of *Proteus mirabilis* associated with alterations in penicillin-binding proteins. J Antimicrob Chemother **1995**; 36:335–42.
227. Shropshire WC, Aitken SL, Pifer R, et al. IS26-mediated amplification of blaOXA-1 and blaCTX-M-15 with concurrent outer membrane porin disruption associated with de novo carbapenem resistance in a recurrent bacteraemia cohort. J Antimicrob Chemother **2021**; 76:385–95.
228. Hamzaoui Z, Ocampo-Sosa A, Fernandez Martinez M, et al. Role of association of OmpK35 and OmpK36 alteration and bla(ESBL) and/or bla(AmpC) genes in conferring carbapenem resistance among non-carbapenemase-producing *Klebsiella pneumoniae*. Int J Antimicrob Agents **2018**; 52:898–905.
229. van Duin D, Arias CA, Komarow L, et al. Molecular and clinical epidemiology of carbapenem-resistant Enterobacterales in the USA (CRACKLE-2): a prospective cohort study. Lancet Infect Dis **2020**; 20:731–41.
230. Sader HS, Mendes RE, Carvalhaes CG, Kimbrough JH, Castanheira M. Changing epidemiology of carbapenemases among carbapenem-resistant Enterobacterales from United States hospitals and the activity of aztreonam-avibactam against contemporary Enterobacterales (2019–2021). Open Forum Infect Dis **2023**; 10:ofad046.
231. Rankin DA, Stahl A, Sabour S, et al. Changes in carbapenemase-producing carbapenem-resistant Enterobacterales, 2019 to 2023. Ann Intern Med **2025**; 178:1818–21.
232. Karlsson M, Lutgring JD, Ansari U, et al. Molecular characterization of carbapenem-resistant Enterobacterales collected in the United States. Microb Drug Resist **2022**; 28:389–97.
233. Devinney K, Burton N, Alroy KA, et al. Notes from the field: increase in New Delhi metallo-β-lactamase-producing carbapenem-resistant Enterobacterales —New York City, 2019–2024. MMWR Morb Mortal Wkly Rep **2025**; 74:401–3.
234. Warecki BA, Tomatis PE, Mojica MF, et al. Cefiderocol "under siege"? Understanding the rise of NDM-mediated resistance to novel agents. Chem Sci **2025**; 16:12519–33.
235. Satlin MJ, Chen L, Gomez-Simmonds A, et al. Impact of a rapid molecular test for *Klebsiella pneumoniae* carbapenemase and ceftazidime-avibactam use on outcomes after bacteremia caused by carbapenem-resistant Enterobacterales. Clin Infect Dis **2022**; 75:2066–75.
236. Ren H, Teng Y, Xu Z, et al. Enhanced outcomes in bacteremia through rapid phenotype identification of carbapenemase-producing organisms. Surg Infect (Larchmt) **2025**; 27:279–88.
237. Blosser SJ, Bhatnagar AS, Bollinger S, Giffen SR. Carbapenemase mechanism testing to support treatment, infection control, and public health: the role of the clinical laboratory. Clin Lab Med **2025**; 45:51–61.
238. Simner PJ, Pitout JDD, Dingle TC. Laboratory detection of carbapenemases among Gram-negative organisms. Clin Microbiol Rev **2024**; 37:e0005422.
239. Pérez-Galera S, Bravo-Ferrer JM, Paniagua M, et al. Risk factors for infections caused by carbapenem-resistant Enterobacterales: an international matched case-control-control study (EURECA). EClinicalMedicine **2023**; 57:101871.

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240. Sader HS, Mendes RE, Kimbrough JH, Kantro V, Castanheira M. Impact of the recent Clinical and Laboratory Standards Institute breakpoint changes on the antimicrobial spectrum of aminoglycosides and the activity of plazomicin against multidrug-resistant and carbapenem-resistant Enterobacterales from United States medical centers. Open Forum Infect Dis **2023**; 10:ofad058.
241. Bland CM, Pai MP, Lodise TP. Reappraisal of contemporary pharmacokinetic and pharmacodynamic principles for informing aminoglycoside dosing. Pharmacotherapy **2018**; 38:1229–38.
242. Tamma PD, Bergman Y, Jacobs EB, et al. Comparing the activity of novel antibiotic agents against carbapenem-resistant Enterobacterales clinical isolates. Infect Control Hosp Epidemiol **2023**; 44:762–7.
243. Falagas ME, Kontogiannis DS, Sargianou M, Kakoullis SA. In vitro antimicrobial activity and clinical use of trimethoprim-sulfamethoxazole for infections due to KPC-producing Gram-negative pathogens: a review. Eur J Clin Microbiol Infect Dis **2025**; 44:2559–68.
244. Livermore DM, Warner M, Mushtaq S, Doumith M, Zhang J, Woodford N. What remains against carbapenem-resistant Enterobacteriaceae? Evaluation of chloramphenicol, ciprofloxacin, colistin, fosfomycin, minocycline, nitrofurantoin, temocillin and tigecycline. Int J Antimicrob Agents **2011**; 37:415–9.
245. Amladi AU, Abirami B, Devi SM, et al. Susceptibility profile, resistance mechanisms & efficacy ratios of fosfomycin, nitrofurantoin & colistin for carbapenem-resistant Enterobacteriaceae causing urinary tract infections. Indian J Med Res **2019**; 149:185–91.
246. Sandri AM, Landersdorfer CB, Jacob J, et al. Population pharmacokinetics of intravenous polymyxin B in critically ill patients: implications for selection of dosage regimens. Clin Infect Dis **2013**; 57:524–31.
247. López Montesinos I, Gómez-Zorrilla S, Palacios-Baena ZR, et al. Aminoglycoside or polymyxin monotherapy for treating complicated urinary tract infections caused by extensively drug-resistant *Pseudomonas aeruginosa*: a propensity score-adjusted and matched cohort study. Infect Dis Ther **2022**; 11:335–50.
248. Satlin MJ, Kubin CJ, Blumenthal JS, et al. Comparative effectiveness of aminoglycosides, polymyxin B, and tigecycline for clearance of carbapenem-resistant *Klebsiella pneumoniae* from urine. Antimicrob Agents Chemother **2011**; 55: 5893–9.
249. Nation RL, Rigatto MHP, Falci DR, Zavascki AP. Polymyxin acute kidney injury: dosing and other strategies to reduce toxicity. Antibiotics (Basel) **2019**; 8:24.
250. Sadyrbaeva-Dolgova S, García-Fumero R, Exposito-Ruiz M, Pasquau-Liaño J, Jiménez-Morales A, Hidalgo-Tenorio C. Incidence of nephrotoxicity associated with intravenous colistimethate sodium administration for the treatment of multidrug-resistant gram-negative bacterial infections. Sci Rep **2022**; 12:15261.
251. Shields RK, Anand R, Clarke LG, et al. Defining the incidence and risk factors of colistin-induced acute kidney injury by KDIGO criteria. PLoS One **2017**; 12: e0173286.
252. Miano TA, Lautenbach E, Wilson FP, Guo W, Borovskiy Y, Hennessy S. Attributable risk and time course of colistin-associated acute kidney injury. Clin J Am Soc Nephrol **2018**; 13:542–50.
253. Kaase M, Szabados F, Anders A, Gatermann SG. Fosfomycin susceptibility in carbapenem-resistant Enterobacteriaceae from Germany. J Clin Microbiol **2014**; 52:1893–7.
254. Falagas ME, Romanos LT, Katritsi ED, Tzvetanova ID. Resistance of uropathogens to gepotidacin: a systematic review of in vitro antimicrobial susceptibility studies. Int J Antimicrob Agents **2026**; 67:107723.
255. Emeraud C, Godmer A, Girlich D, et al. Activity of mecillinam against carbapenem-resistant Enterobacterales. J Antimicrob Chemother **2022**; 77: 2835–9.
256. Fuchs F, Ahmadzada A, Plambeck L, Wille T, Hamprecht A. Susceptibility of clinical Enterobacterales isolates with common and rare carbapenemases to mecillinam. Front Microbiol **2020**; 11:627267.
257. Zykov IN, Frimodt-Møller N, Småbrekke L, Sundsfjord A, Samuelsen O. Efficacy of mecillinam against clinical multidrug-resistant *Escherichia coli* in a murine urinary tract infection model. Int J Antimicrob Agents **2020**; 55:105851.
258. Wagenlehner FM, Sobel JD, Newell P, et al. Ceftazidime-avibactam versus doripenem for the treatment of complicated urinary tract infections, including acute pyelonephritis: RECAPTURE, a phase 3 randomized trial program. Clin Infect Dis **2016**; 63:754–62.
259. Kaye KS, Bhowmick T, Metallidis S, et al. Effect of meropenem-vaborbactam vs piperacillin-tazobactam on clinical cure or improvement and microbial eradication in complicated urinary tract infection: the TANGO I randomized clinical trial. JAMA **2018**; 319:788–99.
260. Portsmouth S, van Veenhuyzen D, Echols R, et al. Cefiderocol versus imipenem-cilastatin for the treatment of complicated urinary tract infections caused by Gram-negative uropathogens: a phase 2, randomised, double-blind, non-inferiority trial. Lancet Infect Dis **2018**; 18:1319–28.
261. Bassetti M, Echols R, Matsunaga Y, et al. Efficacy and safety of cefiderocol or best available therapy for the treatment of serious infections caused by carbapenem-resistant Gram-negative bacteria (CREDIBLE-CR): a randomised, open-label, multicentre, pathogen-focused, descriptive, phase 3 trial. Lancet Infect Dis **2021**; 21:226–40.
262. Sader HS, Mendes RE, Arends SJR, Carvalhaes CG, Castanheira M. Antimicrobial activities of aztreonam-avibactam and comparator agents tested against Enterobacterales from European hospitals analysed by geographic region and infection type (2019–2020). Eur J Clin Microbiol Infect Dis **2022**; 41:477–87.
263. Sader HS, Kimbrough JH, Winkler ML, Castanheira M, Mendes RE. Aztreonam/avibactam activity against Enterobacterales from European medical centres: summary of 5 years of surveillance prior to approval for clinical use (2019–2023). J Antimicrob Chemother **2025**; 80:2070–9.
264. Sims M, Mariyanovski V, McLeroth P, et al. Prospective, randomized, double-blind, Phase 2 dose-ranging study comparing efficacy and safety of imipenem/cilastatin plus relebactam with imipenem/cilastatin alone in patients with complicated urinary tract infections. J Antimicrob Chemother **2017**; 72:2616–26.
265. Vardakas KZ, Legakis NJ, Triarides N, Falagas ME. Susceptibility of contemporary isolates to fosfomycin: a systematic review of the literature. Int J Antimicrob Agents **2016**; 47:269–85.
266. Falagas ME, Karageorgopoulos DE, Dimopoulos G. Clinical significance of the pharmacokinetic and pharmacodynamic characteristics of tigecycline. Curr Drug Metab **2009**; 10:13–21.
267. Zhou C, Jin L, Wang Q, et al. Bloodstream infections caused by carbapenem-resistant Enterobacterales: risk factors for mortality, antimicrobial therapy and treatment outcomes from a prospective multicenter study. Infect Drug Resist **2021**; 14:731–42.
268. Adelman MW, Bower CW, Grass JE, et al. Distinctive features of ertapenem-mono-resistant carbapenem-resistant Enterobacterales in the United States: a cohort study. Open Forum Infect Dis **2022**; 9:ofab643.
269. Shropshire WC, Amiji H, Bremer J, et al. Genetic determinants underlying the progressive phenotype of beta-lactam/beta-lactamase inhibitor resistance in *Escherichia coli*. Microbiol Spectr **2023**; 11:e0222123.
270. Shortridge D, Deshpande LM, Streit JM, Castanheira M. Activity of meropenem/vaborbactam and comparators against non-carbapenemase-producing carbapenem-resistant Enterobacterales isolates from Europe. JAC Antimicrob Resist **2022**; 4:dlac097.
271. Bonnin RA, Bernabeu S, Emeraud C, et al. In vitro activity of imipenem-relebactam, meropenem-vaborbactam, ceftazidime-avibactam and comparators on carbapenem-resistant non-carbapenemase-producing Enterobacterales. Antibiotics (Basel) **2023**; 12:102.
272. Le Terrier C, Mlynarcik P, Sadek M, Nordmann P, Poirel L. Relative inhibitory activities of newly developed diazabicyclooctanes, boronic acid derivatives, and penicillin-based sulfone β-lactamase inhibitors against broad-spectrum AmpC β-lactamases. Antimicrob Agents Chemother **2024**; 68:e0077524.
273. Chen J, Hu Q, Zhou P, Deng S. Ceftazidime-avibactam versus polymyxins in treating patients with carbapenem-resistant Enterobacteriaceae infections: a systematic review and meta-analysis. Infection **2024**; 52:19–28.
274. Shields RK, Nguyen MH, Chen L, et al. Ceftazidime-avibactam is superior to other treatment regimens against carbapenem-resistant *Klebsiella pneumoniae* bacteremia. Antimicrob Agents Chemother **2017**; 61:e00883–17.
275. Wunderink RG, Giamarellos-Bourboulis EJ, Rahav G, et al. Effect and safety of meropenem-vaborbactam versus best-available therapy in patients with carbapenem-resistant Enterobacteriaceae infections: the TANGO II randomized clinical trial. Infect Dis Ther **2018**; 7:439–55.
276. Motsch J, Murta de Oliveira C, Stus V, et al. RESTORE-IMI 1: a multicenter, randomized, double-blind trial comparing efficacy and safety of imipenem/relebactam vs colistin plus imipenem in patients with imipenem-nonsusceptible bacterial infections. Clin Infect Dis **2020**; 70:1799–808.
277. van Duin D, Lok JJ, Earley M, et al. Colistin versus ceftazidime-avibactam in the treatment of infections due to carbapenem-resistant Enterobacteriaceae. Clin Infect Dis **2018**; 66:163–71.
278. Sader HS, Mendes RE, Duncan L, Kimbrough JH, Carvalhaes CG, Castanheira M. Ceftazidime-avibactam, meropenem-vaborbactam, and imipenem-relebactam activities against multidrug-resistant Enterobacterales from United States Medical Centers (2018–2022). Diagn Microbiol Infect Dis **2023**; 106: 115945.
279. Marino A, Maraolo AE, Mazzitelli M, et al. Head-to-head: meropenem/vaborbactam versus ceftazidime/avibactam in ICUs patients with KPC-producing *K. pneumoniae* infections—results from a retrospective multicentre study. Infection **2025**; 53:2645–58.
280. Karaba SM, Hareza DA, Munita JM, et al. Comparative outcomes of ceftazidime-avibactam versus meropenem-vaborbactam for KPC-producing

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Enterobacterales infections. Antimicrobial Agents and Chemotherapy **2025**; 70: e0160225.
281. Ackley R, Roshdy D, Meredith J, et al. Meropenem-vaborbactam versus ceftazidime-avibactam for treatment of carbapenem-resistant Enterobacteriaceae infections. Antimicrob Agents Chemother **2020**; 64: e02313–19.
282. Leanza C, Mascellino MT, Volpicelli L, et al. Real-world use of imipenem/cilastatin/relebactam for the treatment of KPC-producing *Klebsiella pneumoniae* complex and difficult-to-treat resistance (DTR) *Pseudomonas aeruginosa* infections: a single-center preliminary experience. Front Microbiol **2024**; 15:1432296.
283. Papp-Wallace KM, Barnes MD, Alsop J, et al. Relebactam is a potent inhibitor of the KPC-2 β-lactamase and restores imipenem susceptibility in KPC-producing Enterobacteriaceae. Antimicrob Agents Chemother **2018**; 62:e00174–18.
284. Papp-Wallace KM, Mack AR, Taracila MA, Bonomo RA. Resistance to novel β-lactam-β-lactamase inhibitor combinations: the "price of progress". Infect Dis Clin North Am **2020**; 34:773–819.
285. Shields RK, Potoski BA, Haidar G, et al. Clinical outcomes, drug toxicity, and emergence of ceftazidime-avibactam resistance among patients treated for carbapenem-resistant Enterobacteriaceae infections. Clin Infect Dis **2016**; 63: 1615–8.
286. Castón JJ, Gallo M, García M, et al. Ceftazidime-avibactam in the treatment of infections caused by KPC-producing *Klebsiella pneumoniae*: factors associated with clinical efficacy in a single-center cohort. Int J Antimicrob Agents **2020**; 56:106075.
287. Tumbarello M, Raffaelli F, Giannella M, et al. Outcomes and predictors of mortality in patients with KPC-Kp infections treated with meropenem-vaborbactam: an observational multicenter study. Open Forum Infect Dis **2024**; 11:ofae273.
288. Di Bella S, Giacobbe DR, Maraolo AE, et al. Resistance to ceftazidime/avibactam in infections and colonisations by KPC-producing Enterobacterales: a systematic review of observational clinical studies. J Glob Antimicrob Resist **2021**; 25: 268–81.
289. Hobson CA, Pierrat G, Tenaillon O, et al. *Klebsiella pneumoniae* carbapenemase variants resistant to ceftazidime-avibactam: an evolutionary overview. Antimicrob Agents Chemother **2022**; 66:e0044722.
290. Hamza M, Traglia GM, Maccari L, et al. Emerging resistance to novel beta-lactam beta-lactamase inhibitor combinations in Klebsiella pneumoniae bearing KPC variants. J Glob Antimicrob Resis **2025**; 44:297–305.
291. Winkler ML, Papp-Wallace KM, Bonomo RA. Activity of ceftazidime/avibactam against isogenic strains of *Escherichia coli* containing KPC and SHV β-lactamases with single amino acid substitutions in the Ω-loop. J Antimicrob Chemother **2015**; 70:2279–86.
292. Livermore DM, Warner M, Jamrozy D, et al. In vitro selection of ceftazidime-avibactam resistance in Enterobacteriaceae with KPC-3 carbapenemase. Antimicrob Agents Chemother **2015**; 59:5324–30.
293. Shields RK, Chen L, Cheng S, et al. Emergence of ceftazidime-avibactam resistance due to plasmid-borne bla(KPC-3) mutations during treatment of carbapenem-resistant *Klebsiella pneumoniae* infections. Antimicrob Agents Chemother **2017**; 61:e02097-16.
294. Sun D, Rubio-Aparicio D, Nelson K, Dudley MN, Lomovskaya O. Meropenem-vaborbactam resistance selection, resistance prevention, and molecular mechanisms in mutants of KPC-producing *Klebsiella pneumoniae*. Antimicrob Agents Chemother **2017**; 61:e01694–17.
295. Findlay J, Rens C, Poirel L, Nordmann P. In vitro mechanisms of resistance development to imipenem-relebactam in KPC-producing *Klebsiella pneumoniae*. Antimicrob Agents Chemother **2022**; 66:e0091822.
296. Shields RK, McCreary EK, Marini RV, et al. Early experience with meropenem-vaborbactam for treatment of carbapenem-resistant Enterobacteriaceae infections. Clin Infect Dis **2020**; 71:667–71.
297. Wilson WR, Kline EG, Jones CE, et al. Effects of KPC variant and porin genotype on the in vitro activity of meropenem-vaborbactam against carbapenem-resistant Enterobacteriaceae. Antimicrob Agents Chemother **2019**; 63: e02048–18.
298. Balabanian G, Rose M, Manning N, Landman D, Quale J. Effect of porins and bla(KPC) expression on activity of imipenem with relebactam in *Klebsiella pneumoniae*: can antibiotic combinations overcome resistance? Microb Drug Resist **2018**; 24:877–81.
299. Gaibani P, Bovo F, Bussini L, et al. Dynamic evolution of imipenem/relebactam resistance in a KPC-producing *Klebsiella pneumoniae* from a single patient during ceftazidime/avibactam-based treatments. J Antimicrob Chemother **2022**; 77: 1570–7.
300. Yasmin M, Marshall SH, Chen L, et al. A molecular analysis of meropenem-vaborbactam non-susceptible KPC-producing *Klebsiella pneumoniae*. Antimicrob Agents Chemother **2024**; 68:e0020824.
301. Wise MG, Karlowsky JA, Mohamed N, Kamat S, Sahm DF. In vitro activity of aztreonam-avibactam against Enterobacterales isolates collected in Latin America, Africa/Middle East, Asia, and Eurasia for the ATLAS Global Surveillance Program in 2019–2021. Eur J Clin Microbiol Infect Dis **2023**; 42: 1135–43.
302. Wise MG, Karlowsky JA, Hackel MA, et al. In vitro activity of cefiderocol against meropenem-nonsusceptible gram-negative bacilli with defined β-lactamase carriage: SIDERO-WT Surveillance Studies, 2014–2019. Microb Drug Resist **2023**; 29:360–70.
303. Papp-Wallace KM, Sader HS, Maher JM, Kimbrough JH, Castanheira M. Aztreonam-avibactam demonstrates potent activity against carbapenem-resistant Enterobacterales collected from US medical centers over a 6-year period (2017–2022). Open Forum Infect Dis **2025**; 12:ofaf250.
304. Carmeli Y, Cisneros JM, Paul M, et al. Aztreonam-avibactam versus meropenem for the treatment of serious infections caused by Gram-negative bacteria (REVISIT): a descriptive, multinational, open-label, phase 3, randomised trial. Lancet Infect Dis **2025**; 25:218–30.
305. Marshall S, Hujer AM, Rojas LJ, et al. Can ceftazidime-avibactam and aztreonam overcome β-lactam resistance conferred by metallo-β-lactamases in Enterobacteriaceae? Antimicrob Agents Chemother **2017**; 61:e02243–16.
306. Deshpande LM, Doyle TB, Sader HS, Castanheira M. Activity of aztreonam-avibactam and ceftazidime-avibactam against beta-lactamase-producing Enterobacterales isolates from United States hospitals. J Glob Antimicrob Resist **2025**; 44:103–10.
307. Sader HS, Kimbrough JH, Mendes RE, Castanheira M. Antimicrobial susceptibility of Enterobacterales causing bloodstream infection in United States medical centres: comparison of aztreonam-avibactam with beta-lactams active against carbapenem-resistant Enterobacterales. BMC Infect Dis **2024**; 24:1242.
308. Castanheira M, Kimbrough JH, Morgan GM, Karr M, Sader HS. Activity of aztreonam-avibactam, cefiderocol, and cefepime-taniborbactam against a global collection of genetically characterized metallo-β-lactamase-producing Enterobacterales. Antimicrob Agents Chemother **2026**; 70:e0084225.
309. Crandon JL, Nicolau DP. Human simulated studies of aztreonam and aztreonam-avibactam to evaluate activity against challenging gram-negative organisms, including metallo-beta-lactamase producers. Antimicrob Agents Chemother **2013**; 57:3299–306.
310. Das S, Riccobene T, Carrothers TJ, et al. Dose selection for aztreonam-avibactam, including adjustments for renal impairment, for Phase IIa and Phase III evaluation. Eur J Clin Pharmacol **2024**; 80:529–43.
311. Xie R, Rogers H, Chow JW, Soto E, Raber SR. Population pharmacokinetic/pharmacodynamic modeling to optimize aztreonam-avibactam dose regimens for adult patients. Antimicrob Agents Chemother **2025**; 69:e0195024.
312. Daikos GL, Cisneros JM, Carmeli Y, et al. Aztreonam-avibactam for the treatment of serious infections caused by metallo-beta-lactamase-producing Gram-negative pathogens: a Phase 3 randomized trial (ASSEMBLE). JAC Antimicrob Resist **2025**; 7:dlaf131.
313. Tamma PD, Lodise TP, Mojica MF, Perez F, Bonomo RA. Revisiting REVISIT: the case for ceftazidime/avibactam plus aztreonam. Clin Infect Dis **2025**; 81: e241–2.
314. Falcone M, Daikos GL, Tiseo G, et al. Efficacy of ceftazidime-avibactam plus aztreonam in patients with bloodstream infections caused by metallo-β-lactamase-producing Enterobacterales. Clin Infect Dis **2021**; 72: 1871–8.
315. Falcone M, Giordano C, Leonildi A, et al. Clinical features and outcomes of infections caused by metallo-β-lactamase-producing Enterobacterales: a 3-year prospective study from an endemic area. Clin Infect Dis **2024**; 78:1111–9.
316. Harris H, Tao L, Jacobs EB, et al. Multicenter evaluation of an MIC-based aztreonam and ceftazidime-avibactam broth disk elution test. J Clin Microbiol **2023**; 61:e0164722.
317. Lodise TP, Smith NM, O'Donnell N, et al. Determining the optimal dosing of a novel combination regimen of ceftazidime/avibactam with aztreonam against NDM-1-producing Enterobacteriaceae using a hollow-fibre infection model. J Antimicrob Chemother **2020**; 75:2622–32.
318. Falcone M, Menichetti F, Cattaneo D, et al. Pragmatic options for dose optimization of ceftazidime/avibactam with aztreonam in complex patients. J Antimicrob Chemother **2021**; 76:1025–31.
319. Lodise TP, O'Donnell JN, Raja S, et al. Safety of ceftazidime-avibactam in combination with aztreonam (COMBINE) in a phase I, open-label study in healthy adult volunteers. Antimicrob Agents Chemother **2022**; 66:e0093522.
320. Biagi M, Wu T, Lee M, Patel S, Butler D, Wenzler E. Searching for the optimal treatment for metallo- and serine-β-lactamase producing Enterobacteriaceae: aztreonam in combination with ceftazidime-avibactam or meropenem-vaborbactam. Antimicrob Agents Chemother **2019**; 63:e01426–19.

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321. Biagi M, Lee M, Wu T, et al. Aztreonam in combination with imipenem-relebactam against clinical and isogenic strains of serine and metallo-beta-lactamase-producing Enterobacterales. Diagn Microbiol Infect Dis **2022**; 103:115674.
322. Belati A, Bavaro DF, Diella L, De Gennaro N, Di Gennaro F, Saracino A. Meropenem/vaborbactam plus aztreonam as a possible treatment strategy for bloodstream infections caused by ceftazidime/avibactam-resistant *Klebsiella pneumoniae*: a retrospective case series and literature review. Antibiotics (Basel) **2022**; 11:373.
323. Ito A, Sato T, Ota M, et al. In vitro antibacterial properties of cefiderocol, a novel siderophore cephalosporin, against gram-negative bacteria. Antimicrob Agents Chemother **2018**; 62:e01454–17.
324. Karakonstantis S, Rousaki M, Vassilopoulou L, Kritsotakis EI. Global prevalence of cefiderocol non-susceptibility in Enterobacterales, *Pseudomonas aeruginosa*, *Acinetobacter baumannii*, and *Stenotrophomonas maltophilia*: a systematic review and meta-analysis. Clin Microbiol Infect **2024**; 30:178–88.
325. Kimbrough JH, Maher JM, Sader HS, Castanheira M, Mendes RE. In vitro activity assessment of cefiderocol against Enterobacterales, *Pseudomonas aeruginosa*, and *Acinetobacter* spp., including β-lactam nonsusceptible molecularly characterized isolates, collected from 2020 to 2021 in the United States and European hospitals. Microbiol Spectr **2024**; 12:e0147424.
326. Timsit JF, Paul M, Shields RK, et al. Cefiderocol for the treatment of infections due to metallo-B-lactamase-producing pathogens in the CREDIBLE-CR and APEKS-NP phase 3 randomized studies. Clin Infect Dis **2022**; 75:1081–4.
327. Paterson DL, Sulaiman H, Liu PY, et al. Cefiderocol versus standard therapy for hospital-acquired and health-care-associated Gram-negative bacterial bloodstream infection (the GAME CHANGER trial): an open-label, parallel-group, randomised trial. Lancet Infect Dis **2026**; 26:148–59.
328. Sadek M, Juhas M, Poirel L, Nordmann P. Genetic features leading to reduced susceptibility to aztreonam-avibactam among metallo-β-lactamase-producing *Escherichia coli* isolates. Antimicrob Agents Chemother **2020**; 64:e01659–20.
329. Helsens N, Sadek M, Le Terrier C, Poirel L, Nordmann P. Reduced susceptibility to aztreonam-avibactam conferred by acquired AmpC-type β-lactamases in PBP3-modified *Escherichia coli* [manuscript published online ahead of print 6 February 2024]. Eur J Clin Microbiol Infect Dis **2024**. doi:10.1007/s10096-024-04769-z
330. Livermore DM, Mushtaq S, Vickers A, Woodford N. Activity of aztreonam/avibactam against metallo-β-lactamase-producing Enterobacterales from the UK: impact of penicillin-binding protein-3 inserts and CMY-42 β-lactamase in *Escherichia coli*. Int J Antimicrob Agents **2023**; 61:106776.
331. Long H, Zhao F, Feng Y, Zong Z. Global distribution of bla(CMY-42), a gene mediating reduced susceptibility to aztreonam-avibactam and ceftazidime-avibactam, in *Escherichia coli*. Int J Antimicrob Agents **2024**; 63: 107141.
332. Simner PJ, Bergman Y, Conzemius R, et al. An NDM-producing *Escherichia coli* clinical isolate exhibiting resistance to cefiderocol and the combination of ceftazidime-avibactam and aztreonam: another step toward pan-β-lactam resistance. Open Forum Infect Dis **2023**; 10:ofad276.
333. Eklund E, Maruca T, Kruczynski K, et al. High in vitro activity of aztreonam/avibactam against NDM-producing Enterobacterales with emerging resistance in *Escherichia coli* in the United States Mid-Atlantic region. Microbiol Spectr **2026**; 14:e0397125.
334. Mushtaq S, Vickers A, Ellaby N, Woodford N, Livermore DM. Selection and characterization of mutational resistance to aztreonam/avibactam in beta-lactamase-producing Enterobacterales. J Antimicrob Chemother **2021**; 77:98–111.
335. Niu S, Wei J, Zou C, et al. In vitro selection of aztreonam/avibactam resistance in dual-carbapenemase-producing *Klebsiella pneumoniae*. J Antimicrob Chemother **2020**; 75:559–65.
336. Karakonstantis S, Rousaki M, Kritsotakis EI. Cefiderocol: systematic review of mechanisms of resistance, heteroresistance and in vivo emergence of resistance. Antibiotics (Basel) **2022**; 11:723.
337. Lan P, Lu Y, Chen Z, et al. Emergence of high-level cefiderocol resistance in carbapenem-resistant *Klebsiella pneumoniae* from bloodstream infections in patients with hematologic malignancies in China. Microbiol Spectr **2022**; 10: e0008422.
338. McElheny CL, Fowler EL, Iovleva A, Shields RK, Doi Y. In vitro evolution of cefiderocol resistance in an NDM-producing *Klebsiella pneumoniae* due to functional loss of CirA. Microbiol Spectr **2021**; 9:e0177921.
339. Simner PJ, Beisken S, Bergman Y, Ante M, Posch AE, Tamma PD. Defining baseline mechanisms of cefiderocol resistance in the Enterobacterales. Microb Drug Resist **2022**; 28:161–70.
340. Frohlich C, Sorum V, Tokuriki N, Johnsen PJ, Samuelsen O. Evolution of beta-lactamase-mediated cefiderocol resistance. J Antimicrob Chemother **2022**; 77:2429–36.
341. Simner PJ, Mostafa HH, Bergman Y, et al. Progressive development of cefiderocol resistance in *Escherichia coli* during therapy is associated with an increase in blaNDM-5 copy number and gene expression. Clin Infect Dis **2022**; 75:47–54.
342. Castanheira M, Doyle TB, Collingsworth TD, Sader HS, Mendes RE. Increasing frequency of OXA-48-producing Enterobacterales worldwide and activity of ceftazidime/avibactam, meropenem/vaborbactam and comparators against these isolates. J Antimicrob Chemother **2021**; 76:3125–34.
343. Sader HS, Carvalhaes CG, Kimbrough JH, Mendes RE, Castanheira M. Activity of aztreonam-avibactam against Enterobacterales resistant to recently approved beta-lactamase inhibitor combinations collected in Europe, Latin America, and the Asia-Pacific Region (2020–2022). Int J Antimicrob Agents **2024**; 63:107113.
344. Alqahtani H, Alghamdi A, Alobaidallah N, et al. Evaluation of ceftazidime/avibactam for treatment of carbapenemase-producing carbapenem-resistant Enterobacterales with OXA-48 and/or NDM genes with or without combination therapy. JAC Antimicrob Resist **2022**; 4:dlac104.
345. Lima O, Sousa A, Longueira-Suárez R, et al. Ceftazidime-avibactam treatment in bacteremia caused by OXA-48 carbapenemase-producing *Klebsiella pneumoniae*. Eur J Clin Microbiol Infect Dis **2022**; 41:1173–82.
346. Canver MC, Satlin MJ, Westblade LF, et al. Activity of imipenem-relebactam and comparator agents against genetically characterized isolates of carbapenem-resistant enterobacteriaceae. Antimicrob Agents Chemother **2019**; 63: e00672–19.
347. Asempa TE, Kois AK, Gill CM, Nicolau DP. Phenotypes, genotypes and breakpoints: an assessment of β-lactam/β-lactamase inhibitor combinations against OXA-48. J Antimicrob Chemother **2023**; 78:636–45.
348. Outeda-García M, Arca-Suárez J, Lence E, et al. Advancements in the fight against globally distributed OXA-48 carbapenemase: evaluating the new generation of carbapenemase inhibitors. Antimicrob Agents Chemother **2025**; 69: e0161424.
349. Bonnin RA, Bernabeu S, Emeraud C, et al. Susceptibility of OXA-48-producing Enterobacterales to imipenem/relebactam, meropenem/vaborbactam and ceftazidime/avibactam. Int J Antimicrob Agents **2022**; 60:106660.
350. Blanco-Martín T, Guzmán-Puche J, Hernández-García M, et al. Mechanisms of resistance to ceftazidime/avibactam in mutants derived in vitro from *Klebsiella pneumoniae* producing OXA-48-like enzymes. J Antimicrob Chemother **2025**; 80:3265–72.
351. Fröhlich C, Sørum V, Thomassen AM, Johnsen PJ, Leiros HS, Samuelsen O. OXA-48-mediated ceftazidime-avibactam resistance is associated with evolutionary trade-offs. mSphere **2019**; 4:e00024–19.
352. Longshaw C, Roger E, Santerre Henriksen A, Baba T, Nguyen S, Yamano Y. Evidence for efficacy of cefiderocol against OXA-48-containing isolates from the APEKS-NP and CREDIBLE-CR trials. Antimicrob Agents Chemother **2022**; 66:e0110022.
353. García P, Guijarro-Sánchez P, Lasarte-Monterrubio C, et al. Activity and resistance mechanisms of the third generation tetracyclines tigecycline, eravacycline and omadacycline against nationwide Spanish collections of carbapenemase-producing Enterobacterales and *Acinetobacter baumannii*. Biomed Pharmacother **2024**; 181:117666.
354. Ni W, Han Y, Liu J, et al. Tigecycline treatment for carbapenem-resistant Enterobacteriaceae infections: a systematic review and meta-analysis. Medicine (Baltimore) **2016**; 95:e3126.
355. Zha L, Pan L, Guo J, French N, Villanueva EV, Tefsen B. Effectiveness and safety of high dose tigecycline for the treatment of severe infections: a systematic review and meta-analysis. Adv Ther **2020**; 37:1049–64.
356. De Pascale G, Montini L, Pennisi M, et al. High dose tigecycline in critically ill patients with severe infections due to multidrug-resistant bacteria. Crit Care **2014**; 18:R90.
357. Karampatakis T, Tsergouli K, Lowrie K. Efficacy and safety of ceftazidime-avibactam compared to other antimicrobials for the treatment of infections caused by carbapenem-resistant *Klebsiella pneumoniae* strains, a systematic review and meta-analysis. Microb Pathog **2023**; 179:106090.
358. United States Food and Drug Administration. Antibacterial susceptibility test interpretive criteria. Available at: https://www.fda.gov/drugs/development-resources/antibacterial-susceptibility-test-interpretive-criteria. Accessed 31 December 2025.
359. Amann LF, Broeker A, Riedner M, et al. Pharmacokinetic/pharmacodynamic evaluation of tigecycline dosing in a hollow fiber infection model against clinical bla-KPC producing *Klebsiella Pneumoniae* isolates. Diagn Microbiol Infect Dis **2024**; 108:116153.
360. Solomkin J, Evans D, Slepavicius A, et al. Assessing the efficacy and safety of eravacycline vs ertapenem in complicated intra-abdominal infections in the

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investigating gram-negative infections treated with eravacycline (IGNITE 1) trial: a randomized clinical trial. JAMA Surg **2017**; 152:224–32.
361. Solomkin JS, Gardovskis J, Lawrence K, et al. IGNITE4: results of a phase 3, randomized, multicenter, prospective trial of eravacycline vs meropenem in the treatment of complicated intraabdominal infections. Clin Infect Dis **2019**; 69: 921–9.
362. Patino K, Vathy R, Albrecht B, Suchindran S, Green SB. Combating resistant pathogens: exploring the efficacy of eravacycline utilization in multidrug-resistant infections. Microbiol Spectr **2025**; 13:e0131425.
363. Wang Z, Gan G, Yao H. Gastrointestinal risk profile of tigecycline, omadacycline and eravacycline: evidence from the FDA adverse event reporting system. Acta Pharm **2025**;76:1–20.
364. Chen A, Smith KP, Whitfield BA, et al. Activity of minocycline against *Klebsiella pneumoniae* carbapenemase (KPC)-producing Enterobacteriaceae clinical isolates, with comparison to doxycycline and tigecycline. Diagn Microbiol Infect Dis **2017**; 88:365–7.
365. Crowley PD, Ranganath N, Mira P, et al. Show me the "mino:" in vitro efficacy of minocycline on clinical gram-negative bacterial isolates. Microb Drug Resist **2026**;32:128–34.
366. Pogue JM, Neelakanta A, Mynatt RP, Sharma S, Lephart P, Kaye KS. Carbapenem resistance in gram-negative bacilli and intravenous minocycline: an antimicrobial stewardship approach at the Detroit Medical Center. Clin Infect Dis **2014**; 59 Suppl 6:S388–93.
367. Mirza HC, Güçlü AU, Ince Ceviz G, Başustaoğlu A. Comparative in vitro activities of omadacycline, eravacycline and tigecycline against non-ESBL-producing, ESBL-producing and carbapenem-resistant isolates of *K. pneumoniae*. J Med Microbiol **2022**; 71.
368. Lee YL, Ko WC, Lee WS, et al. In-vitro activity of cefiderocol, cefepime/zidebactam, cefepime/enmetazobactam, omadacycline, eravacycline and other comparative agents against carbapenem-nonsusceptible Enterobacterales: results from the Surveillance of Multicenter Antimicrobial Resistance in Taiwan (SMART) in 2017–2020. Int J Antimicrob Agents **2021**; 58:106377.
369. Tamma PD, Cosgrove SE, Maragakis LL. Combination therapy for treatment of infections with gram-negative bacteria. Clin Microbiol Rev **2012**; 25:450–70.
370. Pérez-Nadales E, Fernández-Ruiz M, Natera AM, et al. Efficacy of ceftazidime-avibactam in solid organ transplant recipients with bloodstream infections caused by carbapenemase-producing *Klebsiella pneumoniae*. Am J Transplant **2023**; 23:1022–34.
371. Tumbarello M, Raffaelli F, Giannella M, et al. Ceftazidime-avibactam use for *Klebsiella pneumoniae* carbapenemase-producing *K. pneumoniae* infections: a retrospective observational multicenter study. Clin Infect Dis **2021**; 73:1664–76.
372. Meschiari M, Faltoni M, Kaleci S, et al. Intravenous fosfomycin in combination regimens as a treatment option for difficult-to-treat infections due to multi-drug-resistant Gram-negative organisms: a real-life experience. Int J Antimicrob Agents **2024**; 63:107134.
373. Anastasia A, Bonura S, Rubino R, et al. The use of intravenous fosfomycin in clinical practice: a 5-year retrospective study in a tertiary hospital in Italy. Antibiotics (Basel) **2023**; 12:971.
374. Önal U, Tüzemen U, Küçükdemirci Kaya P, et al. A comparative study of ceftazidime/avibactam-based and fosfomycin plus meropenem-based regimens for managing infections caused by carbapenem-resistant *Klebsiella pneumoniae* in critically ill patients. J Chemother **2025**; 37:1–9.
375. Belati A, Diella L, Bavaro DF, et al. Intravenous fosfomycin as adjunctive therapy for gram-negative bacteria bloodstream infections: a propensity score adjusted retrospective cohort study. Int J Antimicrob Agents **2024**; 64:107247.
376. Magiorakos AP, Srinivasan A, Carey RB, et al. Multidrug-resistant, extensively drug-resistant and pandrug-resistant bacteria: an international expert proposal for interim standard definitions for acquired resistance. Clin Microbiol Infect **2012**; 18:268–81.
377. Kadri SS, Adjemian J, Lai YL, et al. Difficult-to-treat resistance in gram-negative bacteremia at 173 US hospitals: retrospective cohort analysis of prevalence, predictors, and outcome of resistance to all first-line agents. Clin Infect Dis **2018**; 67: 1803–14.
378. Oliver A, Arca-Suárez J, Gomis-Font MA, González-Pinto L, López-Causapé C. Emerging resistance mechanisms to newer β-lactams in *Pseudomonas aeruginosa*. Clin Microbiol Infect **2025**; 31:1790–6.
379. Tekes-Manuva D, Babich T, Kozlovski D, et al. What is the most effective antibiotic monotherapy for severe *Pseudomonas aeruginosa* infection? A systematic review and meta-analysis of randomized controlled trials. Clin Microbiol Infect **2025**; 31:740–52.
380. Babich T, Naucler P, Valik JK, et al. Ceftazidime, carbapenems, or piperacillin-tazobactam as single definitive therapy for *Pseudomonas aeruginosa* bloodstream infection: a multisite retrospective study. Clin Infect Dis **2020**; 70:2270–80.
381. European Committee on Antimicrobial Susceptibility Testing. Available at: https://www.eucast.org/bacteria/clinical-breakpoints-and-interpretation/clinical-breakpoint-tables/. Accessed 31 December 2025.
382. Tamma PD, Harris PNA, Mathers AJ, Wenzler E, Humphries RM. Deconstructing the 2023 Clinical and Laboratory Standards Institute revised piperacillin-tazobactam breakpoints against *Pseudomonas aeruginosa*. Clin Infect Dis **2023**; 76:1868–70.
383. Tam VH, Louie A, Lomaestro BM, Drusano GL. Integration of population pharmacokinetics, a pharmacodynamic target, and microbiologic surveillance data to generate a rational empiric dosing strategy for cefepime against *Pseudomonas aeruginosa*. Pharmacotherapy **2003**; 23:291–5.
384. Cheatham SC, Shea KM, Healy DP, et al. Steady-state pharmacokinetics and pharmacodynamics of cefepime administered by prolonged infusion in hospitalised patients. Int J Antimicrob Agents **2011**; 37:46–50.
385. Reeder JA, Creech CB, Nation RL, et al. Utilizing an opportunistic clinical study and population-based pharmacokinetic models to identify rational empiric dosing regimens for piperacillin-tazobactam in critically ill patients. J Clin Pharmacol **2025**; 65:452–65.
386. Bauer KA, West JE, O'Brien JM, Goff DA. Extended-infusion cefepime reduces mortality in patients with *Pseudomonas aeruginosa* infections. Antimicrob Agents Chemother **2013**; 57:2907–12.
387. Lodise TP Jr, Lomaestro B, Drusano GL. Piperacillin-tazobactam for *Pseudomonas aeruginosa* infection: clinical implications of an extended-infusion dosing strategy. Clin Infect Dis **2007**; 44:357–63.
388. Gill CM, Aktaş E, Alfouzan W, et al. Elevated MICs of susceptible antipseudomonal cephalosporins in non-carbapenemase-producing, carbapenem-resistant *Pseudomonas aeruginosa*: implications for dose optimization. Antimicrob Agents Chemother **2021**; 65:e0120421.
389. Reyes J, Komarow L, Chen L, et al. Global epidemiology and clinical outcomes of carbapenem-resistant *Pseudomonas aeruginosa* and associated carbapenemases (POP): a prospective cohort study. Lancet Microbe **2023**; 4:e159–70.
390. Fuhs DT, Cortés-Lara S, Tait JR, et al. The effects of single and multiple resistance mechanisms on bacterial response to meropenem. Clin Microbiol Infect **2024**; 30:1276–83.
391. Huang H, Hancock RE. Genetic definition of the substrate selectivity of outer membrane porin protein OprD of *Pseudomonas aeruginosa*. J Bacteriol **1993**; 175:7793–800.
392. Supreeyasakon S, Traipattanakul J, Hemapanpairoa J, Juntanawiwat P, Santimaleeworagun W. Evaluation of risk factors for 14-day and 30-day mortality among treatment regimens against *Pseudomonas aeruginosa* resistant to carbapenem but susceptible to traditional antipseudomonal non-carbapenem beta-lactam agents. PLoS One **2024**; 19:e0313944.
393. Aldardeer NF, Labban HM, Alhuthil RT, et al. Effectiveness of traditional non-carbapenem beta-lactams vs. novel beta-lactams for the treatment of carbapenem-resistant *Pseudomonas aeruginosa*: a retrospective cohort study. BMC Infect Dis **2024**; 24:1455.
394. Ng TH, Zhao JJ, Gumbleton R, Olson S, Smith S, Scipione MR. Treatment outcomes for carbapenem-resistant and cephalosporin-susceptible *Pseudomonas aeruginosa* pneumonia. Ann Pharmacother **2024**; 58:581–8.
395. Wagenlehner FM, Umeh O, Steenbergen J, Yuan G, Darouiche RO. Ceftolozane-tazobactam compared with levofloxacin in the treatment of complicated urinary-tract infections, including pyelonephritis: a randomised, double-blind, phase 3 trial (ASPECT-cUTI). Lancet **2015**; 385:1949–56.
396. Castanheira M, Davis AP, Mendes RE, Serio AW, Krause KM, Flamm RK. In vitro activity of plazomicin against gram-negative and gram-positive isolates collected from U.S. hospitals and comparative activities of aminoglycosides against carbapenem-resistant Enterobacteriaceae and isolates carrying carbapenemase genes. Antimicrob Agents Chemother **2018**; 62:e00313–18.
397. Sorlí L, Luque S, Li J, et al. Colistin for the treatment of urinary tract infections caused by extremely drug-resistant *Pseudomonas aeruginosa*: dose is critical. J Infect **2019**; 79:253–61.
398. Titov I, Wunderink RG, Roquilly A, et al. A randomized, double-blind, multicenter trial comparing efficacy and safety of imipenem/cilastatin/relebactam versus piperacillin/tazobactam in adults with hospital-acquired or ventilator-associated bacterial pneumonia (RESTORE-IMI 2 study). Clin Infect Dis **2021**; 73:e4539–48.
399. Pogue JM, Kaye KS, Veve MP, et al. Ceftolozane/tazobactam vs polymyxin or aminoglycoside-based regimens for the treatment of drug-resistant *Pseudomonas aeruginosa*. Clin Infect Dis **2020**; 71:304–10.
400. Long WM, Xu WX, Hu Q, et al. The efficacy and safety of ceftazidime/avibactam or polymyxin B based regimens for carbapenem-resistant *Pseudomonas aeruginosa* infection: a multicenter real-world and propensity score-matched study. Front Pharmacol **2025**; 16:1533952.

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401. Lodise TP, Min J, Nathanson BH, Yücel E. Comparison of early treatment with ceftolozane/tazobactam versus polymyxin-based therapy of pneumonia due to MDR *Pseudomonas aeruginosa* (PUMA). Antimicrob Agents Chemother **2025**; 69:e0056925.
402. Liu C, Leng B, Xie M, et al. Comparing ceftazidime/avibactam and polymyxin B for treating carbapenem-resistant organisms infections: a propensity score-matched retrospective cohort study. Int J Antimicrob Agents **2025**; 65:107418.
403. Almangour TA, Alkherb Z, Ghonem L, et al. Ceftazidime-avibactam versus colistin for the treatment of multidrug-resistant *Pseudomonas aeruginosa* infections: a multicenter cohort study. Pharmaceuticals (Basel) **2025**; 18:108.
404. Satlin MJ, Simner PJ, Slover CM, Yamano Y, Nagata TD, Portsmouth S. Cefiderocol treatment for patients with multidrug- and carbapenem-resistant *Pseudomonas aeruginosa* infections in the compassionate use program. Antimicrob Agents Chemother **2023**; 67:e0019423.
405. Soueges S, Faure E, Parize P, et al. Real-world multicentre study of cefiderocol treatment of immunocompromised patients with infections caused by multidrug-resistant Gram-negative bacteria: CEFI-ID. J Infect **2025**; 90:106376.
406. Lombardi A, Giacobbe DR, Mangioni D, et al. Cefiderocol therapy among immunocompromised adult patients: a descriptive analysis from a prospective, multicentre cohort study. BMC Infect Dis **2025**; 25:1339.
407. Almangour TA, Ghonem L, Alassiri D, et al. Ceftolozane-tazobactam versus ceftazidime-avibactam for the treatment of infections caused by multidrug-resistant *Pseudomonas aeruginosa*: a multicenter cohort study. Antimicrob Agents Chemother **2023**; 67:e0040523.
408. Hareza DA, Cosgrove SE, Bonomo RA, et al. Clinical outcomes and emergence of resistance of *Pseudomonas aeruginosa* infections treated with ceftolozane-tazobactam versus ceftazidime-avibactam. Antimicrob Agents Chemother **2024**; 68:e0090724.
409. Shields RK, Abbo LM, Ackley R, et al. Effectiveness of ceftazidime-avibactam versus ceftolozane-tazobactam for multidrug-resistant *Pseudomonas aeruginosa* infections in the USA (CACTUS): a multicentre, retrospective, observational study. Lancet Infect Dis **2025**; 25:574–84.
410. Lodise TP, Obi EN, Watanabe AH, Yucel E, Min J, Nathanson BH. Comparative evaluation of early treatment with ceftolozane/tazobactam versus ceftazidime/avibactam for non-COVID-19 patients with pneumonia due to multidrug-resistant *Pseudomonas aeruginosa*. J Antimicrob Chemother **2024**; 79:2954–64.
411. Caro L, Nicolau DP, De Waele JJ, et al. Lung penetration, bronchopulmonary pharmacokinetic/pharmacodynamic profile and safety of 3 g of ceftolozane/tazobactam administered to ventilated, critically ill patients with pneumonia. J Antimicrob Chemother **2020**; 75:1546–53.
412. Nicolau DP, Siew L, Armstrong J, et al. Phase 1 study assessing the steady-state concentration of ceftazidime and avibactam in plasma and epithelial lining fluid following two dosing regimens. J Antimicrob Chemother **2015**; 70:2862–9.
413. Karlowsky JA, Lob SH, Bauer KA, et al. Activity of ceftolozane/tazobactam, imipenem/relebactam and ceftazidime/avibactam against clinical Gram-negative isolates-SMART United States 2019–21. JAC Antimicrob Resist **2024**; 6:dlad152.
414. Gottesdiener LS, Li Y, Greenwood-Quaintance KE, et al. Resistance to novel β-lactam/β-lactamase inhibitors among carbapenem-resistant *Pseudomonas aeruginosa* and clinical implications in the prospective observational *Pseudomonas* study. Antimicrob Agents Chemother **2026**; 70:e0038825.
415. Wangchinda W, Wu JY, Abbo LM, et al. Effectiveness of imipenem-relebactam for multidrug-resistant *Pseudomonas aeruginosa* in pneumonia and bloodstream infections in the United States (MIRAGE). Antimicrob Agents Chemother **2025**; 69:e0132525.
416. Sabour S, Harrington KRV, Martinson E, et al. Characterization of carbapenem-resistant Enterobacterales and *Pseudomonas aeruginosa* carrying multiple carbapenemase genes-Antimicrobial Resistance Laboratory Network, 2018–2022. J Clin Microbiol **2024**; 62:e0122024.
417. PÉrez-VÁzquez M, Sola-Campoy PJ, Zurita AM, et al. Carbapenemase-producing *Pseudomonas aeruginosa* in Spain: interregional dissemination of the high-risk clones ST175 and ST244 carrying bla(VIM-2), bla(VIM-1), bla(IMP-8), bla(VIM-20) and bla(KPC-2). Int J Antimicrob Agents **2020**; 56:106026.
418. Koumaki V, Voudanta E, Michelaki A, et al. Epidemiological trends of carbapenemase-*Producing Pseudomonas* aeruginosa in a tertiary care hospital in Athens, Greece, during 2020–2023. Antibiotics (Basel) **2025**; 14:898.
419. Ramatla T, Nkhebenyane J, Lekota KE, et al. Global prevalence and antibiotic resistance profiles of carbapenem-resistant *Pseudomonas aeruginosa* reported from 2014 to 2024: a systematic review and meta-analysis. Front Microbiol **2025**; 16:1599070.
420. Gill CM, Aktaþ E, Alfouzan W, et al. The ERACE-PA global surveillance program: ceftolozane/tazobactam and ceftazidime/avibactam in vitro activity against a global collection of carbapenem-resistant *Pseudomonas aeruginosa*. Eur J Clin Microbiol Infect Dis **2021**; 40:2533–41.
421. Morelli MK, Kloosterboer A, Fulton SA, et al. Investigating and treating a corneal ulcer due to extensively drug-resistant *Pseudomonas aeruginosa*. Antimicrob Agents Chemother **2023**; 67:e0027723.
422. Sundermann AJ, Rangachar Srinivasa V, Mills EG, et al. Two artificial tears outbreak-associated cases of extensively drug-resistant *Pseudomonas aeruginosa* detected through whole genome sequencing-based surveillance. J Infect Dis **2024**; 229:517–21.
423. Grossman MK, Rankin DA, Maloney M, et al. Extensively drug-resistant *Pseudomonas aeruginosa* outbreak associated with artificial tears. Clin Infect Dis **2024**; 79:6–14.
424. Khan A, Tran TT, Rios R, et al. Extensively drug-resistant *Pseudomonas aeruginosa* ST309 harboring tandem guiana extended spectrum beta-lactamase enzymes: a newly emerging threat in the United States. Open Forum Infect Dis **2019**; 6:ofz273.
425. Hishinuma T, Tada T, Kuwahara-Arai K, Yamamoto N, Shimojima M, Kirikae T. Spread of GES-5 carbapenemase-producing *Pseudomonas aeruginosa* clinical isolates in Japan due to clonal expansion of ST235. PLoS One **2018**; 13:e0207134.
426. Gill CM, Santini D, Nicolau DP; ERACE-PA Global Study Group. In vitro activity of cefiderocol against a global collection of carbapenem-resistant *Pseudomonas aeruginosa* with a high level of carbapenemase diversity. J Antimicrob Chemother **2024**; 79:412–6.
427. Kirkegaard-Biosca C, Del Barrio-Tofiño E, Villamarín M, et al. Cefiderocol for the treatment of infections by VIM-type-producing gram-negative bacteria. Antibiotics (Basel) **2024**; 13:874.
428. Mushtaq S, Sadouki Z, Vickers A, Livermore DM, Woodford N. In vitro activity of cefiderocol, a siderophore cephalosporin, against multidrug-resistant gram-negative Bacteria. Antimicrob Agents Chemother **2020**; 64:e01582–20.
429. Kocer K, My TN, Sy BT, et al. Efficacy of cefiderocol in combination with xeruborbactam versus taniborbactam against cefiderocol-resistant NDM-producing *Pseudomonas aeruginosa*. Antimicrob Agents Chemother **2025**; 69:e0085725.
430. Karlowsky JA, Kazmierczak KM, de Jonge BLM, Hackel MA, Sahm DF, Bradford PA. In vitro activity of aztreonam-avibactam against Enterobacteriaceae and *Pseudomonas aeruginosa* isolated by clinical laboratories in 40 countries from 2012 to 2015. Antimicrob Agents Chemother **2017**; 61:e00472–17.
431. Le Terrier C, Nordmann P, Poirel L. In vitro activity of aztreonam in combination with newly developed β-lactamase inhibitors against MDR Enterobacterales and *Pseudomonas aeruginosa* producing metallo-β-lactamases. J Antimicrob Chemother **2022**; 78:101–7.
432. Nakae T, Nakajima A, Ono T, Saito K, Yoneyama H. Resistance to beta-lactam antibiotics in *Pseudomonas aeruginosa* due to interplay between the MexAB-OprM efflux pump and beta-lactamase. Antimicrob Agents Chemother **1999**; 43:1301–3.
433. Müller SE, Junker S, Rehner J, et al. Comparative in vitro activity of ceftazidime-avibactam plus aztreonam and the fixed combination aztreonam/avibactam against multidrug-resistant *Pseudomonas aeruginosa*. J Antimicrob Chemother **2025**; 80:3352–6.
434. Montero MM, Domene-Ochoa S, Prim N, et al. Synergistic efficacy of ceftazidime/avibactam and aztreonam against carbapenemase-producing *Pseudomonas aeruginosa*: insights from the hollow-fiber infection model. Infect Dis (Lond) **2025**; 57:81–8.
435. Pina-Sánchez M, Rua M, López-Causapé C, et al. Ceftazidime-avibactam plus aztreonam cocktail for the treatment of VIM-producing *Pseudomonas aeruginosa* infections: good enough to have another? J Antimicrob Chemother **2025**; 80:1371–6.
436. Vega AD, DeRonde K, Jimenez A, et al. Difficult-to-treat (DTR) *Pseudomonas aeruginosa* harboring Verona-Integron metallo-β-lactamase (bla(VIM)): infection management and molecular analysis. Antimicrob Agents Chemother **2024**; 68:e0147423.
437. Sempere A, Viñado B, Los-Arcos I, et al. Ceftazidime-avibactam plus aztreonam for the treatment of infections by VIM-type-producing gram-negative bacteria. Antimicrob Agents Chemother **2022**; 66:e0075122.
438. Kimbrough JH, Karr MH, Nguyen ST, et al. Activity of cefiderocol against *Pseudomonas aeruginosa* from the USA and Europe (2020–2023) with difficult-to-treat resistance phenotype, including those nonsusceptible to recently developed β-lactam/β-lactamase inhibitor combinations: results from the SENTRY antimicrobial surveillance program. Microbiol Spectr **2025**; 13:e0207925.
439. Karlowsky JA, Hackel MA, Takemura M, Yamano Y, Echols R, Sahm DF. In vitro susceptibility of gram-negative pathogens to cefiderocol in five consecutive annual multinational SIDERO-WT surveillance studies, 2014 to 2019. Antimicrob Agents Chemother **2022**; 66:e0199021.

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440. Tsai YV, Bookstaver PB, Kohn J, et al. The prevalence of gram-negative bacteria with difficult-to-treat resistance and utilization of novel β-lactam antibiotics in the southeastern United States. Antimicrob Steward Healthc Epidemiol **2024**; 4:e35.
441. Berrazeg M, Jeannot K, Ntsogo Enguéné VY, et al. Mutations in β-lactamase AmpC increase resistance of *Pseudomonas aeruginosa* isolates to antipseudomonal cephalosporins. Antimicrob Agents Chemother **2015**; 59:6248–55.
442. Fraile-Ribot PA, Cabot G, Mulet X, et al. Mechanisms leading to in vivo ceftolozane/tazobactam resistance development during the treatment of infections caused by MDR *Pseudomonas aeruginosa*. J Antimicrob Chemother **2018**; 73: 658–63.
443. Haidar G, Philips NJ, Shields RK, et al. Ceftolozane-tazobactam for the treatment of multidrug-resistant *Pseudomonas aeruginosa* infections: clinical effectiveness and evolution of resistance. Clin Infect Dis **2017**; 65:110–20.
444. Rubio AM, Kline EG, Jones CE, et al. In vitro susceptibility of multidrug-resistant *Pseudomonas aeruginosa* following treatment-emergent resistance to ceftolozane-tazobactam. Antimicrob Agents Chemother **2021**; 65: e00084–21.
445. Tamma PD, Beisken S, Bergman Y, et al. Modifiable risk factors for the emergence of ceftolozane-tazobactam resistance. Clin Infect Dis **2021**; 73:e4599–606.
446. Nguyen HA, Peleg AY, Song J, et al. Complex pathways to ceftolozane-tazobactam resistance in clinical *Pseudomonas aeruginosa* isolates: a genomic epidemiology study. Clin Microbiol Infect **2026**; 32:110–7.
447. Shah S, Kline EG, Haidar G, et al. Rates of resistance to ceftazidime-avibactam and ceftolozane-tazobactam among patients treated for multidrug-resistant *Pseudomonas aeruginosa* bacteremia or pneumonia. Clin Infect Dis **2025**; 80: 24–8.
448. González-Pinto L, Blanco-Martín T, Alonso-García I, et al. Impact of transferable β-lactamases and intrinsic AmpC amino acid substitutions on the activity of cefiderocol against wild-type and iron uptake-deficient mutants of *Pseudomonas aeruginosa*. J Antimicrob Chemother **2024**; 79:3023–8.
449. Sastre-Femenia MA, Fernández-Muñoz A, Gomis-Font MA, et al. Spanish nationwide survey of *Pseudomonas aeruginosa* cefiderocol susceptibility and resistance mechanisms. Int J Antimicrob Agents **2025**; 66:107563.
450. Shields RK, Kline EG, Squires KM, Van Tyne D, Doi Y. In vitro activity of cefiderocol against *Pseudomonas aeruginosa* demonstrating evolved resistance to novel β-lactam/β-lactamase inhibitors. JAC Antimicrob Resist **2023**; 5:dlad107.
451. López-Causapé C, Maruri-Aransolo A, Gomis-Font MA, et al. Cefiderocol resistance genomics in sequential chronic *Pseudomonas aeruginosa* isolates from cystic fibrosis patients. Clin Microbiol Infect **2023**; 29:538.e7–e13.
452. Simner PJ, Beisken S, Bergman Y, Posch AE, Cosgrove SE, Tamma PD. Cefiderocol activity against clinical *Pseudomonas aeruginosa* isolates exhibiting ceftolozane-tazobactam resistance. Open Forum Infect Dis **2021**; 8:ofab311.
453. Gomis-Font MA, Cabot G, Sánchez-Diener I, et al. In vitro dynamics and mechanisms of resistance development to imipenem and imipenem/relebactam in *Pseudomonas aeruginosa*. J Antimicrob Chemother **2020**; 75:2508–15.
454. Gomis-Font MA, Cabot G, López-Argüello S, et al. Comparative analysis of in vitro dynamics and mechanisms of ceftolozane/tazobactam and imipenem/relebactam resistance development in *Pseudomonas aeruginosa* XDR high-risk clones. J Antimicrob Chemother **2022**; 77:957–68.
455. Shields RK, Stellfox ME, Kline EG, Samanta P, Van Tyne D. Evolution of imipenem-relebactam resistance following treatment of multidrug-resistant *Pseudomonas aeruginosa* pneumonia. Clin Infect Dis **2022**; 75:710–4.
456. Piccica M, Spinicci M, Botta A, et al. Cefiderocol use for the treatment of infections by carbapenem-resistant Gram-negative bacteria: an Italian multicentre real-life experience. J Antimicrob Chemother **2023**; 78:2752–61.
457. Corbella L, Boán J, San-Juan R, et al. Effectiveness of ceftazidime-avibactam for the treatment of infections due to *Pseudomonas aeruginosa*. Int J Antimicrob Agents **2022**; 59:106517.
458. Giske CG, Amara M, Cantón R, et al. Aminoglycoside breakpoints-the EUCAST approach and what this means for the clinician. Clin Microbiol Infect **2025**; 31: 1954–7.
459. Bhavnani SM, Hammel JP, Onufrak NJ, et al. Evaluation of aminoglycoside dosing regimens adjusted for renal function and in vitro susceptibility test interpretive criteria for Enterobacterales and *Pseudomonas aeruginosa*. Open Forum Infect Dis **2025**; 12:ofaf426.
460. Akajagbor DS, Wilson SL, Shere-Wolfe KD, Dakum P, Charurat ME, Gilliam BL. Higher incidence of acute kidney injury with intravenous colistimethate sodium compared with polymyxin B in critically ill patients at a tertiary care medical center. Clin Infect Dis **2013**; 57:1300–3.
461. Phe K, Lee Y, McDaneld PM, et al. In vitro assessment and multicenter cohort study of comparative nephrotoxicity rates associated with colistimethate versus polymyxin B therapy. Antimicrob Agents Chemother **2014**; 58:2740–6.
462. Rigatto MH, Oliveira MS, Perdigão-Neto LV, et al. Multicenter prospective cohort study of renal failure in patients treated with colistin versus polymyxin B. Antimicrob Agents Chemother **2016**; 60:2443–9.
463. Rattanaumpawan P, Lorsutthitham J, Ungprasert P, Angkasekwinai N, Thamlikitkul V. Randomized controlled trial of nebulized colistimethate sodium as adjunctive therapy of ventilator-associated pneumonia caused by Gram-negative bacteria. J Antimicrob Chemother **2010**; 65:2645–9.
464. Kollef MH, Ricard JD, Roux D, et al. A randomized trial of the amikacin fosfomycin inhalation system for the adjunctive therapy of gram-negative ventilator-associated pneumonia: IASIS trial. Chest **2017**; 151:1239–46.
465. Niederman MS, Alder J, Bassetti M, et al. Inhaled amikacin adjunctive to intravenous standard-of-care antibiotics in mechanically ventilated patients with Gram-negative pneumonia (INHALE): a double-blind, randomised, placebo-controlled, phase 3, superiority trial. Lancet Infect Dis **2020**; 20:330–40.
466. Qin JP, Huang HB, Zhou H, Zhu Y, Xu Y, Du B. Amikacin nebulization for the adjunctive therapy of gram-negative pneumonia in mechanically ventilated patients: a systematic review and meta-analysis of randomized controlled trials. Sci Rep **2021**; 11:6969.
467. Zhang Z, Li H, Hu Y, et al. The efficacy and safety of inhaled antibiotics for pneumonia: a systematic review and meta-analysis. Pulm Pharmacol Ther **2024**; 86: 102315.
468. Wu M, Yang X, Tian J, Fan H, Zhang Y. Antibiotic treatment of pulmonary infections: an umbrella review and evidence map. Front Pharmacol **2021**; 12: 680178.
469. Sella N, Pettenuzzo T, De Cassai A, et al. Inhaled antibiotics for treating pneumonia in invasively ventilated patients in intensive care unit: a meta-analysis of randomized clinical trials with trial sequential analysis. Crit Care **2024**; 28:387.
470. Boisson M, Jacobs M, Grégoire N, et al. Comparison of intrapulmonary and systemic pharmacokinetics of colistin methanesulfonate (CMS) and colistin after aerosol delivery and intravenous administration of CMS in critically ill patients. Antimicrob Agents Chemother **2014**; 58:7331–9.
471. Rouby JJ, Bouhemad B, Monsel A, et al. Aerosolized antibiotics for ventilator-associated pneumonia: lessons from experimental studies. Anesthesiology **2012**; 117:1364–80.
472. Wenzler E, Fraidenburg DR, Scardina T, Danziger LH. Inhaled antibiotics for gram-negative respiratory infections. Clin Microbiol Rev **2016**; 29:581–632.
473. Biagi M, Butler D, Tan X, Qasmieh S, Wenzler E. A breath of fresh air in the fog of antimicrobial resistance: inhaled polymyxins for gram-negative pneumonia. Antibiotics (Basel) **2019**; 8:27.
474. Kalil AC, Metersky ML, Klompas M, et al. Management of adults with hospital-acquired and ventilator-associated pneumonia: 2016 clinical practice guidelines by the Infectious Diseases Society of America and the American Thoracic Society. Clin Infect Dis **2016**; 63:e61–111.
475. Rello J, Solé-Lleonart C, Rouby JJ, et al. Use of nebulized antimicrobials for the treatment of respiratory infections in invasively mechanically ventilated adults: a position paper from the European Society of Clinical Microbiology and Infectious Diseases. Clin Microbiol Infect **2017**; 23:629–39.
476. Meschiari M, Asquier-Khati A, Tiseo G, et al. Treatment of infections caused by multidrug-resistant Gram-negative bacilli: a practical approach by the Italian (SIMIT) and French (SPILF) Societies of Infectious Diseases. Int J Antimicrob Agents **2024**; 64:107186.
477. Axiotakis LG Jr, Enver N, Keating CL, Pitman MJ. Laryngeal injury due to amikacin inhalation for refractory *Mycobacterium avium* complex infection. Chest **2021**; 159:e185–7.
478. Niederman MS. Adjunctive nebulized antibiotics: what is their place in ICU infections? Front Med (Lausanne) **2019**; 6:99.
479. Dubey V, Reza N, Hope W. Drug-resistant *Acinetobacter baumannii*: mortality, emerging treatments, and future pharmacological targets for a WHO priority pathogen. Clin Microbiol Rev **2025**; 38:e0027924.
480. Vijayakumar S, Biswas I, Veeraraghavan B. Accurate identification of clinically important *Acinetobacter* spp.: an update. Future Sci OA **2019**; 5:FSO395.
481. Iovleva A, Mustapha MM, Griffith MP, et al. Carbapenem-resistant *Acinetobacter baumannii* in U.S. hospitals: diversification of circulating lineages and antimicrobial resistance. mBio **2022**; 13:e0275921.
482. Mack AR, Hujer AM, Mojica MF, et al. β-Lactamase diversity in *Acinetobacter baumannii*. Antimicrob Agents Chemother **2025**; 69:e0078424.
483. Valcek A, Nesporova K, Whiteway C, et al. Genomic analysis of a strain collection containing multidrug-, extensively drug-, pandrug-, and carbapenem-resistant modern clinical isolates of *Acinetobacter baumannii*. Antimicrob Agents Chemother **2022**; 66:e0089222.
484. Turton JF, Ward ME, Woodford N, et al. The role of ISAba1 in expression of OXA carbapenemase genes in *Acinetobacter baumannii*. FEMS Microbiol Lett **2006**; 258:72–7.

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485. Takebayashi Y, Findlay J, Heesom KJ, Warburton PJ, Avison MB, Evans BA. Variability in carbapenemase activity of intrinsic OxaAb (OXA-51-like) β-lactamase enzymes in *Acinetobacter baumannii*. J Antimicrob Chemother **2021**; 76:587–95.
486. Nemec A, Dolzani L, Brisse S, van den Broek P, Dijkshoorn L. Diversity of aminoglycoside-resistance genes and their association with class 1 integrons among strains of pan-European *Acinetobacter baumannii* clones. J Med Microbiol **2004**; 53:1233–40.
487. Catel-Ferreira M, Coadou G, Molle V, et al. Structure-function relationships of CarO, the carbapenem resistance-associated outer membrane protein of *Acinetobacter baumannii*. J Antimicrob Chemother **2011**; 66:2053–6.
488. Snyder DS, McIntosh TJ. The lipopolysaccharide barrier: correlation of antibiotic susceptibility with antibiotic permeability and fluorescent probe binding kinetics. Biochemistry **2000**; 39:11777–87.
489. Yoon EJ, Chabane YN, Goussard S, et al. Contribution of resistance-nodulation-cell division efflux systems to antibiotic resistance and biofilm formation in *Acinetobacter baumannii*. mBio **2015**; 6:e00309–15.
490. Penwell WF, Shapiro AB, Giacobbe RA, et al. Molecular mechanisms of sulbactam antibacterial activity and resistance determinants in *Acinetobacter baumannii*. Antimicrob Agents Chemother **2015**; 59:1680–9.
491. McLeod SM, Shapiro AB, Moussa SH, et al. Frequency and mechanism of spontaneous resistance to sulbactam combined with the novel β-lactamase inhibitor ETX2514 in clinical isolates of *Acinetobacter baumannii*. Antimicrob Agents Chemother **2018**; 62:e01576–17.
492. Seward RJ, Towner KJ. Molecular epidemiology of quinolone resistance in *Acinetobacter* spp. Clin Microbiol Infect **1998**; 4:248–54.
493. Williams JD. Beta-lactamase inhibition and in vitro activity of sulbactam and sulbactam/cefoperazone. Clin Infect Dis **1997**; 24:494–7.
494. Barnes MD, Kumar V, Bethel CR, et al. Targeting multidrug-resistant *Acinetobacter* spp. sulbactam and the diazabicyclooctenone β-lactamase inhibitor ETX2514 as a novel therapeutic agent. mBio **2019**; 10:e00159–19.
495. Palavecino EL, Kilic A. In vitro activity of cefiderocol and sulbactam-durlobactam against carbapenem-resistant *Acinetobacter baumannii* clinical isolates collected between 2019 and 2024. Infect Dis (Lond) **2025**; 58:243–8.
496. Buyukyanbolu E, Argotsinger J, Beck ET, et al. Activity of ampicillin-sulbactam, sulbactam-durlobactam, and comparators against *Acinetobacter baumannii-calcoaceticus* complex strains isolated from respiratory and bloodstream sources: results from ACNBio study. Antimicrob Agents Chemother **2025**; 69:e0037925.
497. Wang M, Ge L, Chen L, et al. Clinical outcomes and bacterial characteristics of carbapenem-resistant *Acinetobacter baumannii* among patients from different global regions. Clin Infect Dis **2024**; 78:248–58.
498. Saikia S, Gogoi I, Oloo A, Sharma M, Puzari M, Chetia P. Co-production of metallo-β-lactamase and OXA-type β-lactamases in carbapenem-resistant *Acinetobacter baumannii* clinical isolates in North East India. World J Microbiol Biotechnol **2024**; 40:167.
499. Amar AK, Panda L, Prasad K, et al. Increasing burden of antibiotic resistance in India due to co-existence of multiple classes of carbapenem and different aminoglycoside resistance genes in clinical isolates of *Acinetobacter baumannii*. Curr Microbiol **2025**; 82:120.
500. AlAmri AM, AlQurayan AM, Sebastian T, AlNimr AM. Molecular surveillance of multidrug-resistant *Acinetobacter baumannii*. Curr Microbiol **2020**; 77: 335–42.
501. Singh S, Verma A, Venkatesh V, Verma S, Reddy DH, Agrawal A. The clinical impression of NDM-producing *Acinetobacter baumannii* in intensive care units of the university referral hospital in North India. Indian J Crit Care Med **2024**; 28:1044–9.
502. Kaye KS, Shorr AF, Wunderink RG, et al. Efficacy and safety of sulbactam-durlobactam versus colistin for the treatment of patients with serious infections caused by *Acinetobacter baumannii*-calcoaceticus complex: a multicentre, randomised, active-controlled, phase 3, non-inferiority clinical trial (ATTACK). Lancet Infect Dis **2023**; 23:1072–84.
503. Shields RK, Paterson DL, Tamma PD. Navigating available treatment options for carbapenem-resistant *Acinetobacter baumannii-calcoaceticus* complex infections. Clin Infect Dis **2023**; 76:S179–93.
504. Tanudra A, McLeod S, Miller A, Tommasi R, O'Donell J. Sulbactam-durlobactam (SUL-DUR) in vitro dose response studies with and without imipenem or meropenem carbapenemase-producing *Acinetobacter baumannii* utilizing the hollow-fiber infection model. In: ECCMID 2022. Lisbon, Portugal, Poster 02037.
505. Choi JY, Park YS, Cho CH, et al. Synergic in-vitro activity of imipenem and sulbactam against *Acinetobacter baumannii*. Clin Microbiol Infect **2004**; 10: 1098–101.
506. Fernández-Cuenca F, Martínez-Martínez L, Conejo MC, Ayala JA, Perea EJ, Pascual A. Relationship between beta-lactamase production, outer membrane protein and penicillin-binding protein profiles on the activity of carbapenems against clinical isolates of *Acinetobacter baumannii*. J Antimicrob Chemother **2003**; 51:565–74.
507. Veeraraghavan B, Shin E, Bakthavatchalam YD, et al. A microbiological and structural analysis of the interplay between sulbactam/durlobactam and imipenem against penicillin-binding proteins (PBPs) of *Acinetobacter* spp. Antimicrob Agents Chemother **2025**; 69:e0162724.
508. O'Donnell J, Tanudra A, Chen A, Miller AA, McLeod SM, Tommasi R. In vitro pharmacokinetics/pharmacodynamics of the β-lactamase inhibitor, durlobactam, in combination with sulbactam against *Acinetobacter baumannii-calcoaceticus* complex. Antimicrob Agents Chemother **2024**; 68:e0031223.
509. Moussa SH, Shapiro AB, McLeod SM, et al. Molecular drivers of resistance to sulbactam-durlobactam in contemporary clinical isolates of *Acinetobacter baumannii*. Antimicrob Agents Chemother **2023**; 67:e0066523.
510. Bianco G, Biffignandi GB, Boattini M, et al. Resistance to sulbactam/durlobactam in carbapenem-resistant *Acinetobacter baumannii* in treatment-naive setting: in vitro data and genomic insights from Italian bloodstream isolates. Int J Antimicrob Agents **2026**; 67:107718.
511. Xu Q, Liu X, Liu H, et al. In vitro activity and resistance mechanisms of sulbactam/durlobactam against *Acinetobacter baumannii* clinical isolates in China (2019–2020). Clin Microbiol Infect **2026**; 32:102–9.
512. Halim J, Keane AP, Bouzo J, et al. Synergistic cefiderocol-containing antibiotic combinations active against highly drug-resistant *Acinetobacter baumannii* patient isolates with diverse resistance mechanisms. J Antimicrob Chemother **2025**; 80:2814–24.
513. Noguchi JK, Gill MA. Sulbactam: a beta-lactamase inhibitor. Clin Pharm **1988**; 7:37–51.
514. Lenhard JR, Smith NM, Bulman ZP, et al. High-dose ampicillin-sulbactam combinations combat polymyxin-resistant *Acinetobacter baumannii* in a hollow-fiber infection model. Antimicrob Agents Chemother **2017**; 61:e01268–16.
515. Beganovic M, Daffinee KE, Luther MK, LaPlante KL. Minocycline alone and in combination with polymyxin B, meropenem, and sulbactam against carbapenem-susceptible and -resistant *Acinetobacter baumannii* in an in vitro pharmacodynamic model. Antimicrob Agents Chemother **2021**; 65:e01680–20.
516. Abdul-Mutakabbir JC, Yim J, Nguyen L, et al. In vitro synergy of colistin in combination with meropenem or tigecycline against carbapenem-resistant *Acinetobacter baumannii*. Antibiotics (Basel) **2021**; 10:880.
517. Rodríguez-Hernández MJ, Cuberos L, Pichardo C, et al. Sulbactam efficacy in experimental models caused by susceptible and intermediate *Acinetobacter baumannii* strains. J Antimicrob Chemother **2001**; 47:479–82.
518. Abouelhassan Y, Kuti JL, Nicolau DP, Abdelraouf K. Ampicillin-sulbactam against *Acinetobacter baumannii* infections: pharmacokinetic/pharmacodynamic appraisal of current susceptibility breakpoints and dosing recommendations. J Antimicrob Chemother **2024**; 79:2227–36.
519. Abouelhassan Y, Nicolau DP, Abdelraouf K. Defining optimal sulbactam regimens for treatment of *Acinetobacter baumannii* pneumonia and impact of blaOXA-23 on efficacy. J Antimicrob Chemother **2024**; 79:2306–16.
520. Liu J, Shu Y, Zhu F, et al. Comparative efficacy and safety of combination therapy with high-dose sulbactam or colistin with additional antibacterial agents for multiple drug-resistant and extensively drug-resistant *Acinetobacter baumannii* infections: a systematic review and network meta-analysis. J Glob Antimicrob Resist **2021**; 24:136–47.
521. Jung SY, Lee SH, Lee SY, et al. Antimicrobials for the treatment of drug-resistant *Acinetobacter baumannii* pneumonia in critically ill patients: a systemic review and Bayesian network meta-analysis. Crit Care **2017**; 21:319.
522. SENTRY Antimicrobial Surveillance Program. **2025**. JMI laboratories; Available at: https://sentry-mvp.jmilabs.com. Accessed 31 December 2025.
523. Pfizer ATLAS Antimicrobial Testing Leadership and Surveillance. **2025**. Available at: https://atlas-surveillance.com/login. Accessed 31 December 2025.
524. Jaruratanasirikul S, Nitchot W, Wongpoowarak W, Samaeng M, Nawakitrangsan M. Population pharmacokinetics and Monte Carlo simulations of sulbactam to optimize dosage regimens in patients with ventilator-associated pneumonia caused by *Acinetobacter baumannii*. Eur J Pharm Sci **2019**; 136: 104940.
525. Jaruratanasirikul S, Wongpoowarak W, Wattanavijitkul T, et al. Population pharmacokinetics and pharmacodynamics modeling to optimize dosage regimens of sulbactam in critically ill patients with severe sepsis caused by *Acinetobacter baumannii*. Antimicrob Agents Chemother **2016**; 60:7236–44.
526. Fernández-Cuenca F, Tomás M, Caballero-Moyano FJ, et al. Reporting antimicrobial susceptibilities and resistance phenotypes in *Acinetobacter* spp: a nationwide proficiency study. J Antimicrob Chemother **2018**; 73:692–7.
527. Viana GF, Saalfeld SM, Moreira RR, et al. Can ampicillin/sulbactam resistance in *Acinetobacter baumannii* be predicted accurately by disk diffusion? J Glob Antimicrob Resist **2013**; 1:221–3.

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528. Khalili H, Shojaei L, Mohammadi M, Beigmohammadi MT, Abdollahi A, Doomanlou M. Meropenem/colistin versus meropenem/ampicillin-sulbactam in the treatment of carbapenem-resistant pneumonia. J Comp Eff Res **2018**; 7: 901–11.
529. Pourheidar E, Haghighi M, Kouchek M, et al. Comparison of intravenous ampicillin-sulbactam plus nebulized colistin with intravenous colistin plus nebulized colistin in treatment of ventilator associated pneumonia caused by multi drug resistant *Acinetobacter baumannii*: randomized open label trial. Iran J Pharm Res **2019**; 18:269–81.
530. Makris D, Petinaki E, Tsolaki V, et al. Colistin versus colistin combined with ampicillin-sulbactam for multiresistant *Acinetobacter baumannii* ventilator-associated pneumonia treatment: an open-label prospective study. Indian J Crit Care Med **2018**; 22:67–77.
531. Mosaed R, Haghighi M, Kouchak M, et al. Interim study: comparison of safety and efficacy of levofloxacin plus colistin regimen with levofloxacin plus high dose ampicillin/sulbactam infusion in treatment of ventilator-associated pneumonia due to multi drug resistant Acinetobacter. Iran J Pharm Res **2018**; 17: 206–13.
532. Asrat H, Samaroo-Campbell J, Ata S, Quale J. Contribution of iron-transport systems and β-lactamases to cefiderocol resistance in clinical isolates of *Acinetobacter baumannii* endemic to New York City. Antimicrob Agents Chemother **2023**; 67:e0023423.
533. Smoke SM, Brophy A, Reveron S, et al. Evolution and transmission of cefiderocol-resistant *Acinetobacter baumannii* during an outbreak in the burn intensive care unit. Clin Infect Dis **2023**; 76:e1261–5.
534. Raro OHF, Findlay J, Poirel L, Decousser JW, Nordmann P. In vitro selection of cefiderocol-resistant mutants in *Acinetobacter baumannii* harbouring the most common carbapenemase genes. J Antimicrob Chemother **2026**; 81:dkaf462.
535. Poirel L, Sadek M, Nordmann P. Contribution of PER-type and NDM-type β-lactamases to cefiderocol resistance in *Acinetobacter baumannii*. Antimicrob Agents Chemother **2021**; 65:e0087721.
536. Nakamura R, Ito-Horiyama T, Takemura M, et al. In vivo pharmacodynamic study of cefiderocol, a novel parenteral siderophore cephalosporin, in murine thigh and lung infection models. Antimicrob Agents Chemother **2019**; 63: e02031–18.
537. Monogue ML, Tsuji M, Yamano Y, Echols R, Nicolau DP. Efficacy of humanized exposures of cefiderocol (S-649266) against a diverse population of gram-negative bacteria in a murine thigh infection model. Antimicrob Agents Chemother **2017**; 61:e01022–17.
538. Stainton SM, Monogue ML, Tsuji M, Yamano Y, Echols R, Nicolau DP. Efficacy of humanized cefiderocol exposures over 72 hours against a diverse group of gram-negative isolates in the neutropenic murine thigh infection model. Antimicrob Agents Chemother **2019**; 63:e01040–18.
539. Matsumoto S, Singley CM, Hoover J, et al. Efficacy of cefiderocol against carbapenem-resistant gram-negative bacilli in immunocompetent-rat respiratory tract infection models recreating human plasma pharmacokinetics. Antimicrob Agents Chemother **2017**; 61:e00700–17.
540. Wunderink RG, Matsunaga Y, Ariyasu M, et al. Cefiderocol versus high-dose, extended-infusion meropenem for the treatment of Gram-negative nosocomial pneumonia (APEKS-NP): a randomised, double-blind, phase 3, non-inferiority trial. Lancet Infect Dis **2021**; 21:213–25.
541. Onorato L, de Luca I, Monari C, Coppola N. Cefiderocol either in monotherapy or combination versus best available therapy in the treatment of carbapenem-resistant *Acinetobacter baumannii* infections: a systematic review and meta-analysis. J Infect **2024**; 88:106113.
542. Zhan Y, Mao W, Zhao C, et al. Comparison of cefiderocol and colistin-based regimens for the treatment of severe infections caused by carbapenem-resistant *Acinetobacter baumannii*: a systematic review with meta-analysis and trial sequential analysis. BMC Infect Dis **2024**; 24:967.
543. Castanheira M, Mendes RE, Jones RN. Update on *Acinetobacter* species: mechanisms of antimicrobial resistance and contemporary in vitro activity of minocycline and other treatment options. Clin Infect Dis **2014**; 59 Suppl 6:S367–73.
544. Lepak AJ, Trang M, Hammel JP, et al. Development of modernized *Acinetobacter baumannii* susceptibility test interpretive criteria for recommended antimicrobial agents using pharmacometric approaches. Antimicrob Agents Chemother **2023**; 67:e0145222.
545. Wang P, Bowler SL, Kantz SF, et al. Comparison of minocycline susceptibility testing methods for carbapenem-resistant *Acinetobacter baumannii*. J Clin Microbiol **2016**; 54:2937–41.
546. Rumbo C, Gato E, López M, et al. Contribution of efflux pumps, porins, and β-lactamases to multidrug resistance in clinical isolates of *Acinetobacter baumannii*. Antimicrob Agents Chemother **2013**; 57:5247–57.
547. Yang YS, Lee Y, Tseng KC, et al. In vivo and in vitro efficacy of minocycline-based combination therapy for minocycline-resistant *Acinetobacter baumannii*. Antimicrob Agents Chemother **2016**; 60:4047–54.
548. Lodise TP, Van Wart S, Sund ZM, et al. Pharmacokinetic and pharmacodynamic profiling of minocycline for injection following a single infusion in critically ill adults in a phase IV open-label multicenter study (ACUMIN). Antimicrob Agents Chemother **2021**; 65:e01809–20.
549. Goff DA, Bauer KA, Mangino JE. Bad bugs need old drugs: a stewardship program's evaluation of minocycline for multidrug-resistant *Acinetobacter baumannii* infections. Clin Infect Dis **2014**; 59 Suppl 6:S381–7.
550. Ritchie DJ, Garavaglia-Wilson A. A review of intravenous minocycline for treatment of multidrug-resistant *Acinetobacter* infections. Clin Infect Dis **2014**; 59 Suppl 6:S374–80.
551. Wood GC, Hanes SD, Boucher BA, Croce MA, Fabian TC. Tetracyclines for treating multidrug-resistant *Acinetobacter baumannii* ventilator-associated pneumonia. Intensive Care Med **2003**; 29:2072–6.
552. Chen Z, Shi X. Adverse events of high-dose tigecycline in the treatment of ventilator-associated pneumonia due to multidrug-resistant pathogens. Medicine (Baltimore) **2018**; 97:e12467.
553. Mei H, Yang T, Wang J, Wang R, Cai Y. Efficacy and safety of tigecycline in treatment of pneumonia caused by MDR *Acinetobacter baumannii*: a systematic review and meta-analysis. J Antimicrob Chemother **2019**; 74:3423–31.
554. Zhang X, Xie Y, Liu F, et al. Simulated lung exposure in pneumonia patients to evaluate the antibacterial capacity and resistance suppression of clinical tigecycline regimens against carbapenem-resistant *Acinetobacter baumannii*. J Antimicrob Chemother **2026**; 81:dkaf477.
555. Luo X, Wang S, Li D, Wen J, Sun N, Fan G. Population pharmacokinetics of tigecycline in critically ill patients. Front Pharmacol **2023**; 14:1083464.
556. Xie J, Roberts JA, Alobaid AS, et al. Population pharmacokinetics of tigecycline in critically ill patients with severe infections. Antimicrob Agents Chemother **2017**; 61:e00345–17.
557. De Pascale G, Lisi L, Ciotti GMP, et al. Pharmacokinetics of high-dose tigecycline in critically ill patients with severe infections. Ann Intensive Care **2020**; 10:94.
558. Noel AR, Attwood M, Bowker KE, MacGowan AP. In vitro pharmacodynamics of omadacycline against *Escherichia coli* and *Acinetobacter baumannii*. J Antimicrob Chemother **2021**; 76:667–70.
559. Vanachayangkul P, Ruamsap N, Caridha D, et al. Preclinical in vitro and in vivo evaluation of omadacycline against multidrug-resistant *Acinetobacter baumannii*. Antimicrob Agents Chemother **2025**; 69:e0093225.
560. Morrisette T, Alosaimy S, Lagnf AM, et al. Real-world, multicenter case series of patients treated with oral omadacycline for resistant gram-negative pathogens. Infect Dis Ther **2022**; 11:1715–23.
561. Chen D, Chen Y, Hu N, et al. A matched pilot cohort study of intravenous omadacycline in the treatment of severe pneumonia associated with carbapenem-resistant *Acinetobacter baumannii*. Front Microbiol **2025**; 16:1597860.
562. Fanning WL, Gump DW. Distressing side-effects of minocycline hydrochloride. Arch Intern Med **1976**; 136:761–2.
563. Minocycline. FDA drug label. Food and Drug Administration. Available at: https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=e415c323-8219-464b-9e3e-72c5a796cdaa. Accessed 31 December 2025.
564. Zilberberg MD, Kollef MH, Shorr AF. Secular trends in *Acinetobacter baumannii* resistance in respiratory and blood stream specimens in the United States, 2003 to 2012: a survey study. J Hosp Med **2016**; 11:21–6.
565. McKay SL, Vlachos N, Daniels JB, et al. Molecular epidemiology of carbapenem-resistant *Acinetobacter baumannii* in the United States, 2013–2017. Microb Drug Resist **2022**; 28:645–53.
566. Nation RL, Velkov T, Li J. Colistin and polymyxin B: peas in a pod, or chalk and cheese? Clin Infect Dis **2014**; 59:88–94.
567. Cheah SE, Wang J, Nguyen VT, Turnidge JD, Li J, Nation RL. New pharmacokinetic/pharmacodynamic studies of systemically administered colistin against *Pseudomonas aeruginosa* and *Acinetobacter baumannii* in mouse thigh and lung infection models: smaller response in lung infection. J Antimicrob Chemother **2015**; 70:3291–7.
568. Landersdorfer CB, Wang J, Wirth V, et al. Pharmacokinetics/pharmacodynamics of systemically administered polymyxin B against *Klebsiella pneumoniae* in mouse thigh and lung infection models. J Antimicrob Chemother **2018**; 73: 462–8.
569. Montero A, Ariza J, Corbella X, et al. Efficacy of colistin versus beta-lactams, aminoglycosides, and rifampin as monotherapy in a mouse model of pneumonia caused by multiresistant *Acinetobacter baumannii*. Antimicrob Agents Chemother **2002**; 46:1946–52.
570. Kaye KS, Marchaim D, Thamlikitkul V, et al. Colistin monotherapy versus combination therapy for carbapenem-resistant organisms. NEJM Evid **2023**; 2:.

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571. Paul M, Daikos GL, Durante-Mangoni E, et al. Colistin alone versus colistin plus meropenem for treatment of severe infections caused by carbapenem-resistant Gram-negative bacteria: an open-label, randomised controlled trial. Lancet Infect Dis **2018**; 18:391–400.
572. Durante-Mangoni E, Signoriello G, Andini R, et al. Colistin and rifampicin compared with colistin alone for the treatment of serious infections due to extensively drug-resistant *Acinetobacter baumannii*: a multicenter, randomized clinical trial. Clin Infect Dis **2013**; 57:349–58.
573. Park HJ, Cho JH, Kim HJ, Han SH, Jeong SH, Byun MK. Colistin monotherapy versus colistin/rifampicin combination therapy in pneumonia caused by colistin-resistant *Acinetobacter baumannii*: a randomised controlled trial. J Glob Antimicrob Resist **2019**; 17:66–71.
574. Aydemir H, Akduman D, Piskin N, et al. Colistin vs. the combination of colistin and rifampicin for the treatment of carbapenem-resistant *Acinetobacter baumannii* ventilator-associated pneumonia. Epidemiol Infect **2013**; 141:1214–22.
575. Betrosian AP, Frantzeskaki F, Xanthaki A, Georgiadis G. High-dose ampicillin-sulbactam as an alternative treatment of late-onset VAP from multidrug-resistant *Acinetobacter baumannii*. Scand J Infect Dis **2007**; 39:38–43.
576. Gröschel MI, Meehan CJ, Barilar I, et al. The phylogenetic landscape and nosocomial spread of the multidrug-resistant opportunist *Stenotrophomonas maltophilia*. Nat Commun **2020**; 11:2044.
577. Wang N, Tang C, Wang L. Risk factors for acquired *Stenotrophomonas maltophilia* pneumonia in intensive care unit: a systematic review and meta-analysis. Front Med (Lausanne) **2021**; 8:808391.
578. Aitken SL, Sahasrabhojane PV, Kontoyiannis DP, et al. Alterations of the oral microbiome and cumulative carbapenem exposure are associated with *Stenotrophomonas maltophilia* infection in patients with acute myeloid leukemia receiving chemotherapy. Clin Infect Dis **2021**; 72:1507–13.
579. Karaba SM, Goodman KE, Amoah J, Cosgrove SE, Tamma PD. StenoSCORE: predicting *Stenotrophomonas maltophilia* bloodstream infections in the hematologic malignancy population. Antimicrob Agents Chemother **2021**; 65: e0079321.
580. Kunz Coyne AJ, Lucas K, Gray R, May E. Propensity-matched comparison of timely vs delayed antibiotic therapy in *Stenotrophomonas maltophilia* pneumonia. Open Forum Infect Dis **2025**; 12:ofaf469.
581. Gezer Y, Tayşi MR, Tarakçı A, et al. Evaluation of clinical outcomes and risk factors associated with mortality in patients with *Stenotrophomonas maltophilia* bloodstream infection: a multicenter study. BMC Infect Dis **2024**; 24:1387.
582. Khunkit P, Siripaitoon P, Lertsrisatit Y, et al. Impact of empirical antimicrobial treatment on patients with ventilator-associated pneumonia due to *Stenotrophomonas maltophilia*. Antibiotics (Basel) **2024**; 13:729.
583. Gajdacs M, Urban E. Prevalence and antibiotic resistance of *Stenotrophomonas maltophilia* in respiratory tract samples: a 10-year epidemiological snapshot. Health Serv Res Manag Epidemiol **2019**; 6:2333392819870774.
584. Cho SY, Lee DG, Choi SM, et al. *Stenotrophomonas maltophilia* bloodstream infection in patients with hematologic malignancies: a retrospective study and in vitro activities of antimicrobial combinations. BMC Infect Dis **2015**; 15:69.
585. Kim SH, Cha MK, Kang CI, et al. Pathogenic significance of hemorrhagic pneumonia in hematologic malignancy patients with *Stenotrophomonas maltophilia* bacteremia: clinical and microbiological analysis. Eur J Clin Microbiol Infect Dis **2019**; 38:285–95.
586. Imoto W, Yamada K, Yamairi K, et al. Clinical characteristics of rapidly progressive fatal hemorrhagic pneumonia caused by *Stenotrophomonas maltophilia*. Intern Med **2020**; 59:193–8.
587. Tada K, Kurosawa S, Hiramoto N, et al. *Stenotrophomonas maltophilia* infection in hematopoietic SCT recipients: high mortality due to pulmonary hemorrhage. Bone Marrow Transplant **2013**; 48:74–9.
588. Araoka H, Fujii T, Izutsu K, et al. Rapidly progressive fatal hemorrhagic pneumonia caused by *Stenotrophomonas maltophilia* in hematologic malignancy. Transpl Infect Dis **2012**; 14:355–63.
589. Crossman LC, Gould VC, Dow JM, et al. The complete genome, comparative and functional analysis of *Stenotrophomonas maltophilia* reveals an organism heavily shielded by drug resistance determinants. Genome Biol **2008**; 9:R74.
590. Mojica MF, Rutter JD, Taracila M, et al. Population structure, molecular epidemiology, and β-lactamase diversity among *Stenotrophomonas maltophilia* isolates in the United States. mBio **2019**; 10:e00405–19.
591. Lambert T, Ploy MC, Denis F, Courvalin P. Characterization of the chromosomal aac(6')-Iz gene of *Stenotrophomonas maltophilia*. Antimicrob Agents Chemother **1999**; 43:2366–71.
592. Gordon NC, Wareham DW. Novel variants of the Smqnr family of quinolone resistance genes in clinical isolates of *Stenotrophomonas maltophilia*. J Antimicrob Chemother **2010**; 65:483–9.
593. Alonso A, Martínez JL. Cloning and characterization of SmeDEF, a novel multidrug efflux pump from *Stenotrophomonas maltophilia*. Antimicrob Agents Chemother **2000**; 44:3079–86.
594. Sánchez MB, Martínez JL. Overexpression of the efflux pumps SmeVWX and SmeDEF is a major cause of resistance to co-trimoxazole in *Stenotrophomonas maltophilia*. Antimicrob Agents Chemother **2018**; 62:e00301–18.
595. Khan A, Pettaway C, Dien Bard J, Arias CA, Bhatti MM, Humphries RM. Evaluation of the performance of manual antimicrobial susceptibility testing methods and disk breakpoints for *Stenotrophomonas maltophilia*. Antimicrob Agents Chemother **2023**; 95:e02631–20.
596. Khan A, Arias CA, Abbott A, Dien Bard J, Bhatti MM, Humphries RM. Evaluation of the Vitek 2, Phoenix, and MicroScan for antimicrobial susceptibility testing of *Stenotrophomonas maltophilia*. J Clin Microbiol **2021**; 59:e0065421.
597. Biagi M, Vialichka A, Jurkovic M, et al. Activity of cefiderocol alone and in combination with levofloxacin, minocycline, polymyxin B, or trimethoprim-sulfamethoxazole against multidrug-resistant *Stenotrophomonas maltophilia*. Antimicrob Agents Chemother **2020**; 64:e00559–20.
598. Shortridge D, Streit JM, Mendes R, Castanheira M. In vitro activity of cefiderocol against U.S. and European gram-negative clinical isolates collected in 2020 as part of the SENTRY antimicrobial surveillance program. Microbiol Spectr **2022**; 10:e0271221.
599. Sakoh T, Miyajima E, Endo Y, et al. Cefiderocol susceptibility of 146 *Stenotrophomonas maltophilia* strains clinically isolated from blood in two Japanese hospitals over a 10-year period. Eur J Clin Microbiol Infect Dis **2024**; 43:2485–8.
600. Méndez-Sotelo BJ, Delgado-Beltrán M, Hernández-Durán M, et al. In vitro activity of ceftazidime/avibactam, cefiderocol, meropenem/vaborbactam and imipenem/relebactam against clinical strains of the *Stenotrophomonas maltophilia* complex. PLoS One **2024**; 19:e0298577.
601. Sakr C, Danjean M, Cizeau F, et al. Unexpected natural cefiderocol resistance in *Stenotrophomonas maltophilia* associated to the genogroup 4 genetic background. Clin Microbiol Infect **2025**; 31:1691–6.
602. Chen IH, Kidd JM, Abdelraouf K, Nicolau DP. Comparative in vivo antibacterial activity of human-simulated exposures of cefiderocol and ceftazidime against *Stenotrophomonas maltophilia* in the murine thigh model. Antimicrob Agents Chemother **2019**; 63:e01558–19.
603. Nakamura R, Oota M, Matsumoto S, Sato T, Yamano Y. In vitro activity and in vivo efficacy of cefiderocol against *Stenotrophomonas maltophilia*. Antimicrob Agents Chemother **2021**; 65:e01436–20.
604. Petraitis V, Petraitiene R, Kavaliauskas P, et al. Efficacy of cefiderocol in experimental *Stenotrophomonas maltophilia* pneumonia in persistently neutropenic rabbits. Antimicrob Agents Chemother **2022**; 66:e0061822.
605. FDA rationale for recognition decision: cefiderocol. Available at: www.fda.gov/drugs/development-resources/fda-rationale-recognition-decision-cefiderocol Accessed 31 December 2025.
606. Torre-Cisneros J, Ferrer R, Martos CF, et al. Cefiderocol treatment for patients infected by *Stenotrophomonas maltophilia*, *Burkholderia cepacia* complex and *Achromobacter* spp.: subgroup analysis from the PERSEUS study. Eur J Clin Microbiol Infect Dis **2025**; 44:1367–74.
607. Vena A, Mezzogori L, Castaldo N, et al. Cefiderocol for the treatment of nosocomial bloodstream infections caused by *Stenotrophomonas maltophilia*: a case series and literature review. Infect Dis Ther **2025**; 14:657–69.
608. Lupia T, Carnevale-Schianca F, Vita D, et al. *Stenotrophomonas maltophilia* infections in haematological malignancies and hematopoietic stem cell transplantation: a case series including cefiderocol-based regimens. Medicina (Kaunas) **2024**; 60:88.
609. Ko JH, Kang CI, Cornejo-Juarez P, et al. Fluoroquinolones versus trimethoprim-sulfamethoxazole for the treatment of *Stenotrophomonas maltophilia* infections: a systematic review and meta-analysis. Clin Microbiol Infect **2019**; 25:546–54.
610. Biagi M, Lamm D, Meyer K, et al. Activity of aztreonam in combination with avibactam, clavulanate, relebactam, and vaborbactam against multidrug-resistant *Stenotrophomonas maltophilia*. Antimicrob Agents Chemother **2020**; 64:e00297–20.
611. Mojica MF, Papp-Wallace KM, Taracila MA, et al. Avibactam restores the susceptibility of clinical isolates of *Stenotrophomonas maltophilia* to aztreonam. Antimicrob Agents Chemother **2017**; 61:e00777–17.
612. Lin Q, Zou H, Chen X, et al. Avibactam potentiated the activity of both ceftazidime and aztreonam against *S. maltophilia* clinical isolates in vitro. BMC Microbiol **2021**; 21:60.
613. Sader HS, Castanheira M, Arends SJR, Doyle TB. Antimicrobial susceptibility of *Stenotrophomonas maltophilia* from United States medical centers (2019–2023). Antimicrob Agents Chemother **2025**; 69:e0012425.
614. Tao L, Dahlquist A, Harris H, et al. Multicenter evaluation of activity of aztreonam in combination with avibactam, relebactam, and vaborbactam against

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metallo-β-lactamase-producing carbapenem-resistant gram-negative bacilli. Antimicrob Agents Chemother **2024**; 68:e0069324.
615. Sader HS, Duncan LR, Arends SJR, Carvalhaes CG, Castanheira M. Antimicrobial activity of aztreonam-avibactam and comparator agents when tested against a large collection of contemporary *Stenotrophomonas maltophilia* isolates from medical centers worldwide. Antimicrob Agents Chemother **2020**; 64:e01433-20.
616. Sader HS, Mendes RE, Doyle TB, Winkler ML, Castanheira M. Antimicrobial susceptibility of clinical isolates of *Stenotrophomonas maltophilia* from Europe, Asia, and Latin America (2018–2023). Int J Infect Dis **2025**; 153:107803.
617. Calvopiña K, Hinchliffe P, Brem J, et al. Structural/mechanistic insights into the efficacy of nonclassical β-lactamase inhibitors against extensively drug resistant *Stenotrophomonas maltophilia* clinical isolates. Mol Microbiol **2017**; 106: 492–504.
618. Fu Y, Fouad A, Roenfanz HF, Nicolau DP, Kuti JL. Efficacy of human-simulated aztreonam-avibactam against *Stenotrophomonas maltophilia* in the neutropenic murine thigh infection model. J Antimicrob Chemother **2025**; 80:3431–7.
619. Mojica MF, Ouellette CP, Leber A, et al. Successful treatment of bloodstream infection due to metallo-β-lactamase-producing *Stenotrophomonas maltophilia* in a renal transplant patient. Antimicrob Agents Chemother **2016**; 60:5130–4.
620. Wong-So J, Magreault S, Carbonnelle E, et al. Aztreonam plus ceftazidime-avibactam for post-neurosurgical meningitis due to *Stenotrophomonas maltophilia*. Antimicrob Agents Chemother **2025**; 69: e0017225.
621. Fan X, Chen F, Ruan Z, Li L. Therapeutic challenges in managing *Stenotrophomonas maltophilia* bloodstream infection in a renal dysfunction patient: a case study and literature review. Front Cell Infect Microbiol **2025**; 15: 1694228.
622. Diarra A, Pascal L, Carpentier B, et al. Successful use of avibactam and aztreonam combination for a multiresistant *Stenotrophomonas maltophilia* bloodstream infection in a patient with idiopathic medullary aplasia. Infect Dis Now **2021**; 51:637–8.
623. Kunz Coyne AJ, Gray R, Gonnabathula P, May ES, Tamma PD, Do A. Comparative in vitro activity of aztreonam–avibactam and aztreonam plus ceftazidime–avibactam against *Stenotrophomonas maltophilia* complex. Antimicrob Agents Chemother **2025**; 70:e0145625.
624. Hölsken O, Sponheuer K, Weber F, et al. First clinical application of aztreonam-avibactam in treating carbapenem-resistant Enterobacterales: insights from therapeutic drug monitoring and pharmacokinetic simulations. J Pers Med **2024**; 14:1135.
625. Sarzynski SH, Lawandi A, Warner S, et al. Association between minimum inhibitory concentration values and mortality risk in patients with *Stenotrophomonas maltophilia* infections: a retrospective cohort study of electronic health records from 148 US hospitals. JAC Antimicrob Resist **2023**; 5:dlad049.
626. Baek JH, Kim CO, Jeong SJ, et al. Clinical factors associated with acquisition of resistance to levofloxacin in *Stenotrophomonas maltophilia*. Yonsei Med J **2014**; 55:987–93.
627. Nys C, Cherabuddi K, Venugopalan V, Klinker KP. Clinical and microbiologic outcomes in patients with monomicrobial *Stenotrophomonas maltophilia* infections. Antimicrob Agents Chemother **2019**; 63:e00788–19.
628. Cho SY, Kang CI, Kim J, et al. Can levofloxacin be a useful alternative to trimethoprim-sulfamethoxazole for treating *Stenotrophomonas maltophilia* bacteremia? Antimicrob Agents Chemother **2014**; 58:581–3.
629. Wang YL, Scipione MR, Dubrovskaya Y, Papadopoulos J. Monotherapy with fluoroquinolone or trimethoprim-sulfamethoxazole for treatment of *Stenotrophomonas maltophilia* infections. Antimicrob Agents Chemother **2014**; 58:176–82.
630. García-León G, Ruiz de Alegría Puig C, Garcia de la Fuente C, Martínez-Martínez L, Martínez JL, Sánchez MB. High-level quinolone resistance is associated with the overexpression of smeVWX in *Stenotrophomonas maltophilia* clinical isolates. Clin Microbiol Infect **2015**; 21:464–7.
631. Wu CJ, Lu HF, Lin YT, Zhang MS, Li LH, Yang TC. Substantial contribution of SmeDEF, SmeVWX, SmQnr, and heat shock response to fluoroquinolone resistance in clinical isolates of *Stenotrophomonas maltophilia*. Front Microbiol **2019**; 10:822.
632. Zhang L, Li XZ, Poole K. Multiple antibiotic resistance in *Stenotrophomonas maltophilia*: involvement of a multidrug efflux system. Antimicrob Agents Chemother **2000**; 44:287–93.
633. Fratoni AJ, Nicolau DP, Kuti JL. Levofloxacin pharmacodynamics against *Stenotrophomonas maltophilia* in a neutropenic murine thigh infection model: implications for susceptibility breakpoint revision. J Antimicrob Chemother **2021**; 77:164–8.
634. Bonfiglio G, Cascone C, Azzarelli C, Cafiso V, Marchetti F, Stefani S. Levofloxacin in vitro activity and time-kill evaluation of *Stenotrophomonas maltophilia* clinical isolates. J Antimicrob Chemother **2000**; 45:115–7.
635. Grillon A, Schramm F, Kleinberg M, Jehl F. Comparative activity of ciprofloxacin, levofloxacin and moxifloxacin against *Klebsiella pneumoniae*, *Pseudomonas aeruginosa* and *Stenotrophomonas maltophilia* assessed by minimum inhibitory concentrations and time-kill studies. PLoS One **2016**; 11:e0156690.
636. Wei C, Ni W, Cai X, Cui J. A Monte Carlo pharmacokinetic/pharmacodynamic simulation to evaluate the efficacy of minocycline, tigecycline, moxifloxacin, and levofloxacin in the treatment of hospital-acquired pneumonia caused by *Stenotrophomonas maltophilia*. Infect Dis (Lond) **2015**; 47:846–51.
637. Imoto W, Kaneko Y, Yamada K, et al. A mouse model of rapidly progressive fatal haemorrhagic pneumonia caused by *Stenotrophomonas maltophilia*. J Glob Antimicrob Resist **2020**; 23:450–5.
638. Sarzynski SH, Warner S, Sun J, et al. Trimethoprim-sulfamethoxazole versus levofloxacin for *Stenotrophomonas maltophilia* infections: a retrospective comparative effectiveness study of electronic health records from 154 US hospitals. Open Forum Infect Dis **2022**; 9:ofab644.
639. Almangour TA, Alkherb Z, Alruwaite S, et al. Trimethoprim-sulfamethoxazole versus levofloxacin for the treatment of *Stenotrophomonas maltophilia* infections: a multicentre cohort study. J Glob Antimicrob Resist **2024**; 38:42–8.
640. Boonmee P, Nasomsong W, Lorchirachoonkul N, et al. The activities of antimicrobials against *Stenotrophomonas maltophilia* isolates and evaluation of clinical outcomes among treatment regimens in patients with *Stenotrophomonas maltophilia* infections: a retrospective multicenter cohort study. Infect Drug Resist **2023**; 16:5173–84.
641. Muder RR, Harris AP, Muller S, et al. Bacteremia due to *Stenotrophomonas* (*Xanthomonas*) *maltophilia*: a prospective, multicenter study of 91 episodes. Clin Infect Dis **1996**; 22:508–12.
642. Graves ET, Wardlow L, Ogake S, et al. Comparison of trimethoprim-sulfamethoxazole versus minocycline monotherapy for treatment of *Stenotrophomonas maltophilia* pneumonia. J Antimicrob Chemother **2025**; 80: 988–95.
643. Hand E, Davis H, Kim T, Duhon B. Monotherapy with minocycline or trimethoprim/sulfamethoxazole for treatment of *Stenotrophomonas maltophilia* infections. J Antimicrob Chemother **2016**; 71:1071–5.
644. Tekçe YT, Erbay A, Cabadak H, Sen S. Tigecycline as a therapeutic option in *Stenotrophomonas maltophilia* infections. J Chemother **2012**; 24:150–4.
645. Lai JJ, Siu LK, Chang FY, et al. Appropriate antibiotic therapy is a predictor of outcome in patients with *Stenotrophomonas maltophilia* blood stream infection in the intensive care unit. J Microbiol Immunol Infect **2023**; 56:624–33.
646. Almangour TA, Alali HA, Alkherb Z, et al. Monotherapy versus combination for the treatment of *Stenotrophomonas maltophilia*: a multicenter cohort study. Expert Rev Anti Infect Ther **2024**; 22:997–1005.
647. Curry H, Casaus D, Lucas K, Kunz Coyne AJ. Comparative evaluation of definitive therapy for deep-seated *Stenotrophomonas maltophilia* infections. JAC Antimicrob Resist **2025**; 7:dlaf215.
648. Gatti M, Bianchin M, Raschi E, De Ponti F. Assessing the association between fluoroquinolones and emerging adverse drug reactions raised by regulatory agencies: an umbrella review. Eur J Intern Med **2020**; 75:60–70.
649. Chien SC, Rogge MC, Gisclon LG, et al. Pharmacokinetic profile of levofloxacin following once-daily 500-milligram oral or intravenous doses. Antimicrob Agents Chemother **1997**; 41:2256–60.
650. Gould VC, Okazaki A, Avison MB. Coordinate hyperproduction of SmeZ and SmeJK efflux pumps extends drug resistance in *Stenotrophomonas maltophilia*. Antimicrob Agents Chemother **2013**; 57:655–7.
651. Wei C, Ni W, Cai X, Zhao J, Cui J. Evaluation of trimethoprim/sulfamethoxazole (SXT), minocycline, tigecycline, moxifloxacin, and ceftazidime alone and in combinations for SXT-susceptible and SXT-resistant *Stenotrophomonas maltophilia* by in vitro time-kill experiments. PLoS One **2016**; 11:e0152132.
652. Fratoni AJ, Nicolau DP, Kuti JL. Minocycline pharmacodynamics against *Stenotrophomonas maltophilia* in the neutropenic murine infection model: implications for susceptibility breakpoints. J Antimicrob Chemother **2022**; 77: 1052–60.
653. Hevia EC, Wooten L, Carr AL. Trimethoprim/sulfamethoxazole vs minocycline for the treatment of nonurinary monomicrobial *Stenotrophomonas maltophilia* infections in hospitalized patients. Ann Pharmacother **2024**; 58:698–704.
654. Junco SJ, Bowman MC, Turner RB. Clinical outcomes of *Stenotrophomonas maltophilia* infection treated with trimethoprim/sulfamethoxazole, minocycline, or fluoroquinolone monotherapy. Int J Antimicrob Agents **2021**; 58:106367.
655. Jacobson S, Junco Noa L, Wallace MR, Bowman MC. Clinical outcomes using minocycline for *Stenotrophomonas maltophilia* infections. J Antimicrob Chemother **2016**; 71:3620.

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656. Wu J, Zhang G, Zhao Q, Wang L, Yang J, Cui J. In vitro antimicrobial activity and dose optimization of eravacycline and other tetracycline derivatives against levofloxacin-non-susceptible and/or trimethoprim-sulfamethoxazole-resistant *Stenotrophomonas maltophilia*. Infect Drug Resist **2023**; 16:6005–15.
657. Biagi M, Tan X, Wu T, et al. Activity of potential alternative treatment agents for *Stenotrophomonas maltophilia* isolates nonsusceptible to levofloxacin and/or trimethoprim-sulfamethoxazole. J Clin Microbiol **2020**; 58:e01603–19.
658. Cai B, Tillotson G, Benjumea D, Callahan P, Echols R. The burden of bloodstream infections due to *Stenotrophomonas Maltophilia* in the United States: a large, retrospective database study. Open Forum Infect Dis **2020**; 7:ofaa141.
659. Hu LF, Chen GS, Kong QX, et al. Increase in the prevalence of resistance determinants to trimethoprim/sulfamethoxazole in clinical *Stenotrophomonas maltophilia* isolates in China. PLoS One **2016**; 11:e0157693.
660. Toleman MA, Bennett PM, Bennett DM, Jones RN, Walsh TR. Global emergence of trimethoprim/sulfamethoxazole resistance in *Stenotrophomonas maltophilia* mediated by acquisition of sul genes. Emerg Infect Dis **2007**; 13:559–65.
661. Lasko MJ, Gethers ML, Tabor-Rennie JL, Nicolau DP, Kuti JL. In vitro time-kill studies of trimethoprim/sulfamethoxazole against *Stenotrophomonas maltophilia* versus *Escherichia coli* using cation-adjusted Mueller-Hinton broth and ISO-sensitest broth. Antimicrob Agents Chemother **2022**; 66:e0216721.
662. Zelenitsky SA, Iacovides H, Ariano RE, Harding GK. Antibiotic combinations significantly more active than monotherapy in an in vitro infection model of *Stenotrophomonas maltophilia*. Diagn Microbiol Infect Dis **2005**; 51:39–43.
663. Lasko MJ, Tabor-Rennie JL, Nicolau DP, Kuti JL. Trimethoprim/sulfamethoxazole pharmacodynamics against *Stenotrophomonas maltophilia* in the in vitro chemostat model. J Antimicrob Chemother **2022**; 77:3187–93.
664. Preyra R, Eddin LE, Ahmadi F, Jafari A, Muanda FT. Safety of sulfamethoxazole-trimethoprim for the treatment of bacterial infection in outpatient settings: a systematic review and meta-analysis with active comparator disproportionality analysis. Br J Clin Pharmacol **2025**; 91:1632–48.
665. Klepser ME, Zhu Z, Nicolau DP, et al. Oral absorption of trimethoprim-sulfamethoxazole in patients with AIDS. Pharmacotherapy **1996**; 16:656–62.

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